LHC Status and plans

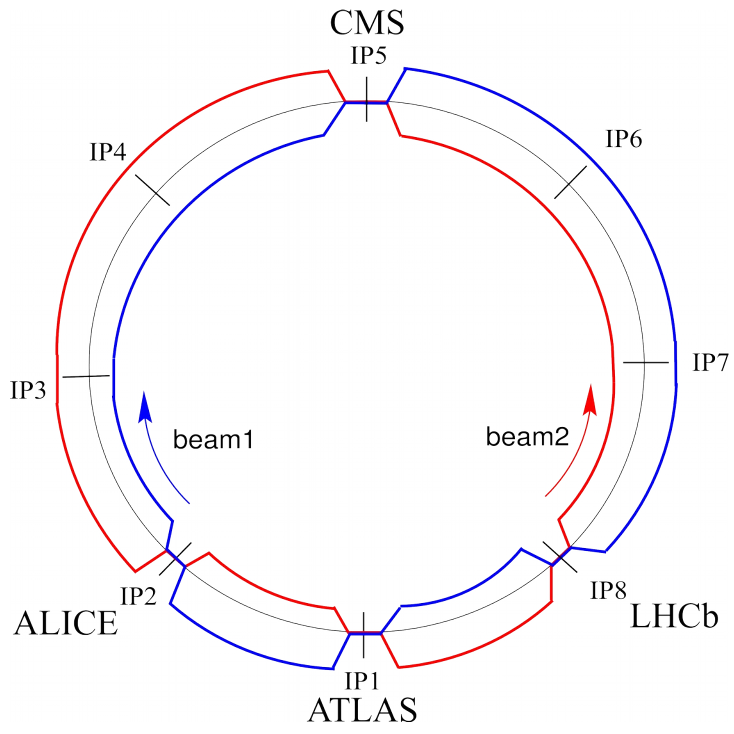
X. Buffat

on behalf of CERN Accelerator and Technology Sector

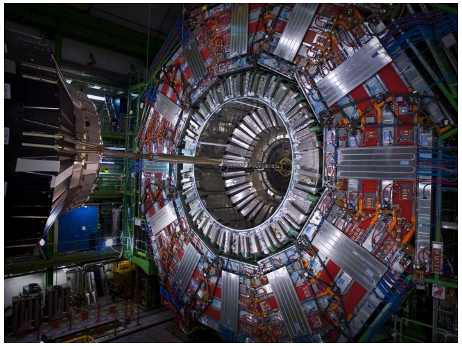
Content

- Introduction
- Highlights from run 2
- The High-Luminosity LHC
- Challenges for run 3

Introduction



Introduction



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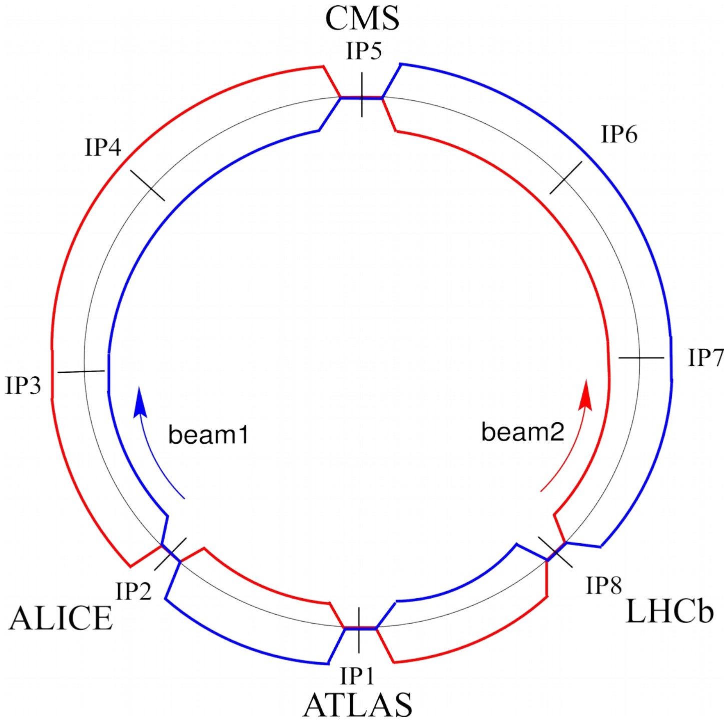
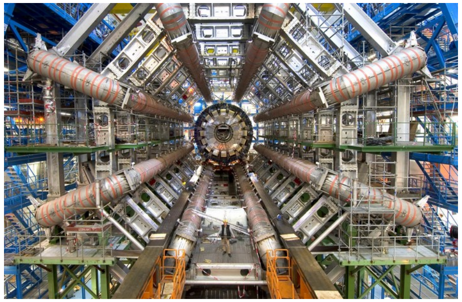


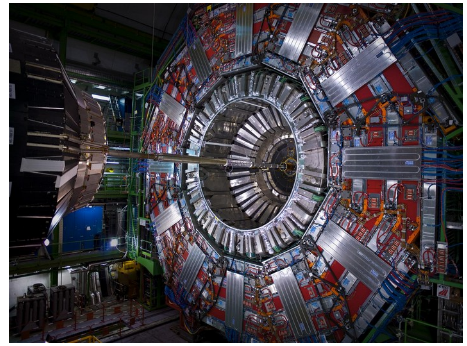


160 fb^{-1}

*Luminosity integrated in run 2

https://lpc.web.cern.ch/

Introduction



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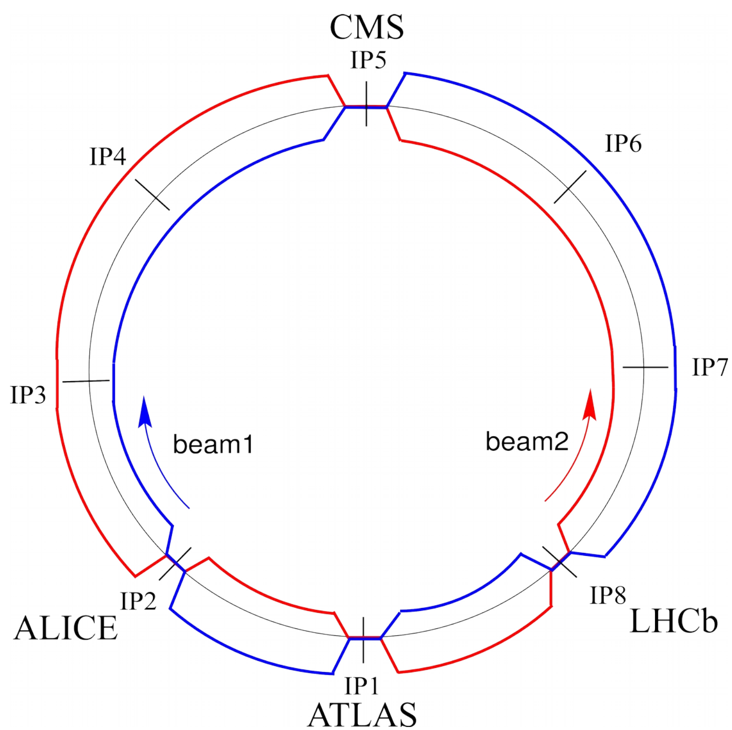
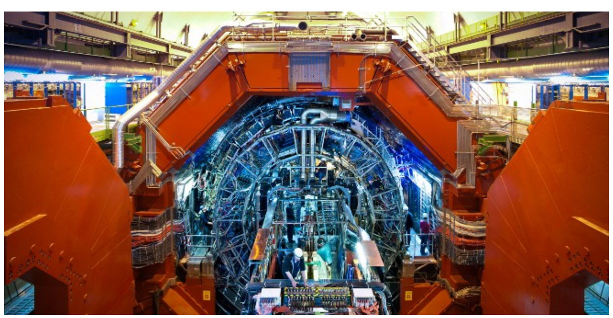
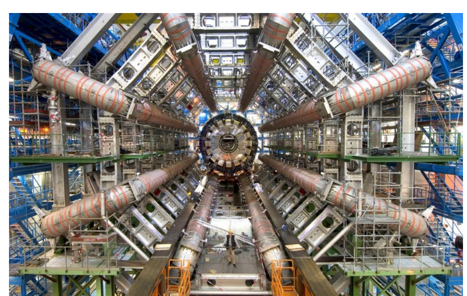




3.3 pb^{-1}



160 fb^{-1}

6.7 fb^{-1}

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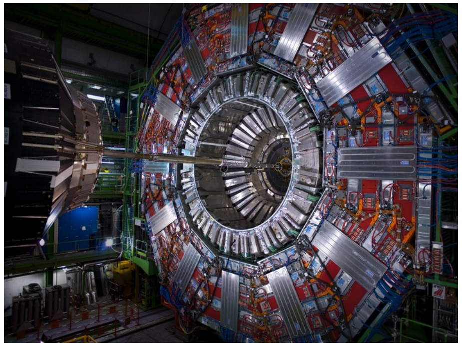
Introduction

	Design	2018 proton run
Beam energy [TeV]	7	6.5
Peak luminosity [10^{34} Hz/cm^2]	1	**2.1**
Number of bunches	2808	2556
Bunch charge [10^{11} p]	1.15	1.2
Stored energy [MJ]	362	320
β* [cm]	55	30 → 25
Data taking time prop.		**49%**



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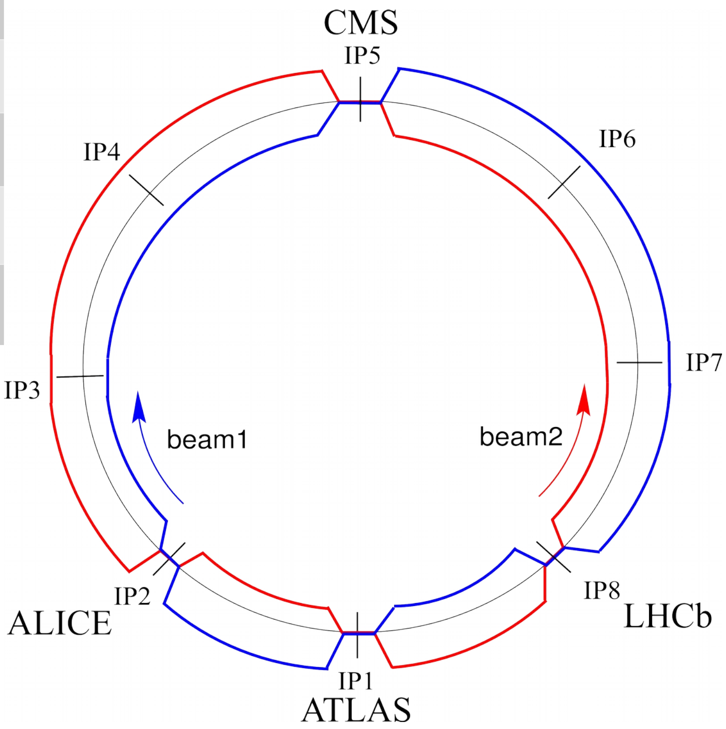

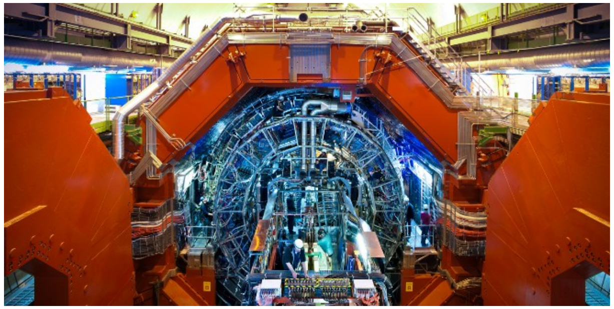

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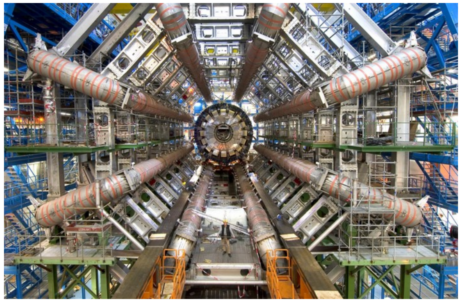

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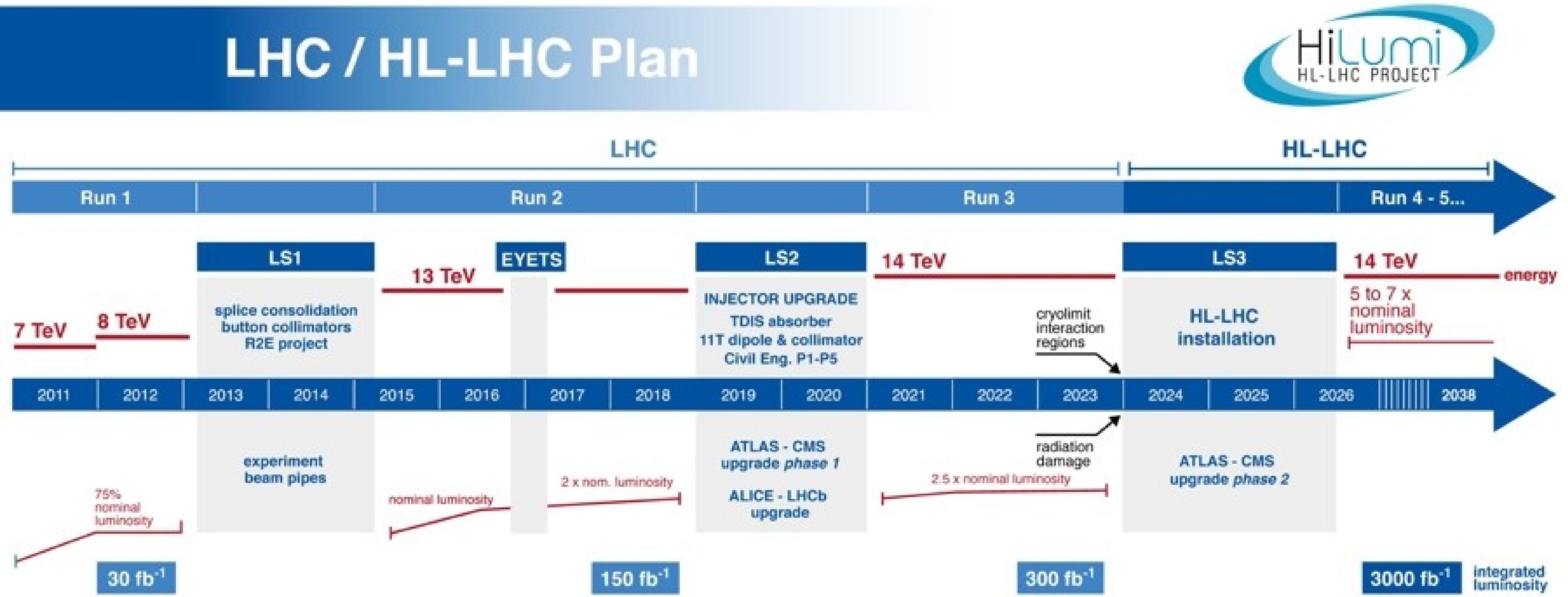
Run 2 highlights



http://hilumilhc.web.cern.ch/

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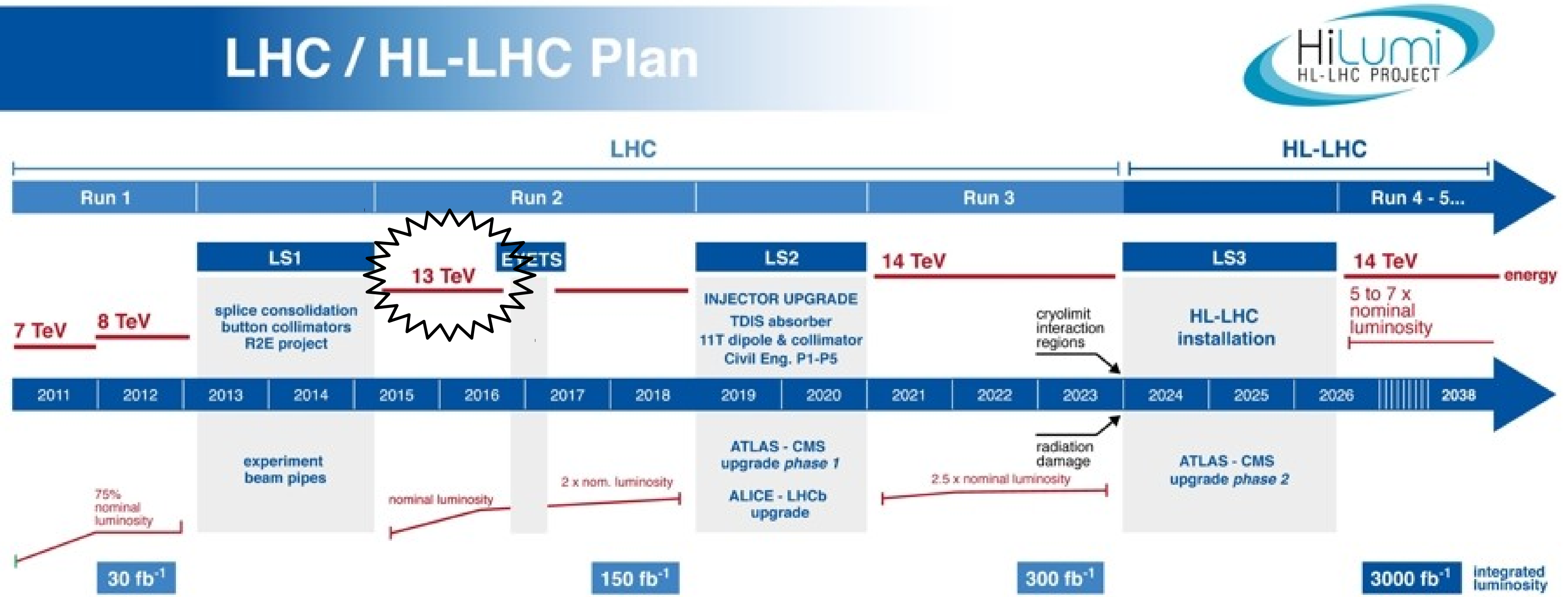

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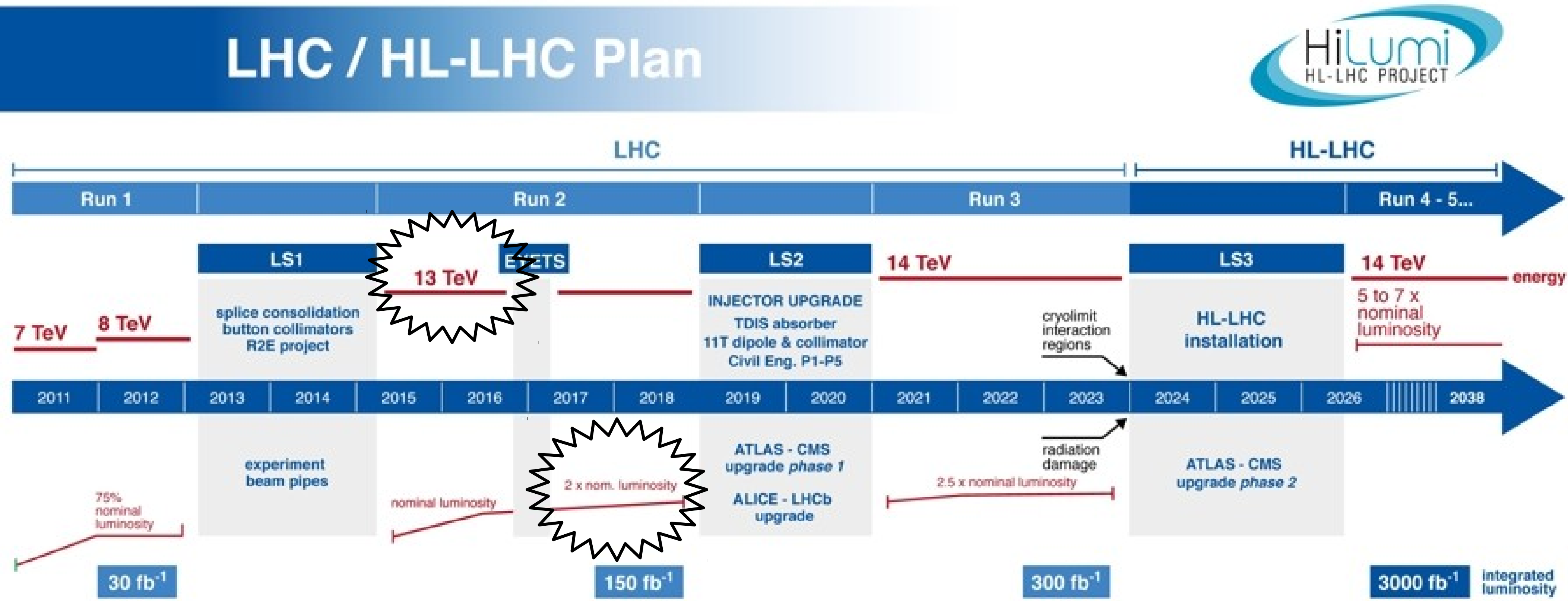

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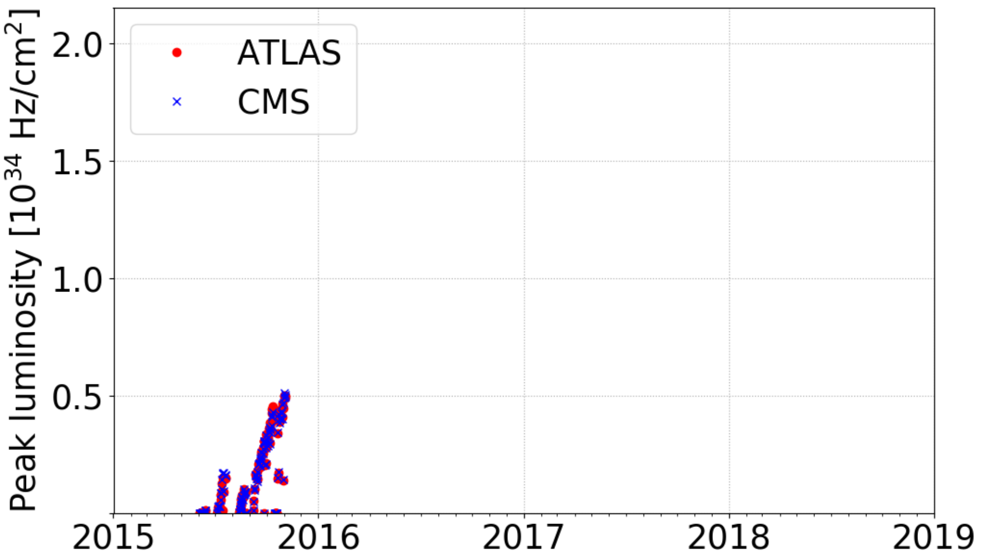
Performance with proton

- Recovery from the first long shut down with
  - Higher energy **4 → 6.5 TeV**
  - Reduced bunch spacing **50 → 25 ns**



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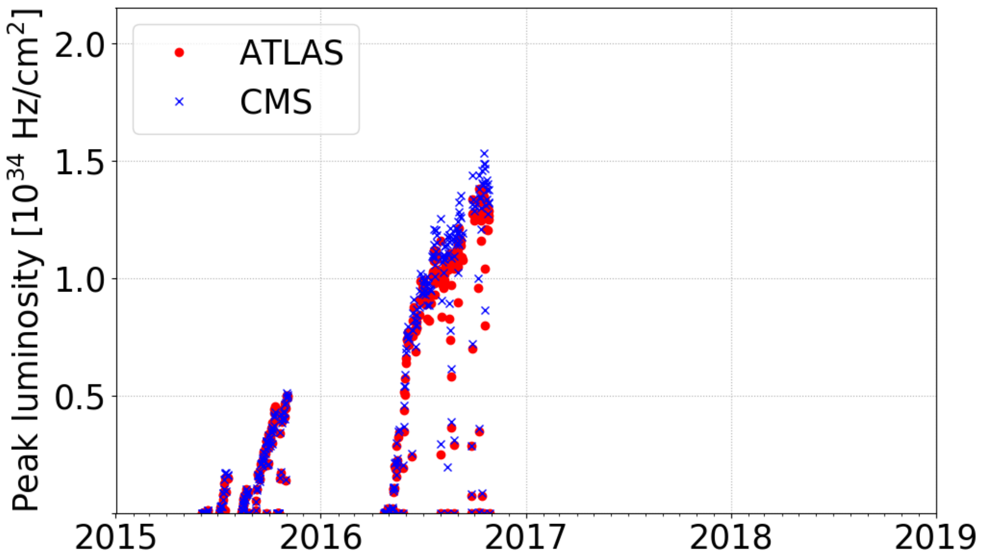

- Production with:
  - Lower β* **80 → 40 cm**
  - Reduced crossing angle
  - Higher brightness from the injectors (**BCMS** beams)
  - Combined ramp and squeeze

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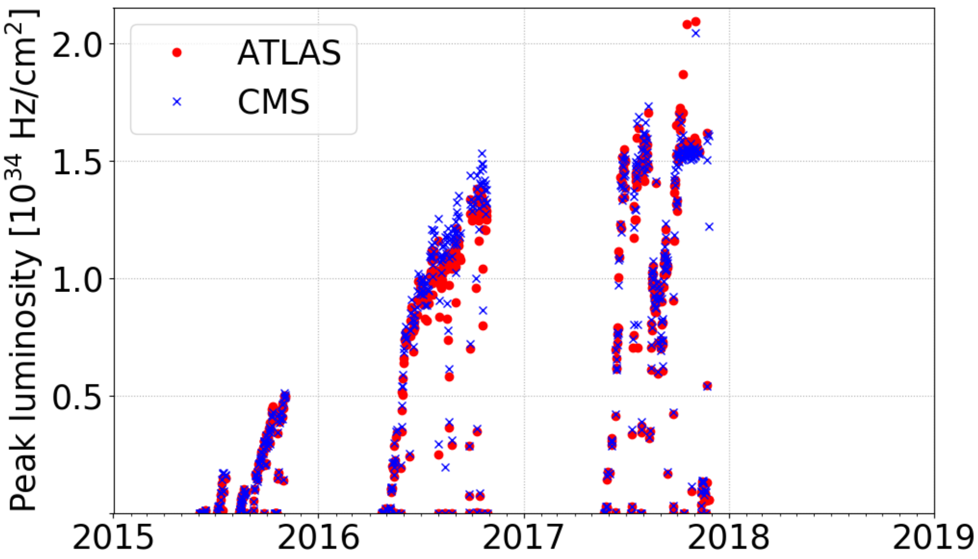

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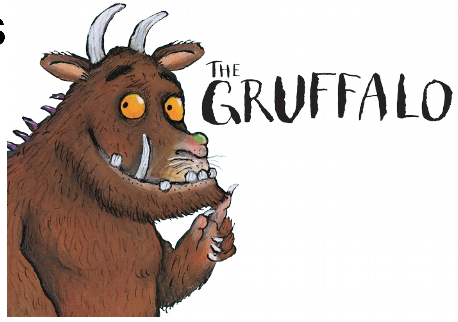
  - Non-linear optics corrections
  - Crossing angle levelling
  - BCMS and 8b4e beams



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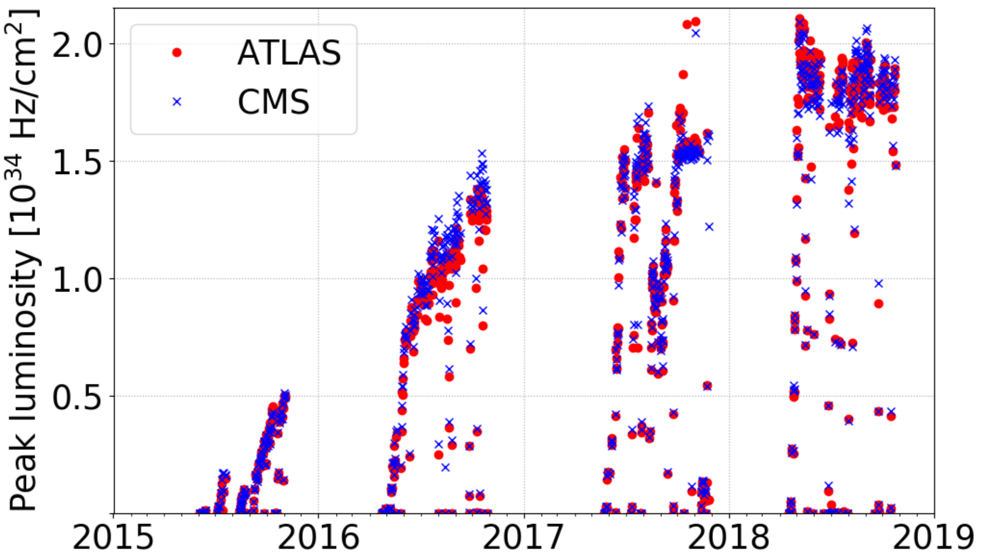

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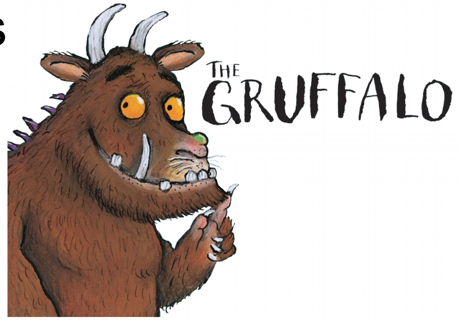
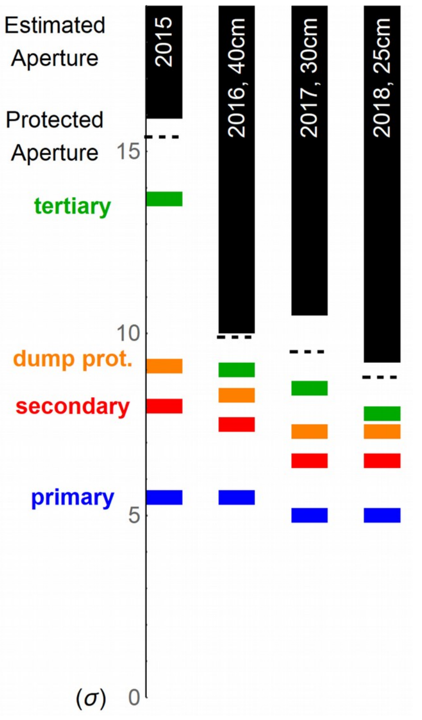
- Production with:
  - Crossing angle and β* levelling → **25 cm**
  - BCMS beam
  - Higher bunch intensity



Achieving low β* : Collimation



- A tight collimation scheme is required to efficiently and safely use the physical aperture of the triplet

  → Machine stability and reproducibility

  → Optics control

R. Bruce, et al., Nucl. Instrum. Methods Phys. Res. A 848, 19 (2017)

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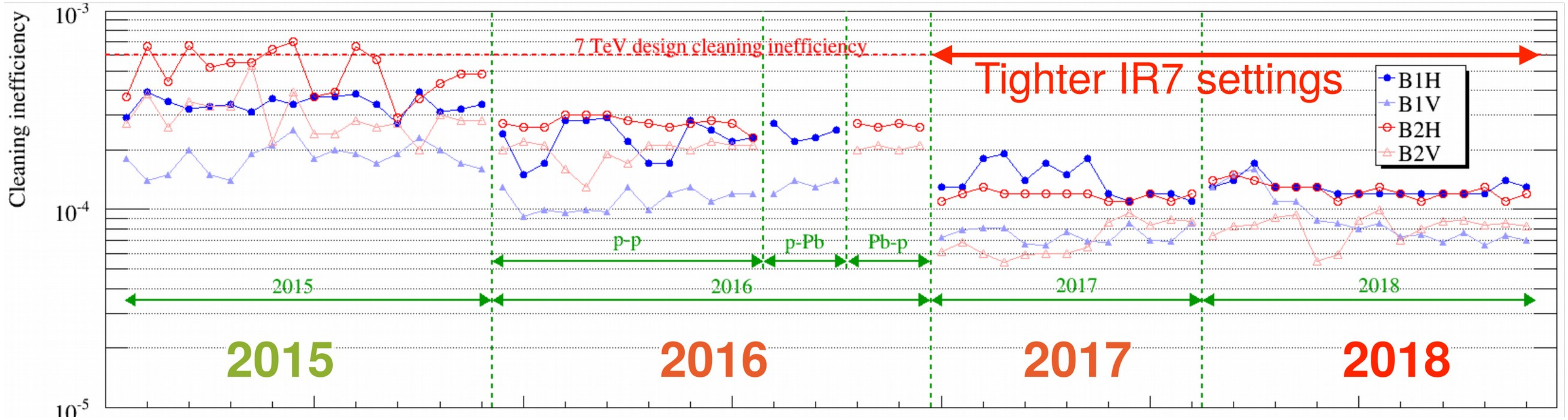

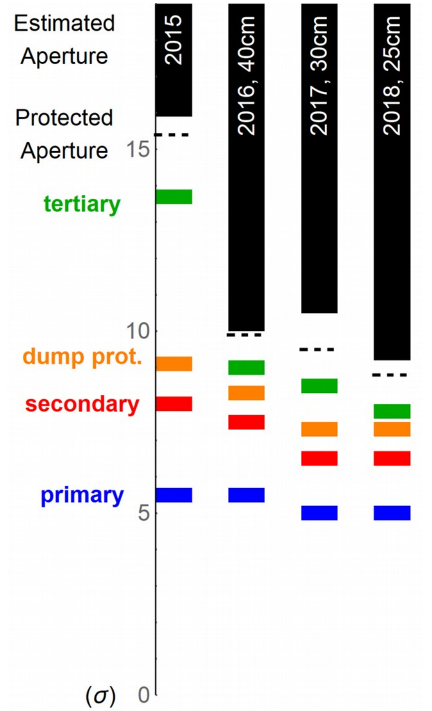

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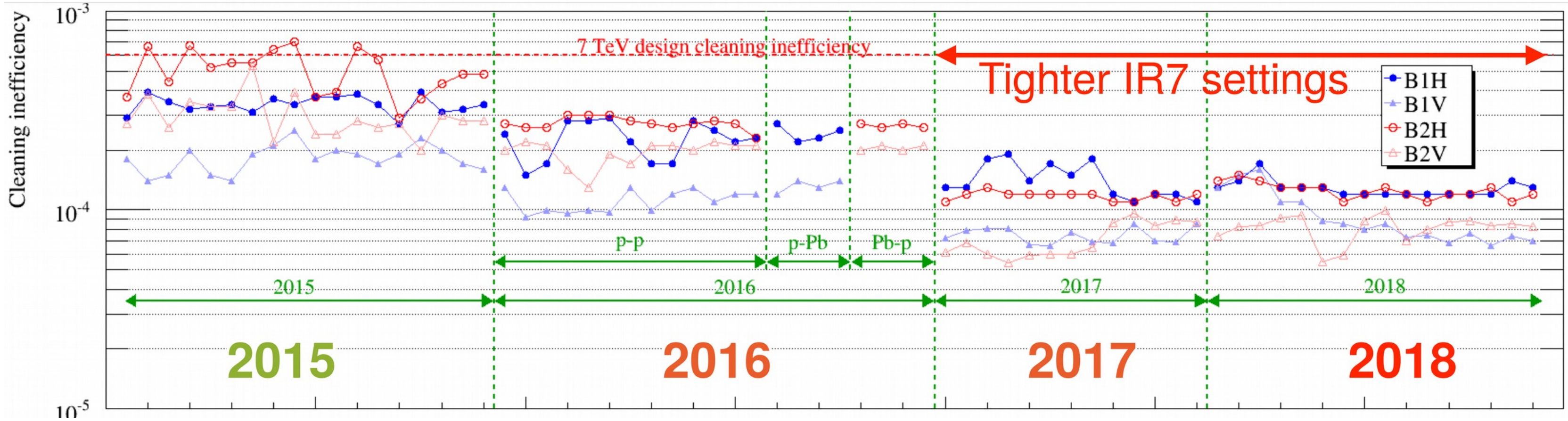
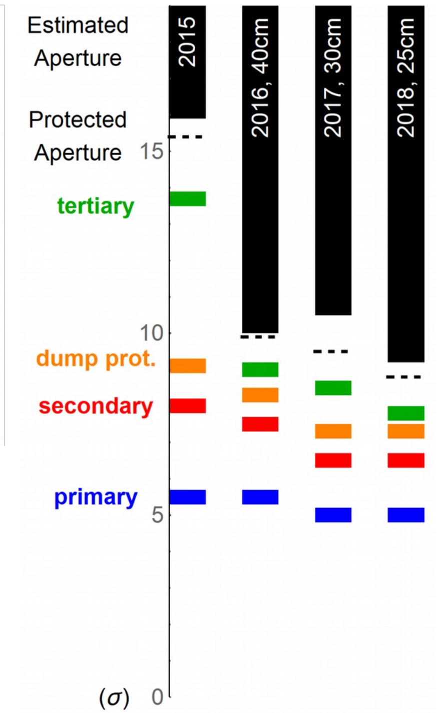




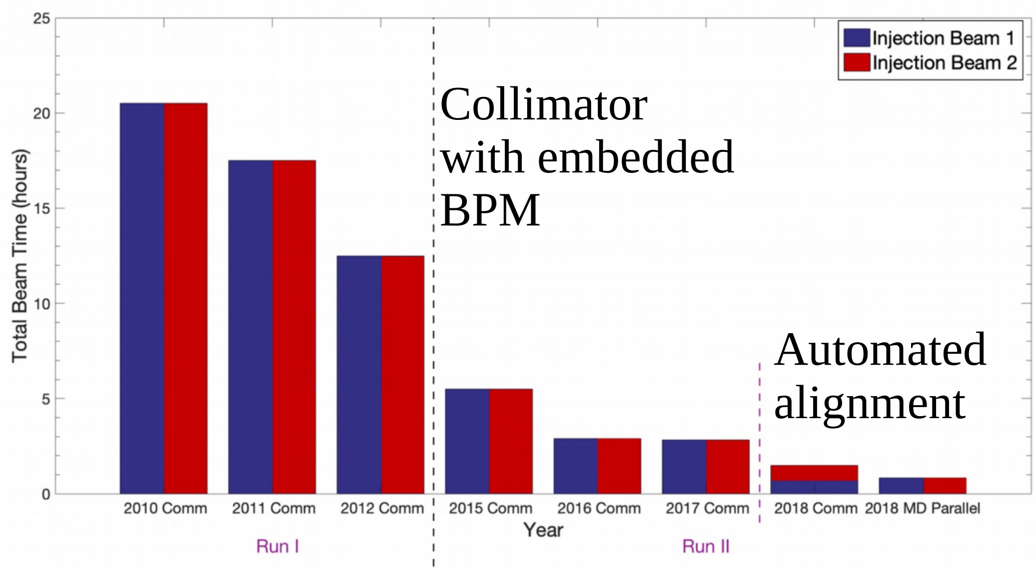
- A tight collimation scheme is required to efficiently and safely use the physical aperture of the triplet

→ Machine stability and reproducibility

→ Optics control

→ Alignment and validation

→ Beam stability (Impedance)



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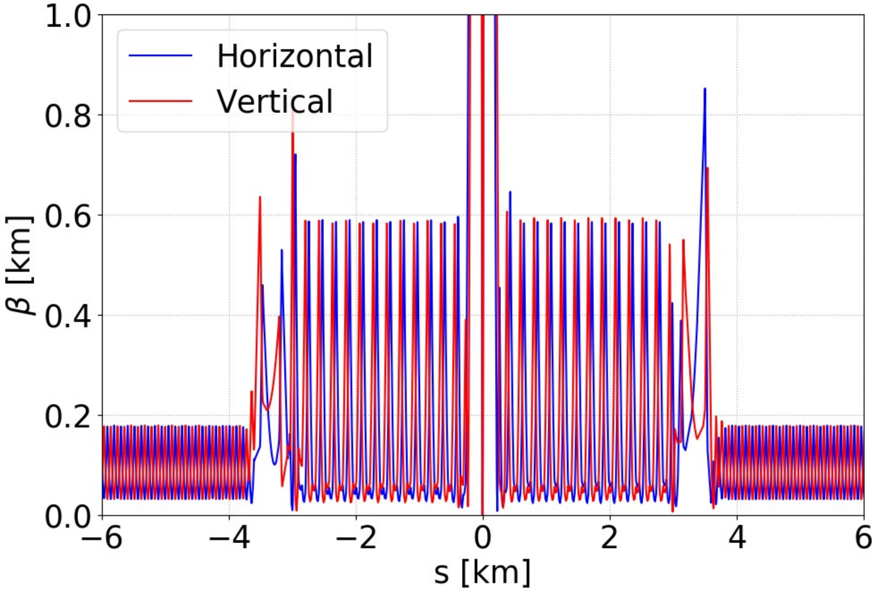
Achieving low β* : Optics

- Chromatic correction with low β* is ensured with the ATS optics



S. Fartoukh, Phys. Rev. ST Accel. Beams 16, 111002 (2013) and CERN-ACC-2018-0032

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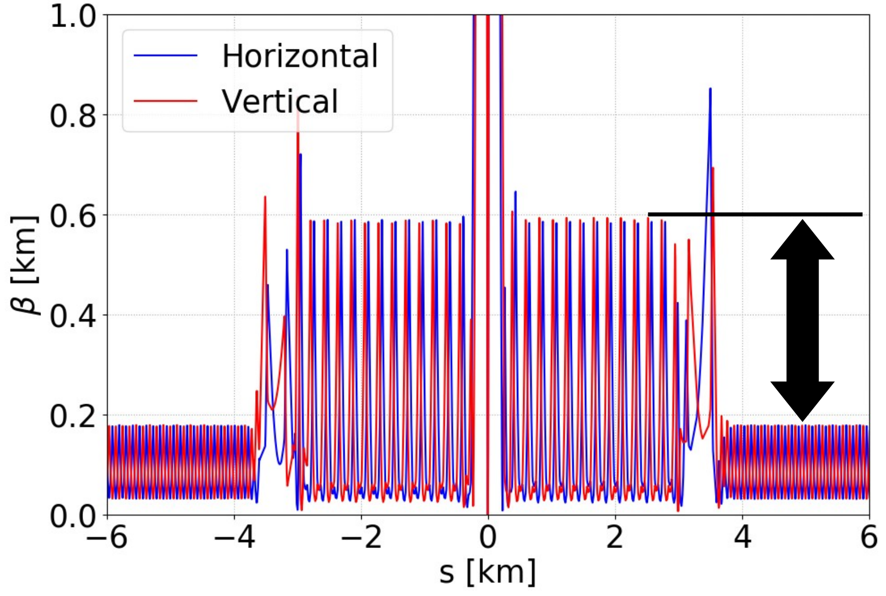

– Demonstrated regular operation a tele-index of **1.6** and up to **3.1** dedicated tests

+ additional benefits :

→ Long-range beam-beam compensation
→ Enhancement of Landau damping

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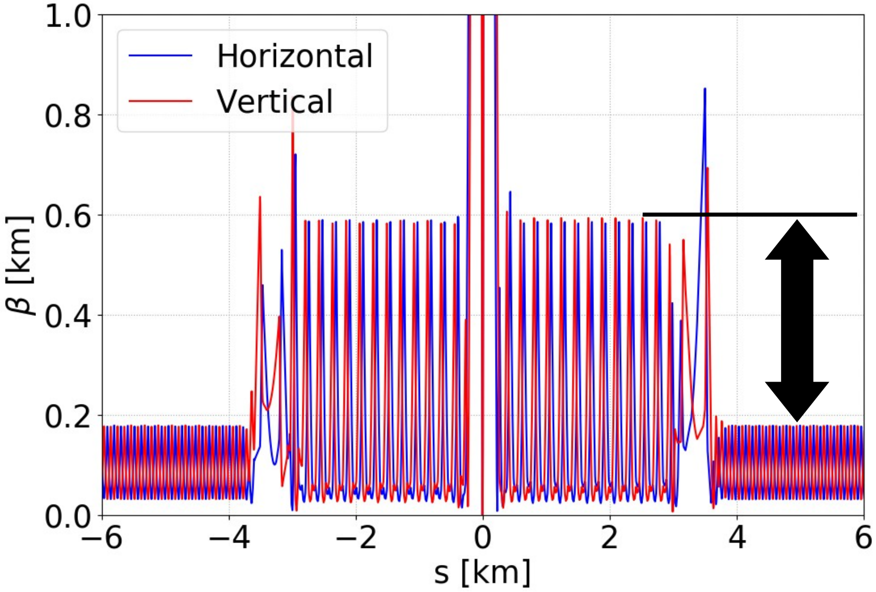
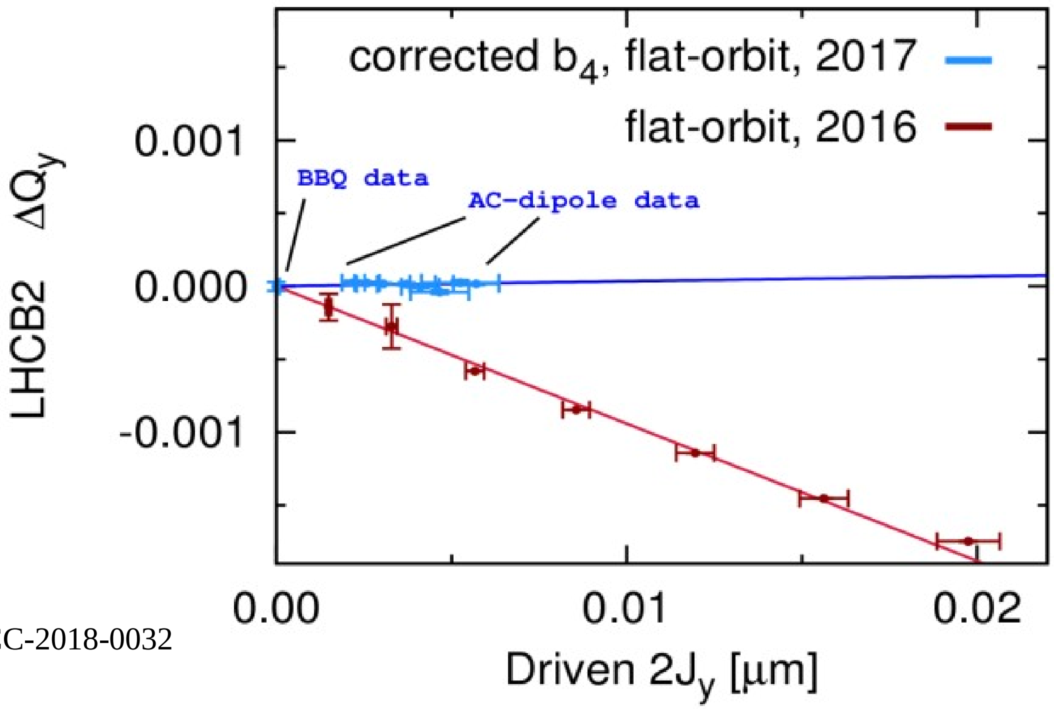


- Demonstrated regular operation a tele-index of **1.6** and up to **3.1** dedicated tests

  + additional benefits :

  → Long-range beam-beam compensation
  → Enhancement of Landau damping

- Introduced local non-linear correction in the IR
  - β-beating **~1%**
  - β* accuracy
  - Landau damping



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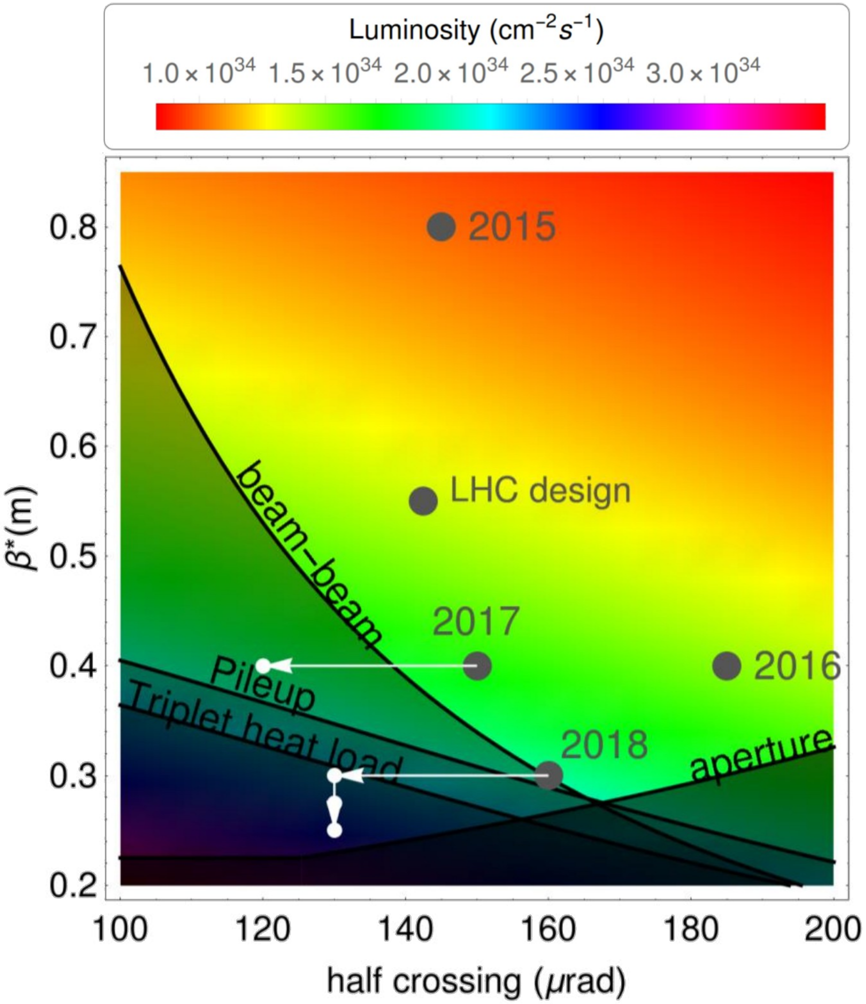
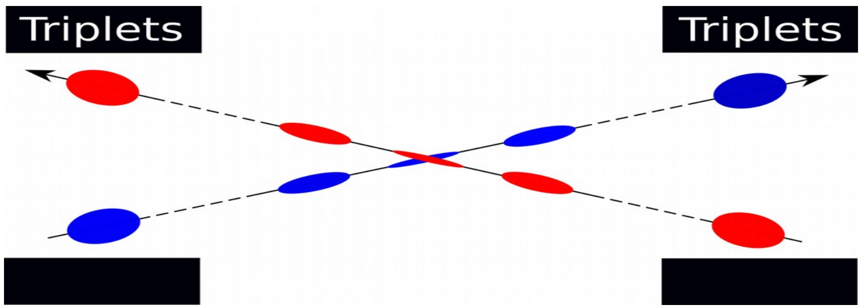
E.H. Maclean, et al. Phys. Rev. Accel. Beams 22, 061004 (2019)
T. Persson, et al. Phys. Rev. Accel. Beams 20, 061002 (2017)

Luminosity levelling





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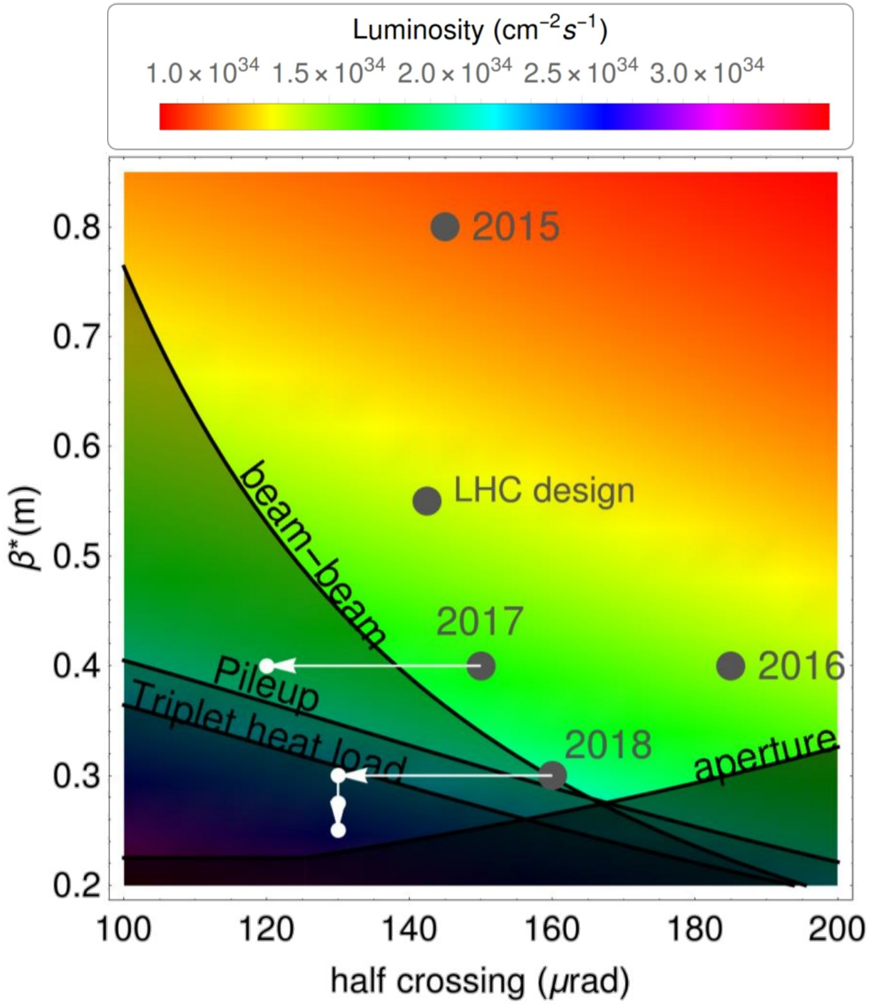
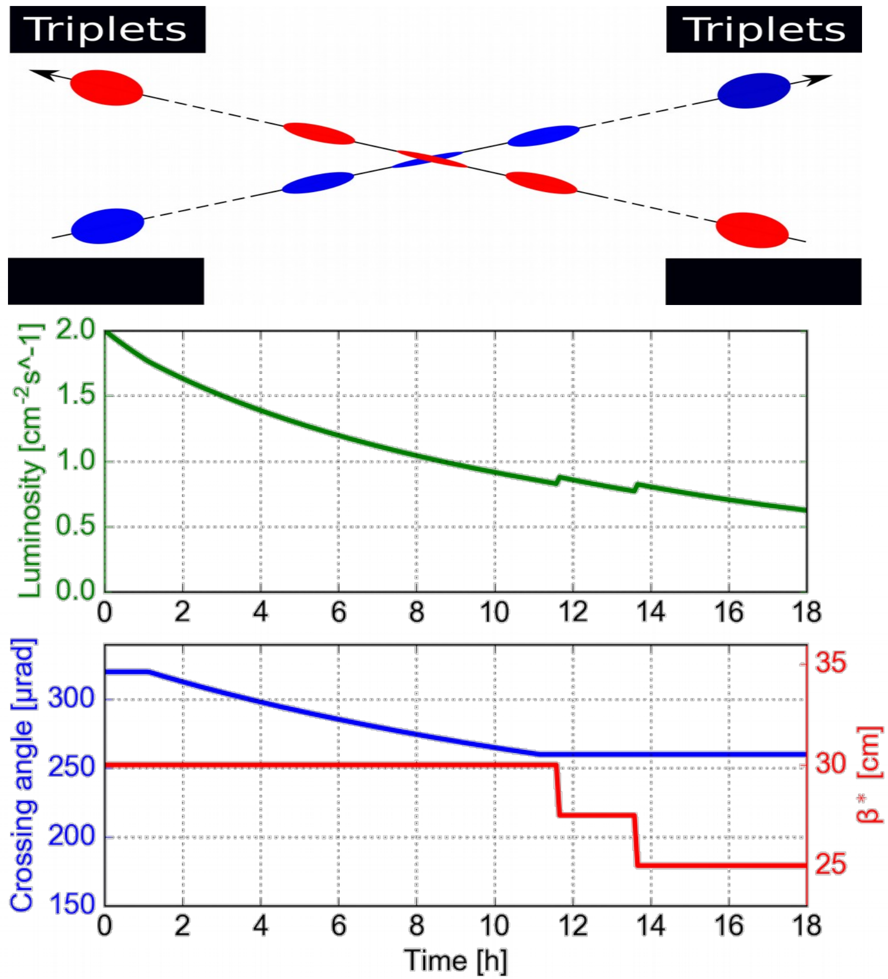
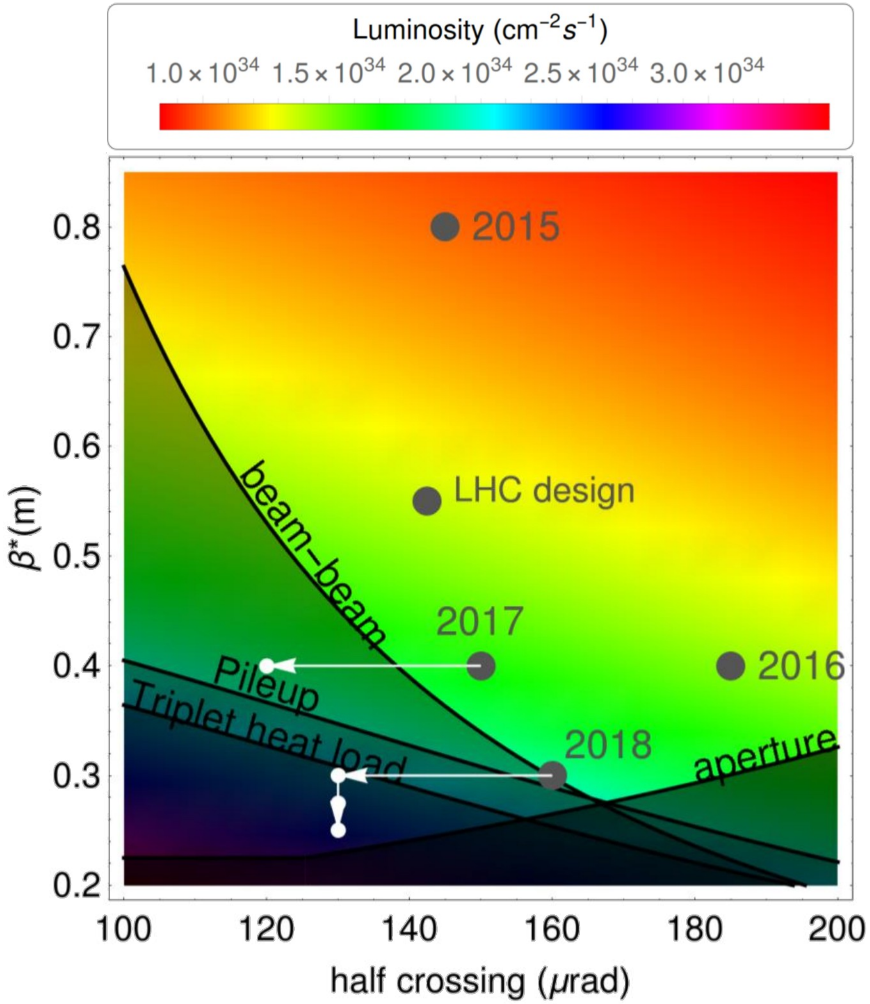


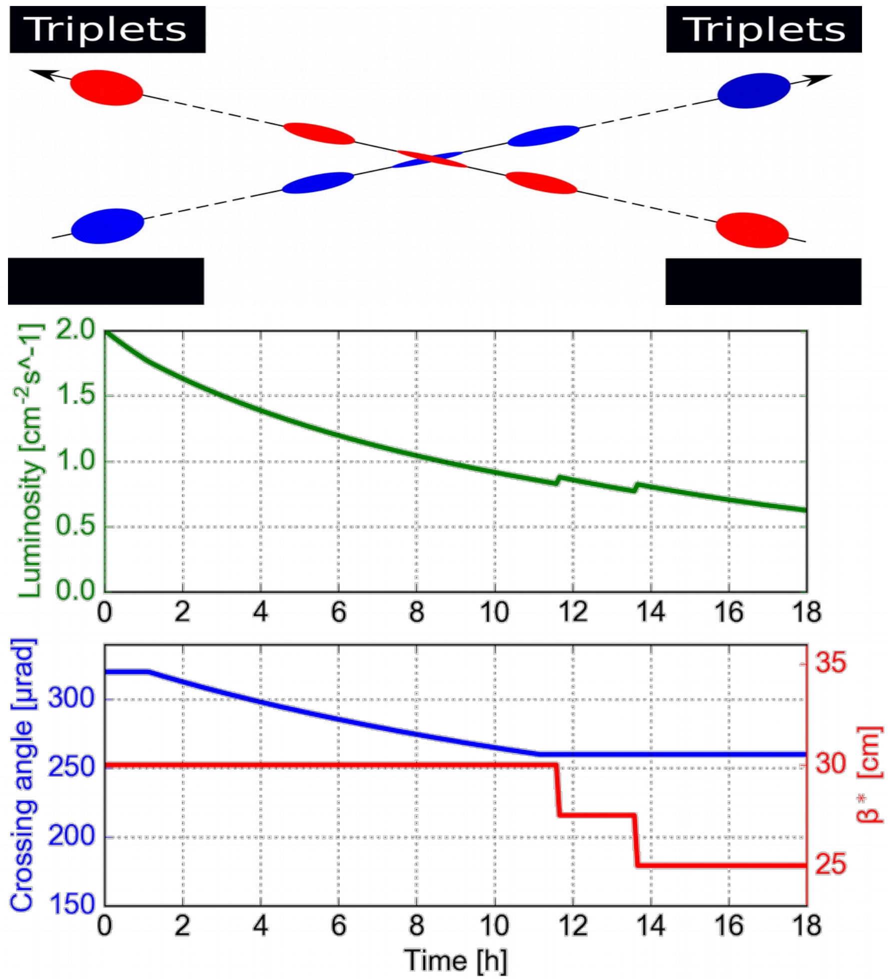


- Full exploitation of the relaxation of the constraints with the beam intensity decay for **integrated** luminosity

R. Bruce, et al., in Proc. Evian 2019

Luminosity levelling





- Full exploitation of the relaxation of the constraints with the beam intensity decay for **integrated** luminosity

  → Given the stored energy, a high operational robustness is required for dynamic orbit and optics changes with detectors on (**STABLE BEAM**)

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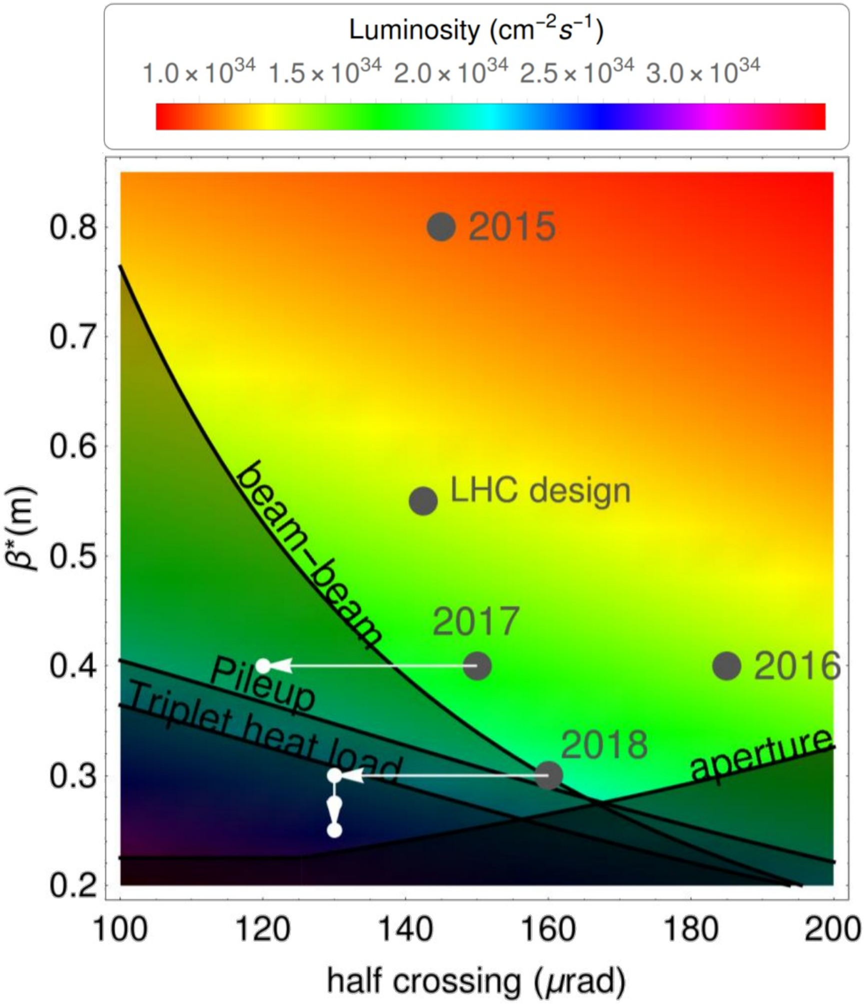
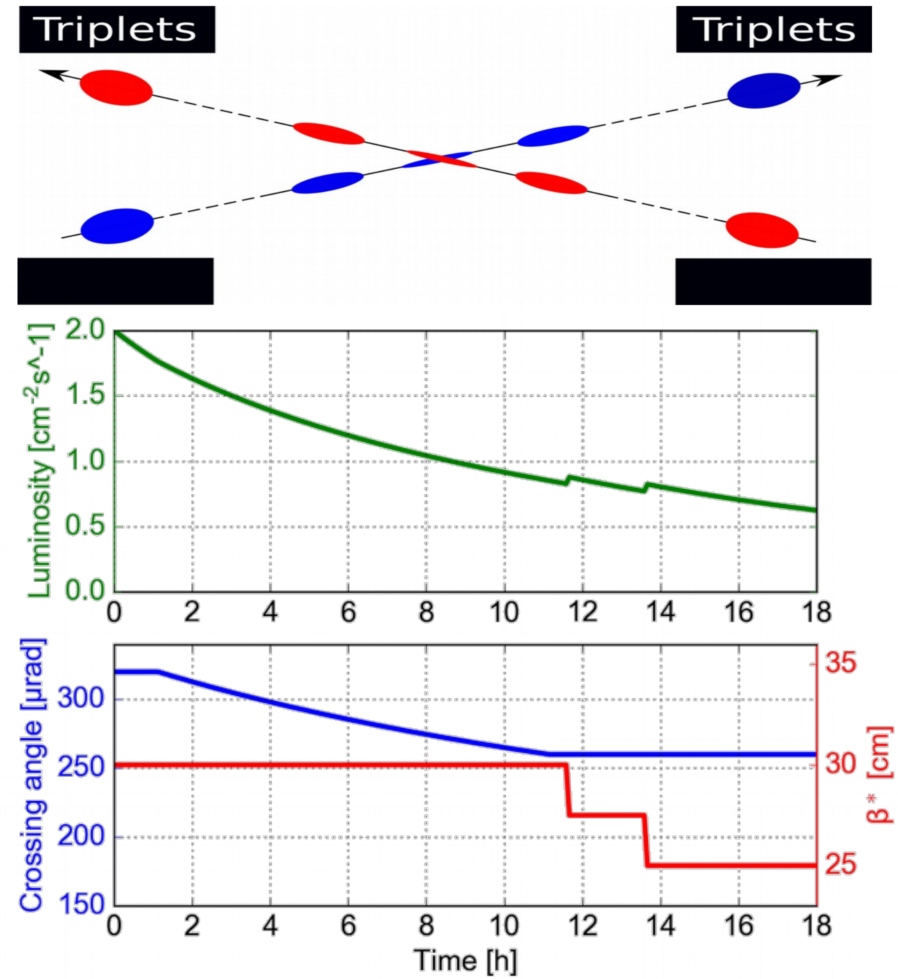


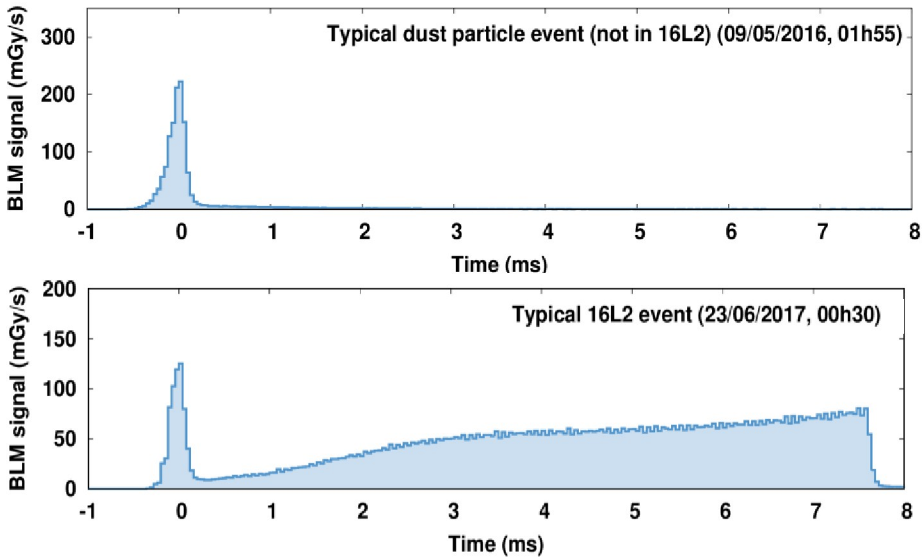


- Full exploitation of the relaxation of the constraints with the beam intensity decay for **integrated** luminosity

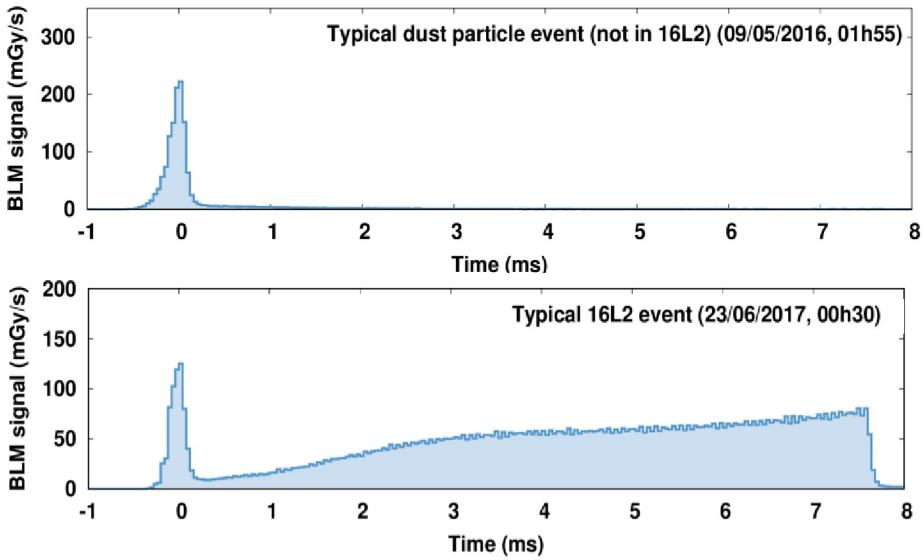
  → Given the stored energy, a high operational robustness is required for dynamic orbit and optics changes with detectors on (**STABLE BEAM**)

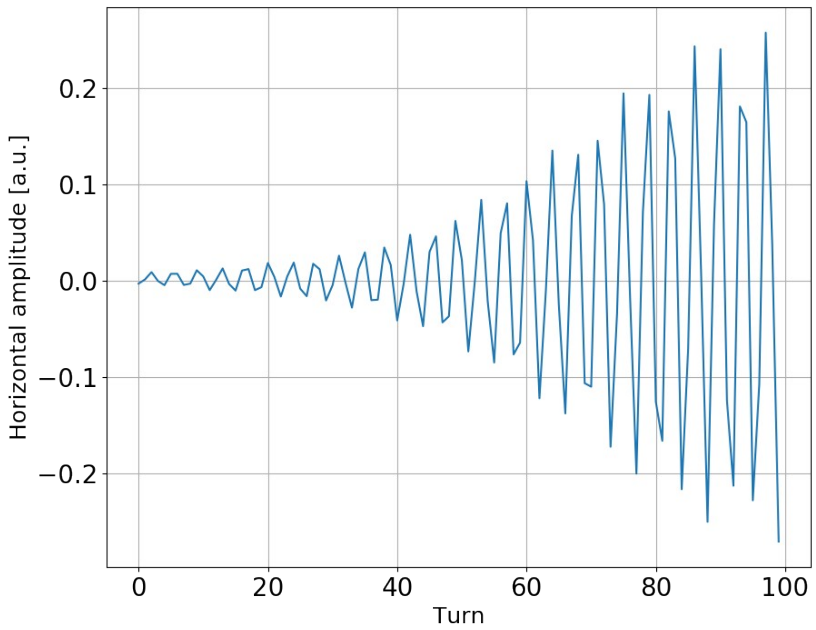
  → Luminosity levelling with transverse offset was also demonstrated in dedicated experiments with high intensity beams

R. Bruce, et al., in Proc. Evian 2019



A. Lechner, et al., in Proc. IPAC'18





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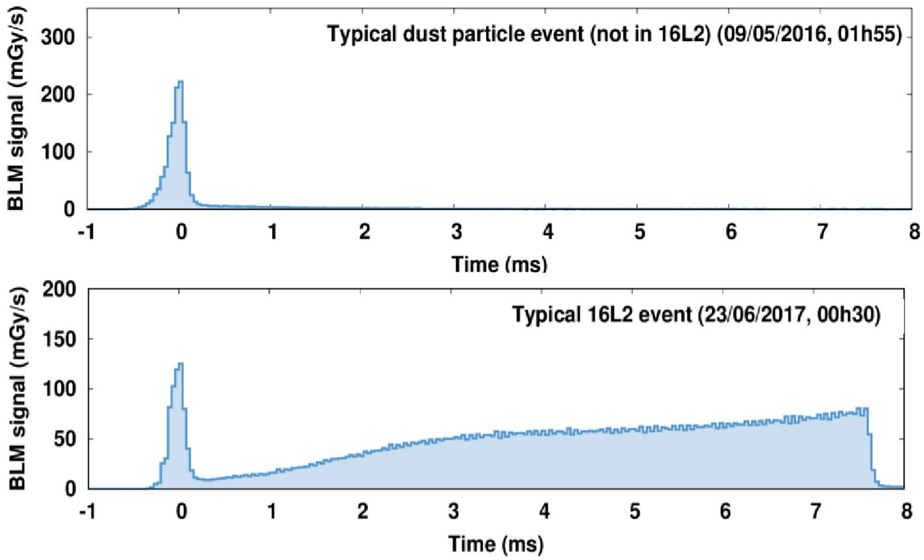
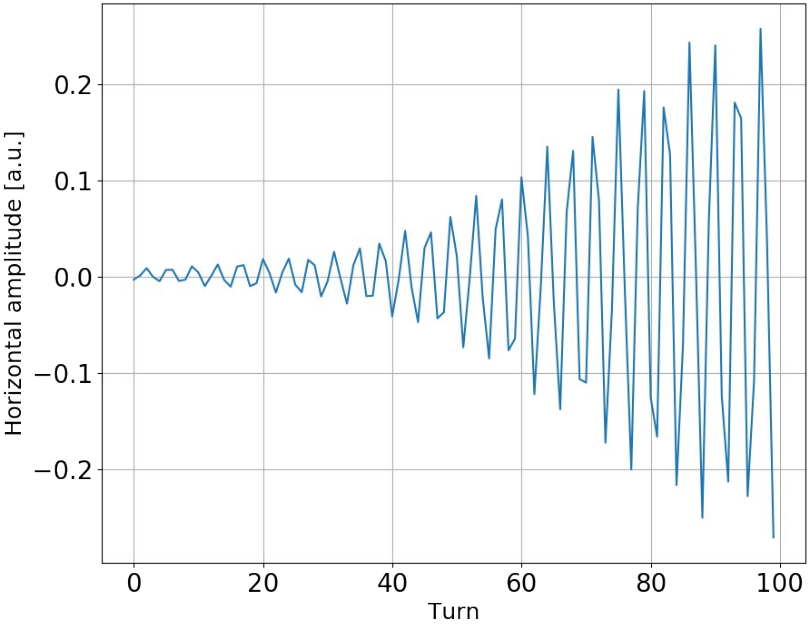
- One magnet needed to be exchanged during the extended year end technical stop due to an inter-turn short
- During re-cool down of the sector, a vacuum leak led to a significant amount of air to freeze in an arc interconnect (16L2)





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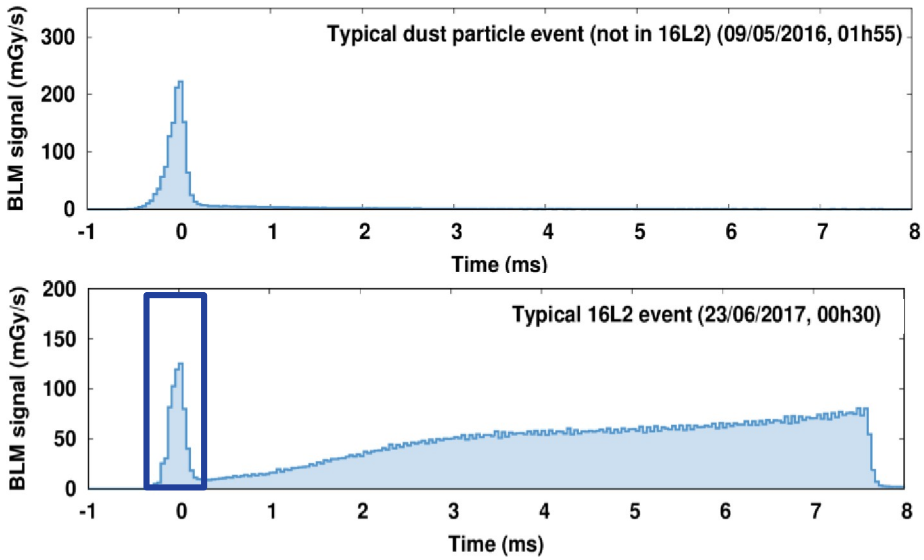

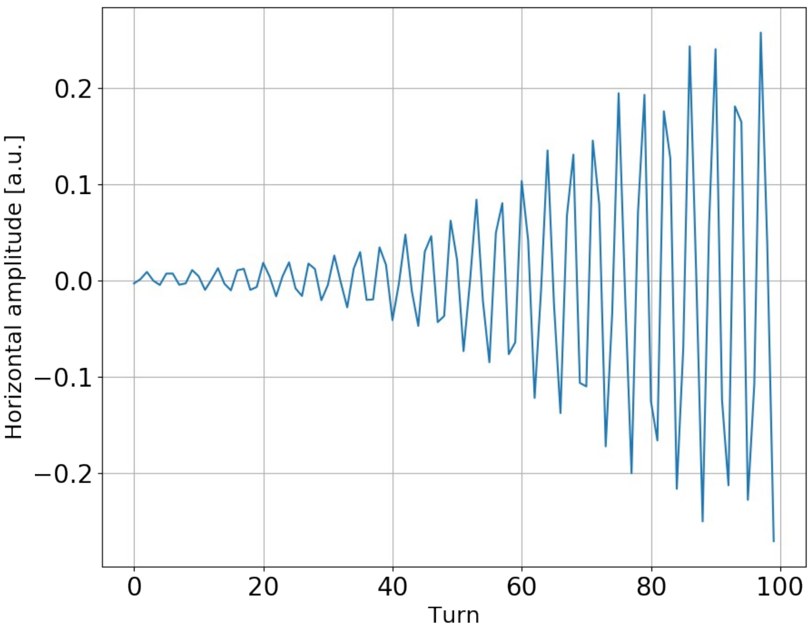

Solid air condensate (N_2, O_2 H_2O,…) detached from the beam screen

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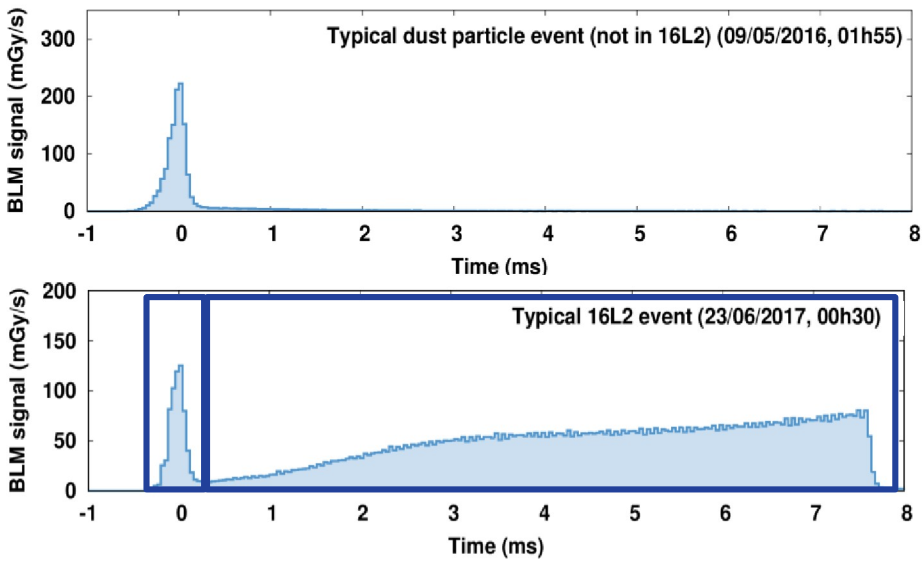

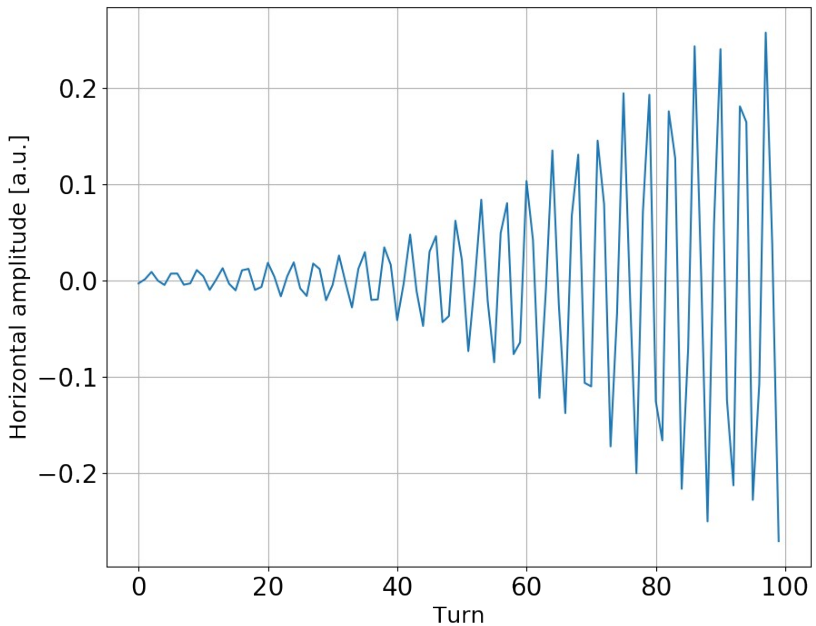

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Sublimated atoms, ions and electrons generate local losses and a strong coherent beam instability

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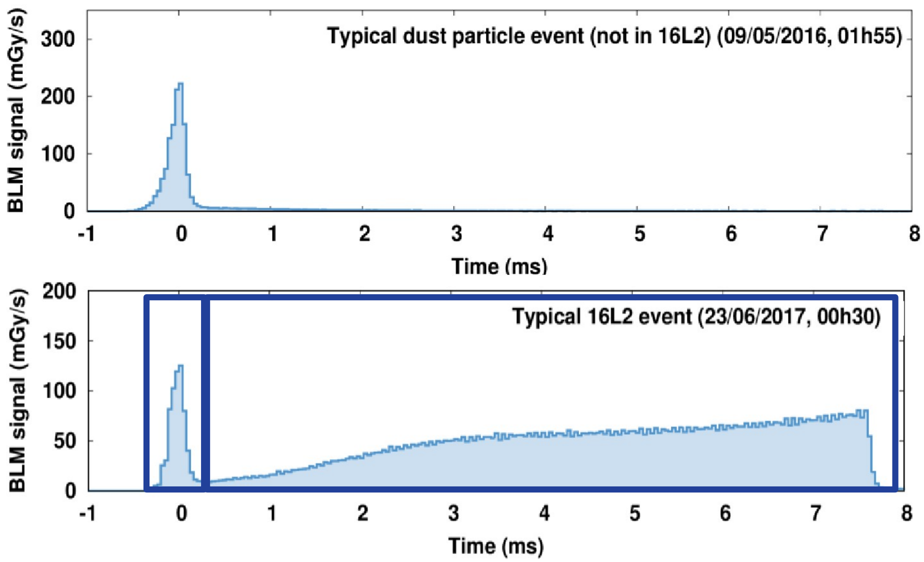
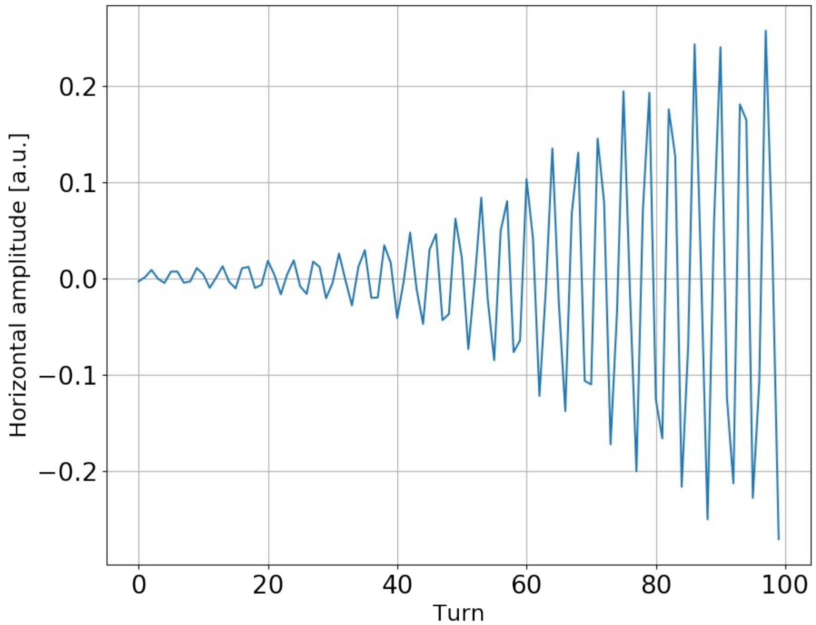




Solid air condensate (N_2, O_2 H_2O,…) detached from the beam screen

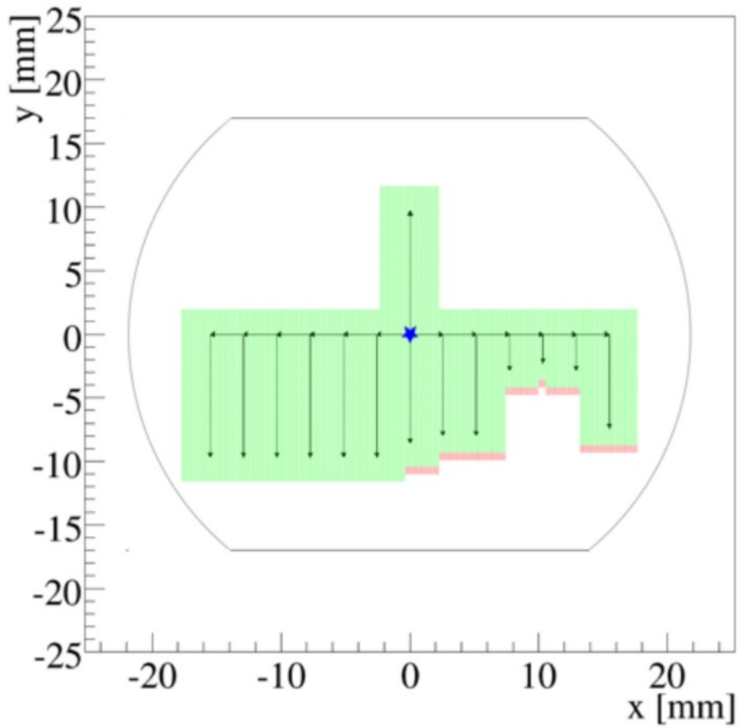
Sublimated atoms, ions and electrons generate local losses and a strong coherent beam instability

→ Electron cloud reduction measures seemed effective (8b4e scheme, local corrector field, solenoid)

→ The bunch intensity with 25ns trains remained limited to ~$1.2 \cdot 10^{11}$ for the rest of the run

A. Lechner, et al., in Proc. IPAC'18 / B. Salvant, et al., in Proc. IPAC'18 / L. Mether, et al., in Proc. IPAC'18 / J.M. Jimenez, et al., in Proc. IPAC'18

Unidentified lying object



- An physical aperture restriction was observed in 2015 in an arc

  → A local orbit bump was implemented to avoid it

D. Mirarchi, et al., in Proc. Evian 2019

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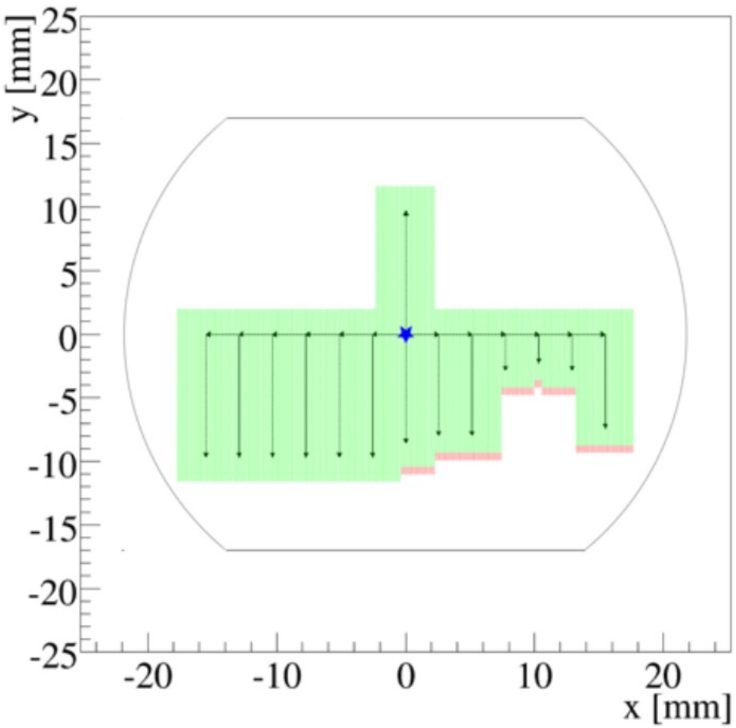

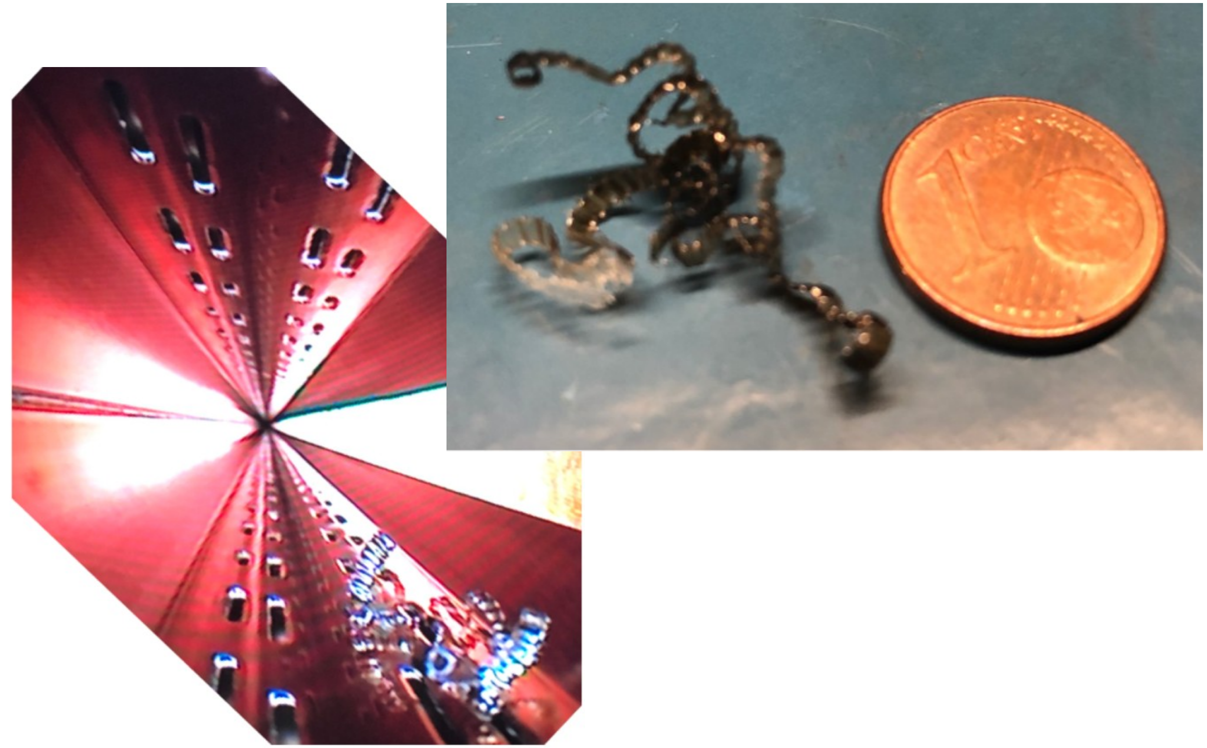

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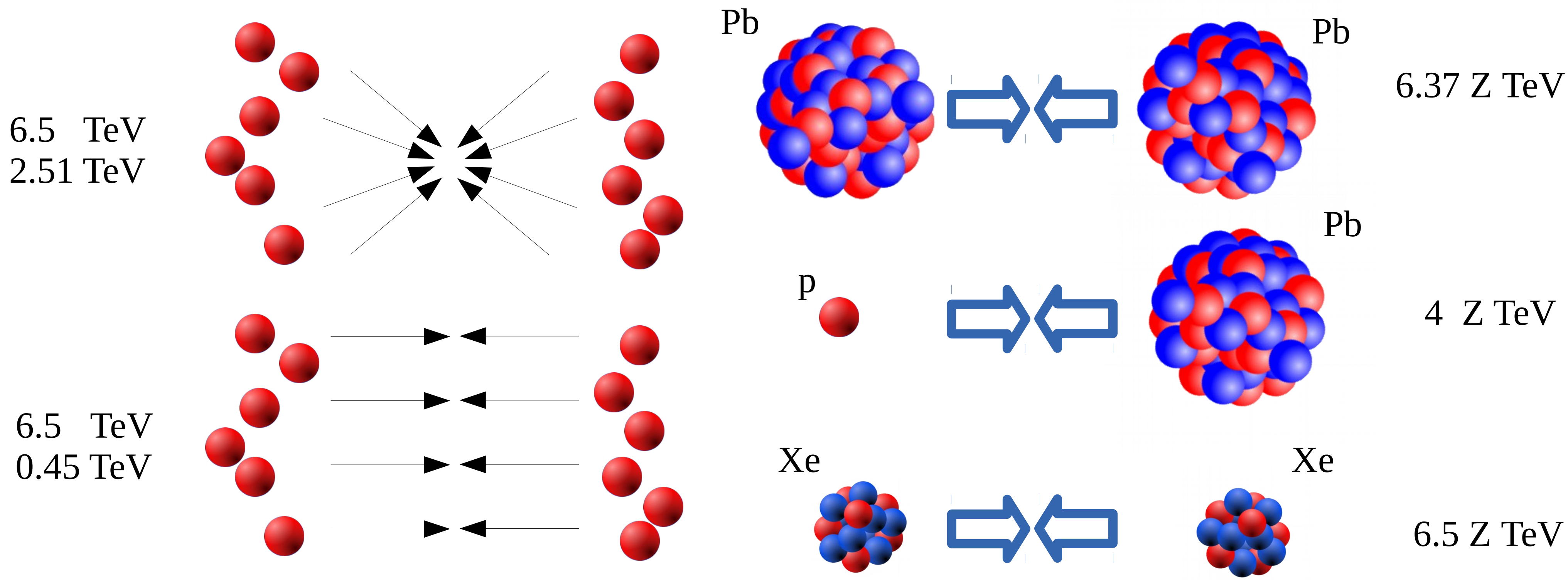
- A piece of plastic was taken out from the location inferred from beam based measurement during LS2

  → Probably from the beam screen wrapping ripped off during installation

D. Mirarchi, et al., in Proc. Evian 2019

LHC beam menu



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- Exceeded design performance for Pb-Pb operation
- First operational usage of crystal collimation

J. Jowett, et al., in Proc IPAC'18 and Proc IPAC'19

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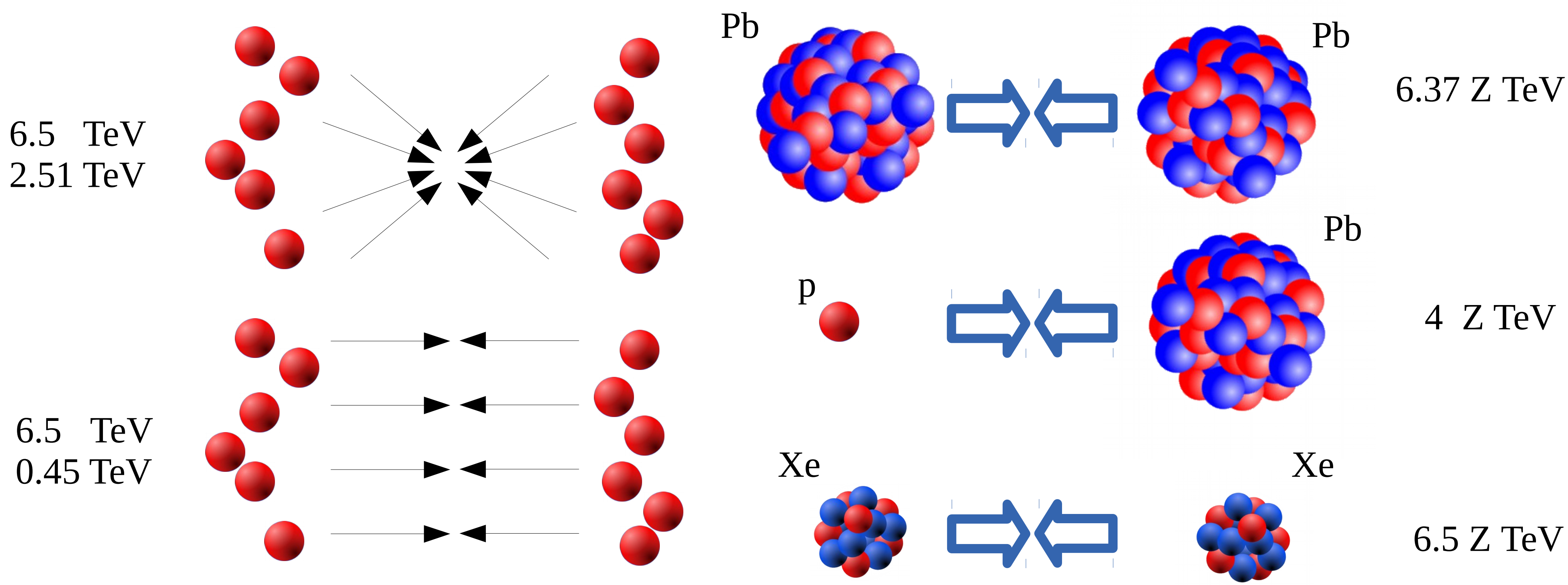

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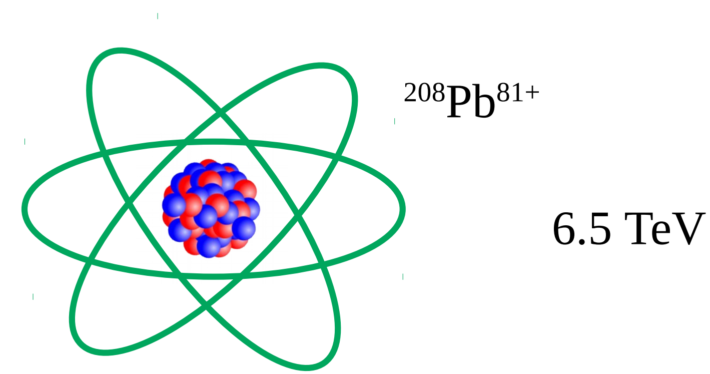

J. Jowett, et al., in Proc IPAC’18 and Proc IPAC’19
M. Schaumann, et al., in Proc. IPAC’19
M. Krasny, arXiv:1511.07794

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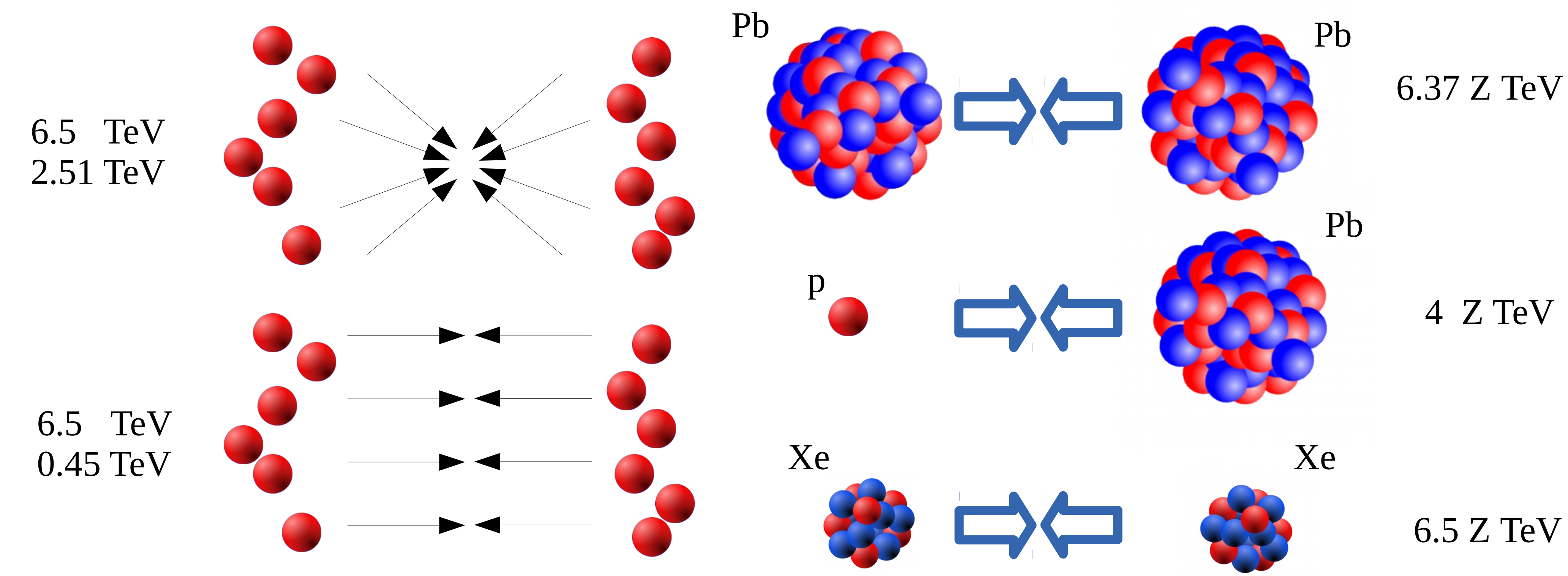

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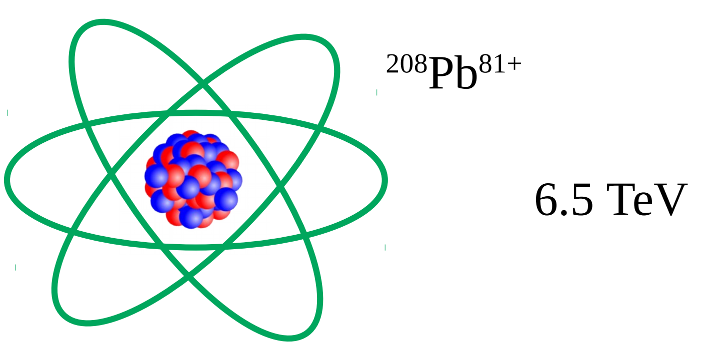
- Exceeded design performance for Pb-Pb operation
- First operational usage of crystal collimation
- First acceleration of partially stripped ions in 2018 (→ γ-factory, electron-ion collision)
- Towards a more diverse program : O, Ar, Kr



J. Jowett, et al., in Proc IPAC'18 and Proc IPAC'19
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The High Luminosity LHC baseline

High integrated luminosity **300 → 3000 fb^{-1}**
with limited pile-up increase **27 → 130 events / bunch crossing**

High intensity **1.15 → 2.3 10^{11}p**

Low β* **55 → 15 cm**

http://hilumilhc.web.cern.ch/

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- Injector upgrade (LIU project)
- Collimation
  - Mitigation of losses in the DS (11T dipoles)
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  - Local cleaning in the IR
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  - Beam aperture + shielding
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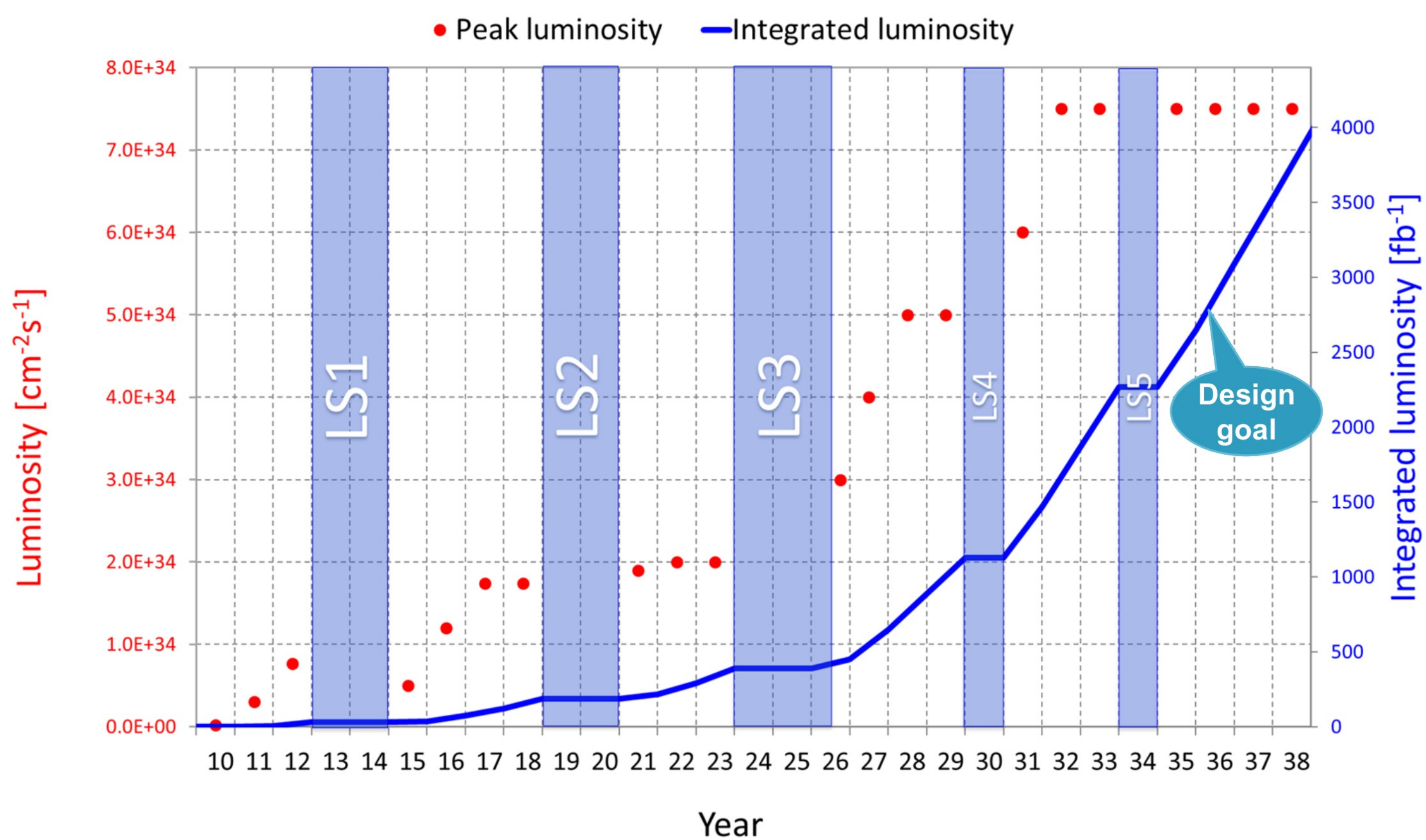
Implemented in run 2

Tested with beam in run 2

The High Luminosity LHC ultimate design goal



- The machine is designed for a levelled luminosity up to $7.5 \cdot 10^{34}$ Hz/cm^2, corresponding to a pile up of 197 events per bunch crossing
  - Up to 4000 fb^{-1} could be delivered if implemented in 2032

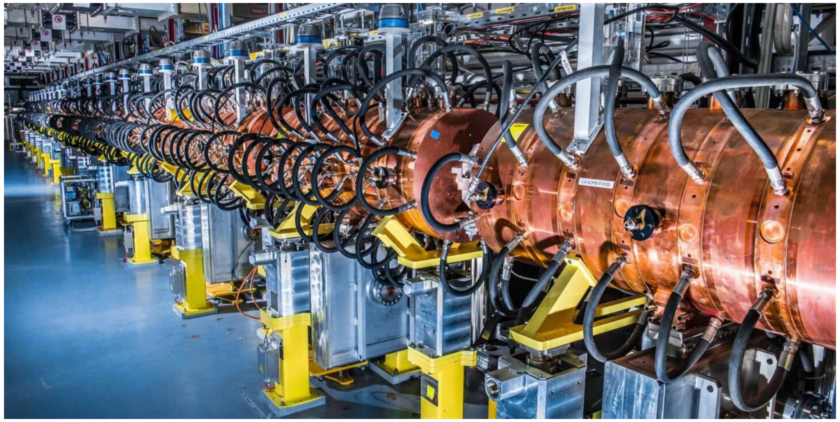
L. Rossi, in Proc. IPAC'19

The LHC injector upgrade

LHC Injectors Upgrade

https://espace.cern.ch/liu-project/default.aspx

The LHC injector upgrade



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- Connection to PS Booster ongoing

LHC Injectors Upgrade

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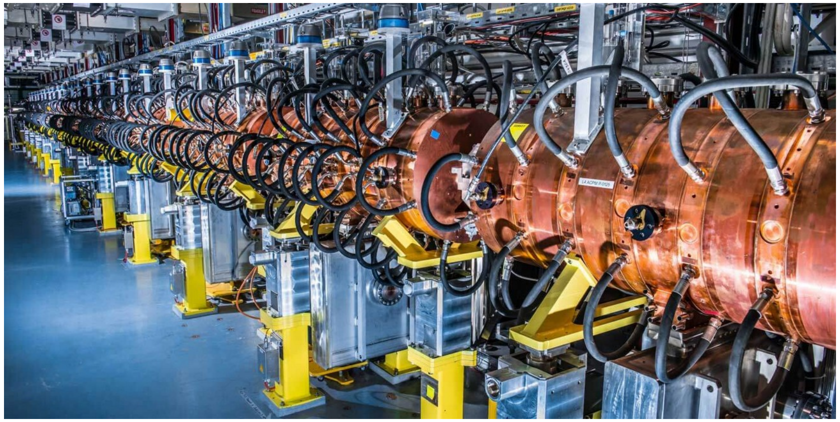

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LHC Injectors Upgrade

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- 1.4 → **2 GeV** extraction
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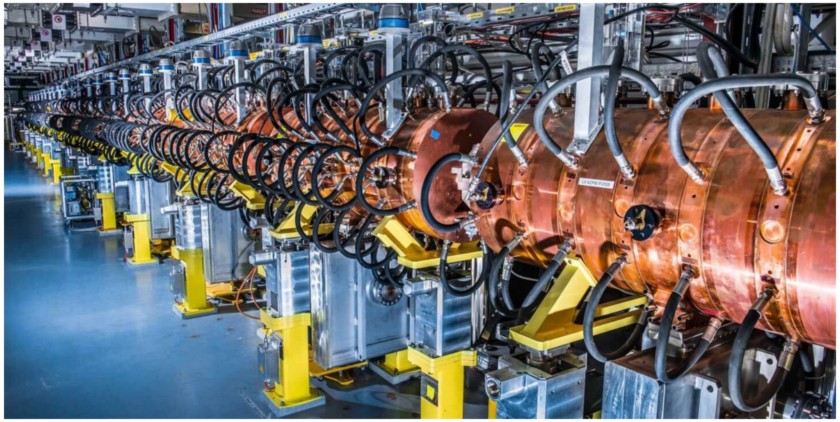

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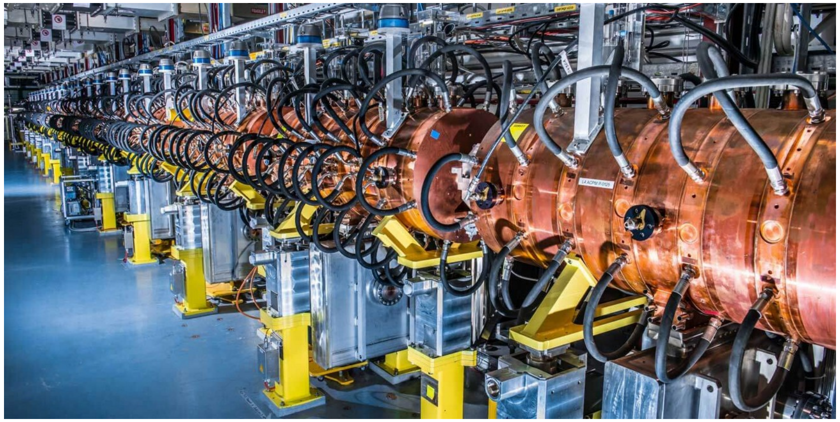

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- Transverse feedback
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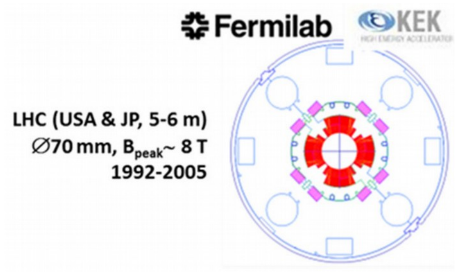
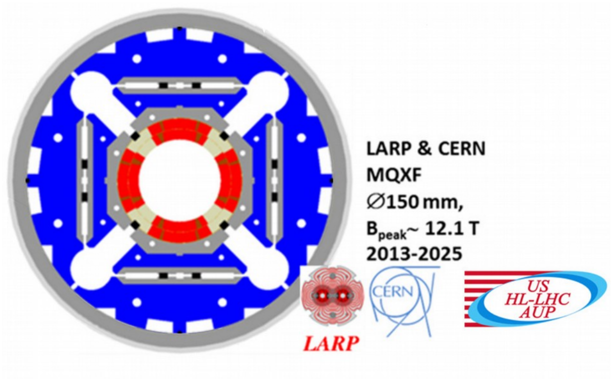
- RF upgrade (beam stability, power **2·1.6MW**)
- Impedance reduction
- Electron cloud mitigation (aC coating)
- Installation ongoing

https://espace.cern.ch/liu-project/default.aspx

Magnets





https://espace.cern.ch/HiLumi/wp3/SitePages/Home.aspx

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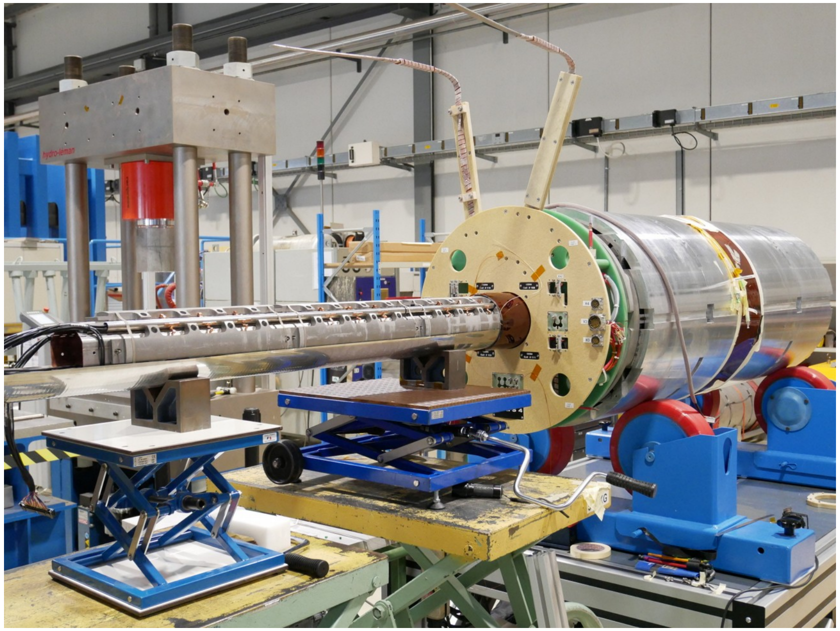

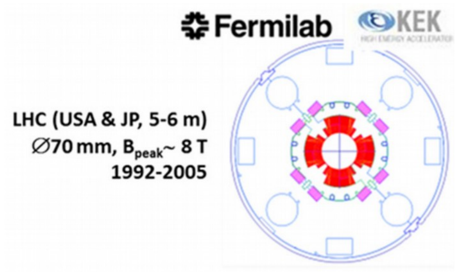

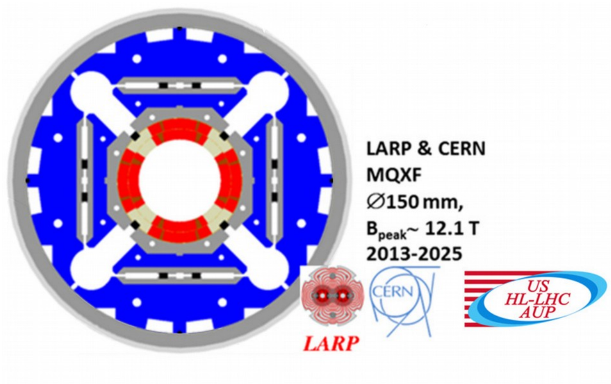

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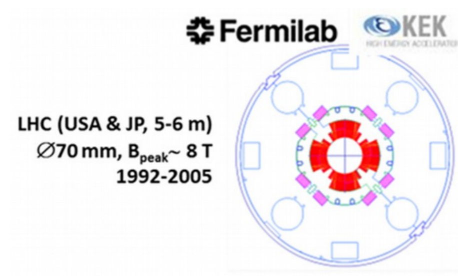

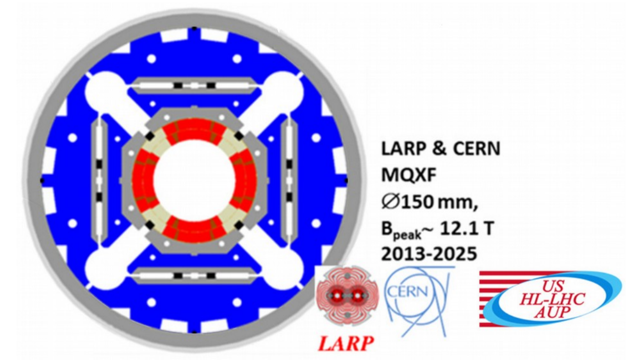

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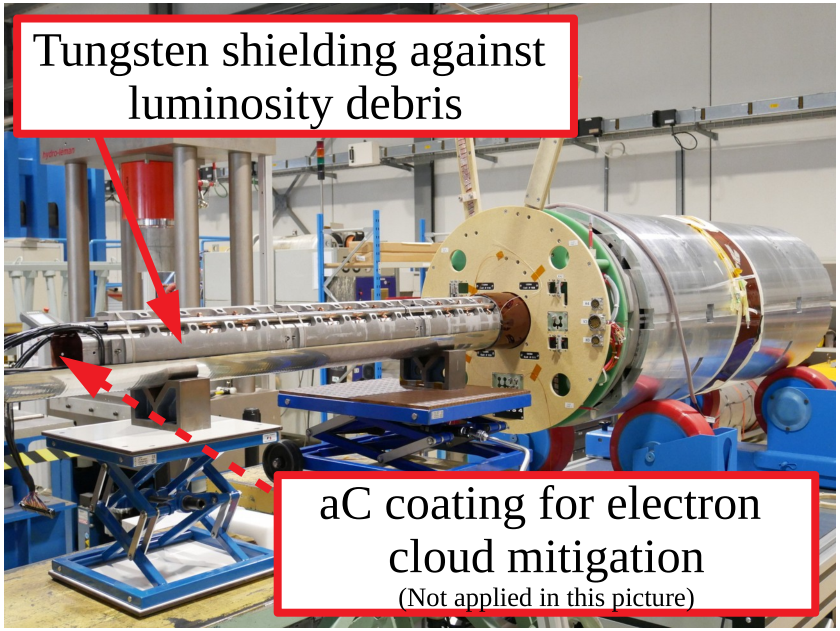

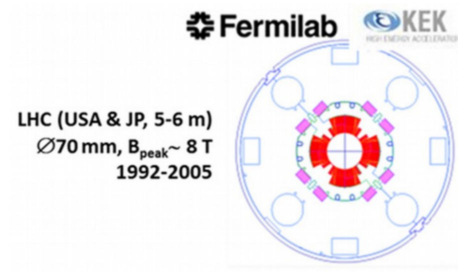

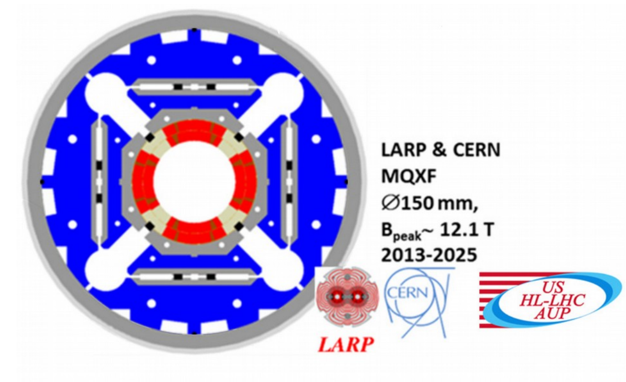

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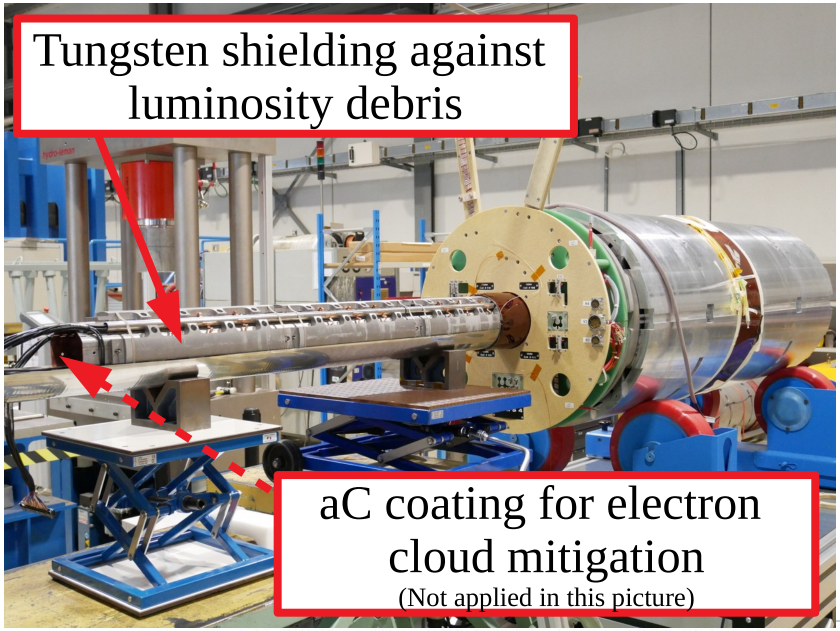

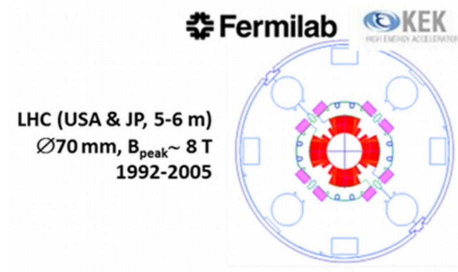

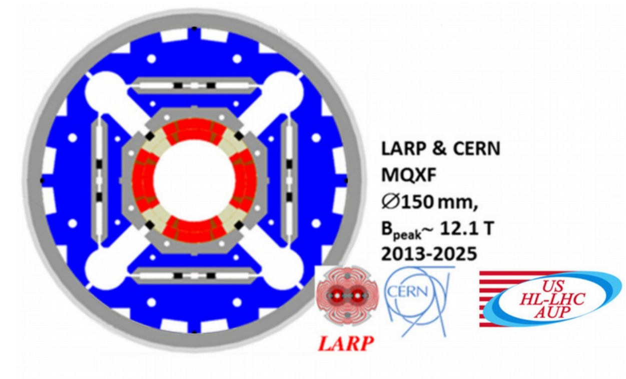

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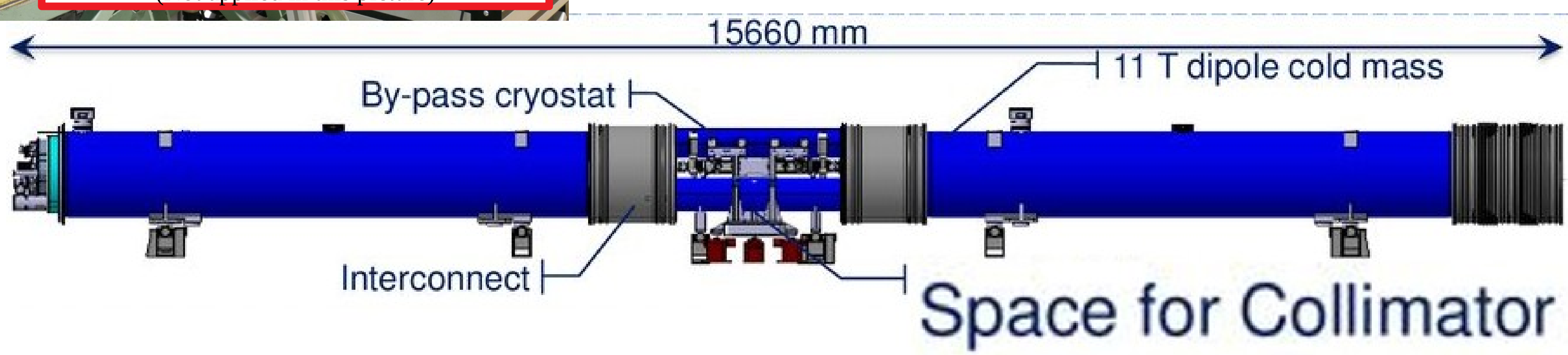

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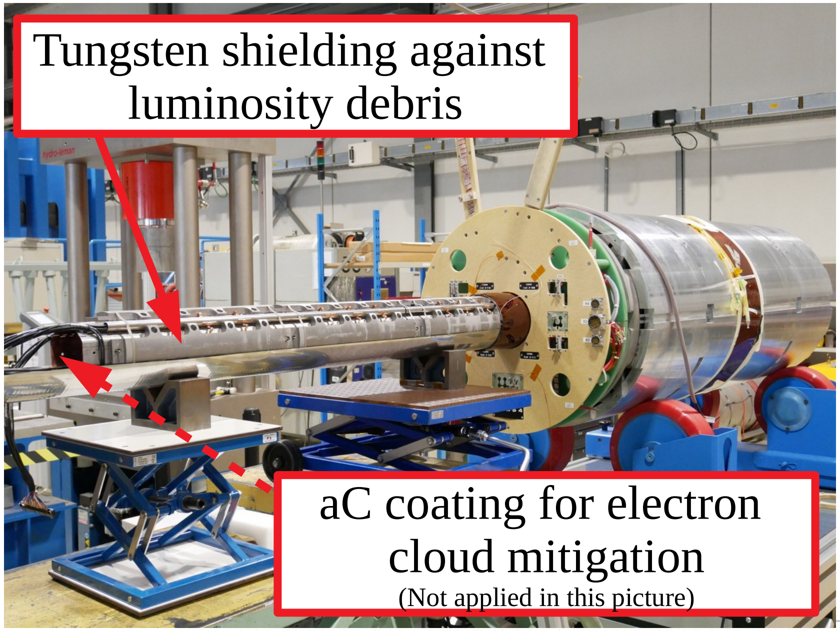

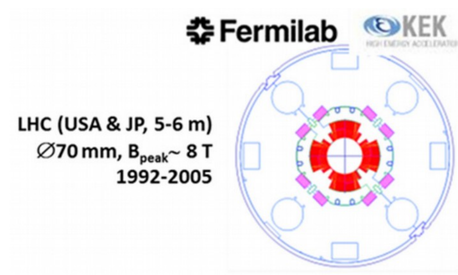

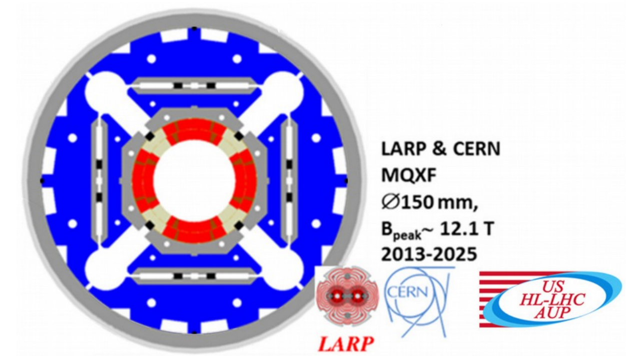

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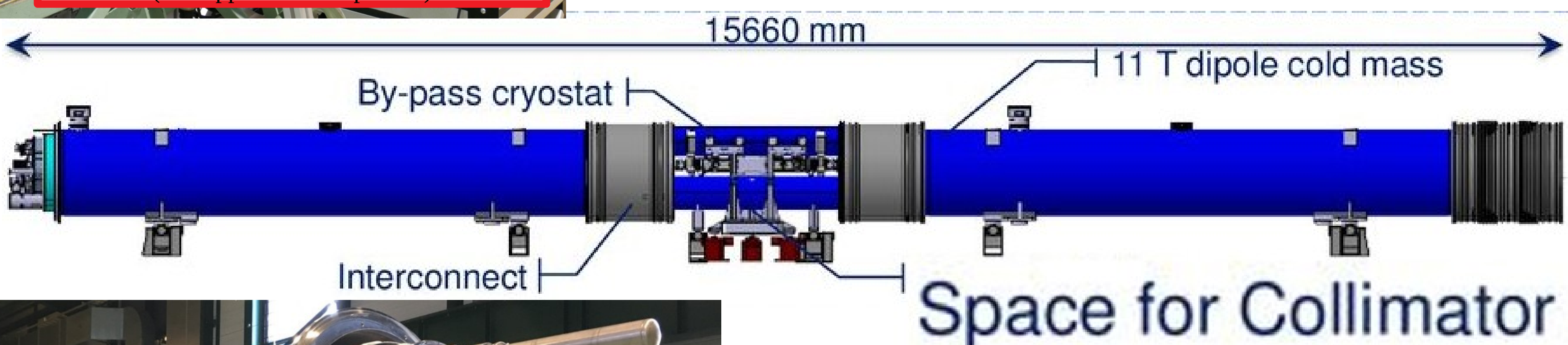
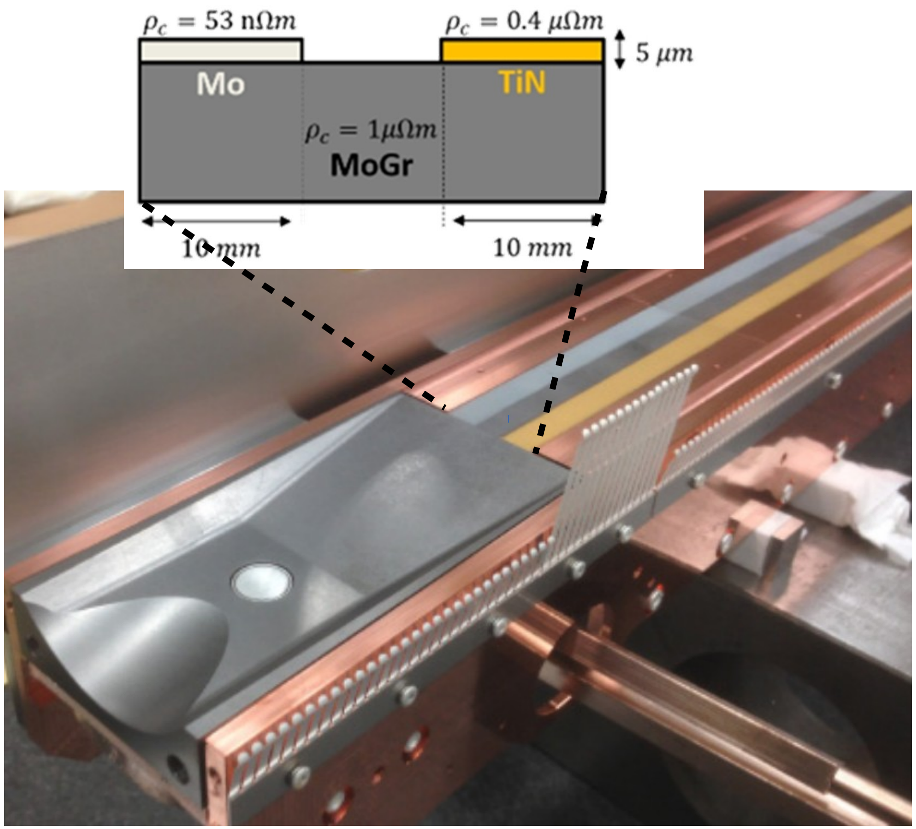


- Long prototypes of the new Nb_3Sn **11T** dipole are under test

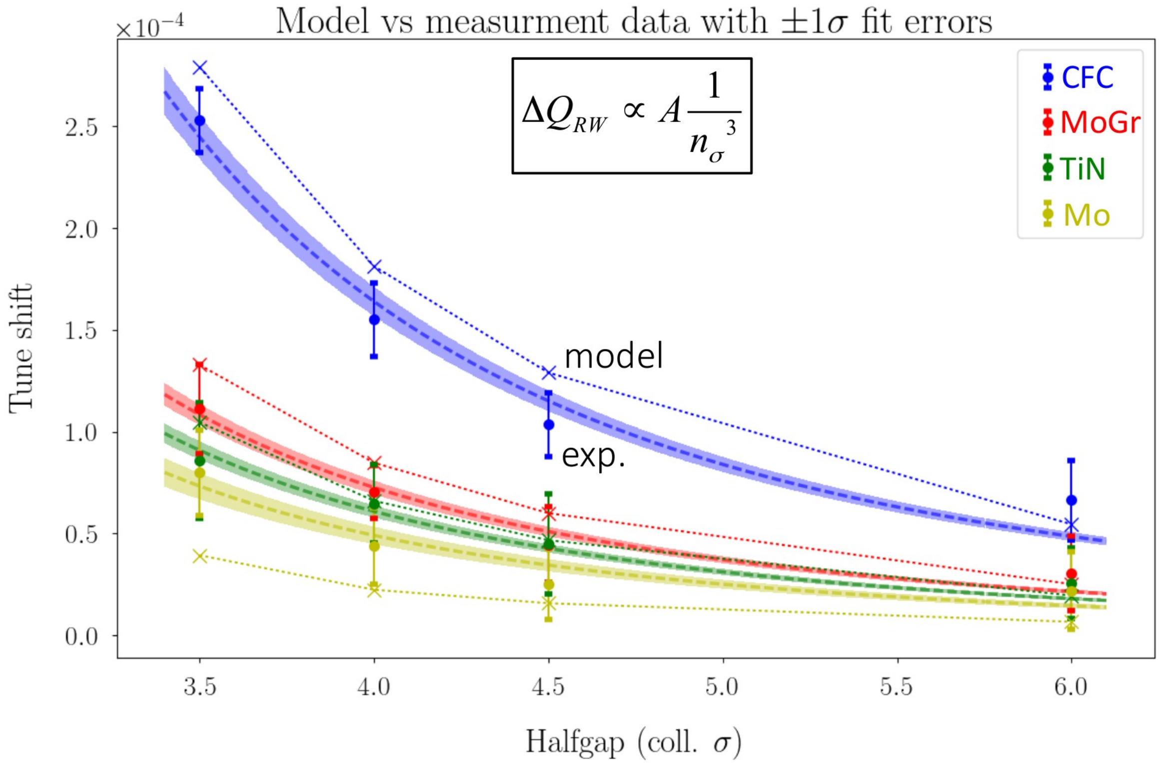
  → Installation planned during LS2, with a tight schedule

https://espace.cern.ch/HiLumi/wp3/SitePages/Home.aspx

Low impedance collimators



Prototype TCSPM for beam based assessment of the different coatings

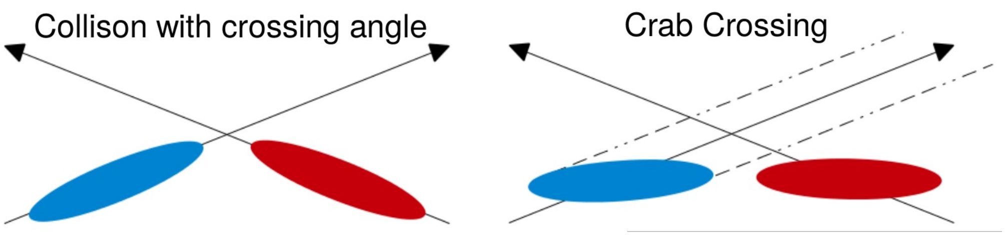


- The low impedance is achieved with Mo coating on MoGr
  - Slightly higher resistivity measured, possibly due to surface roughness

→ The replacement of 2 primary (MoGr) and 4 secondary collimators (Mo+MoGr) is taking place during LS2

S.A. Antipov, et al., in Proc. IPAC'18

Crab cavity test



R. Calaga, et al., in Proc. IPAC’18

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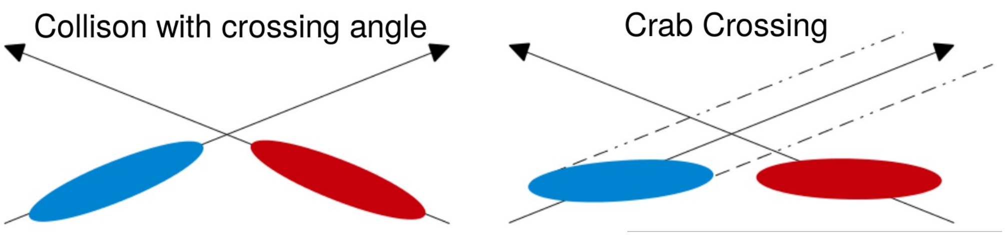

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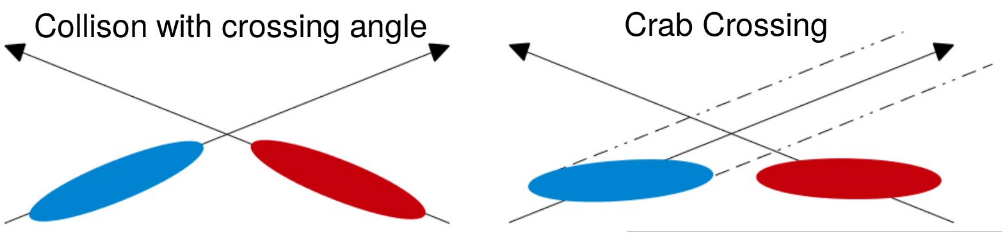
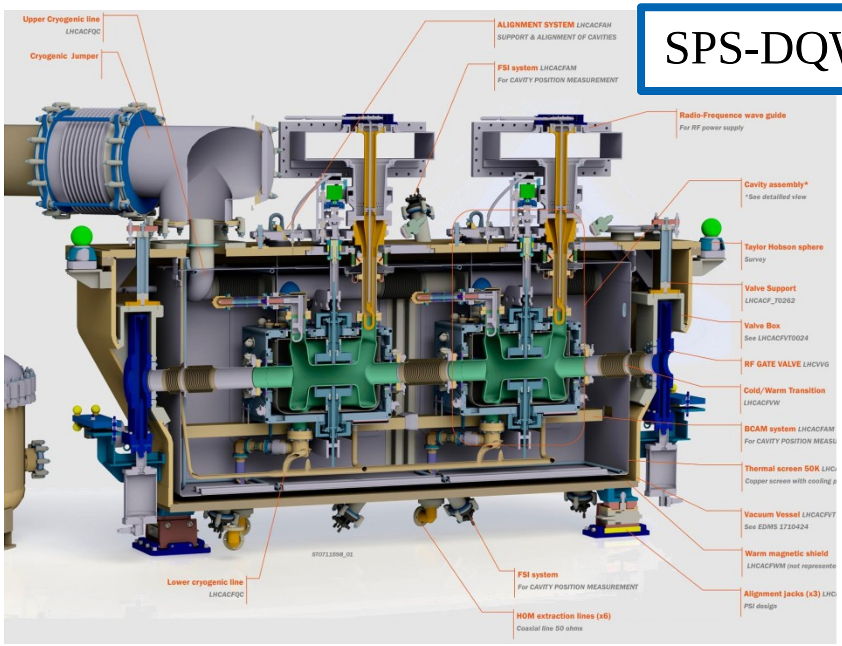


SPS-DQW



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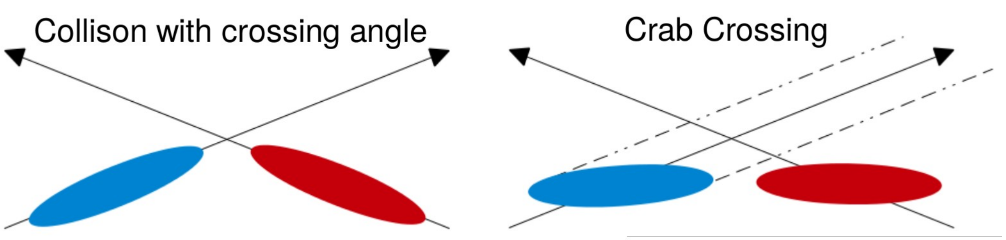

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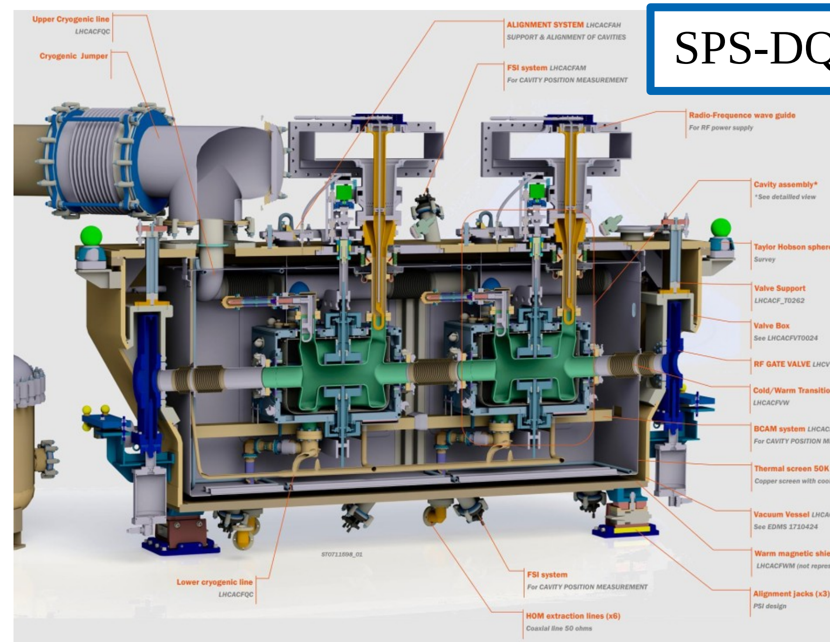

- Demonstrated crabbing with the DQW design in the SPS
  - Operation, synchronisation, transparency test
  - Alignment
  - Transvese emittance growth
  - RF multipoles
  - HOM

R. Calaga, et al., in Proc. IPAC'18

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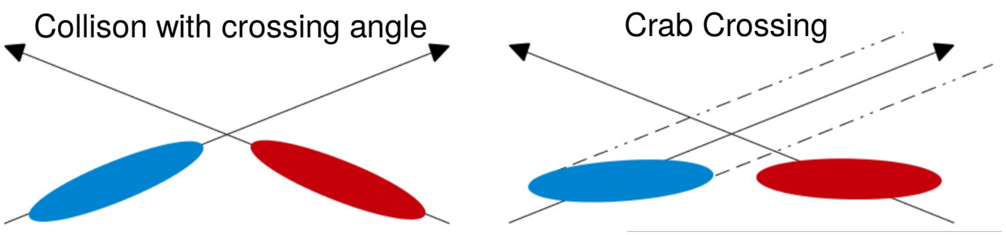

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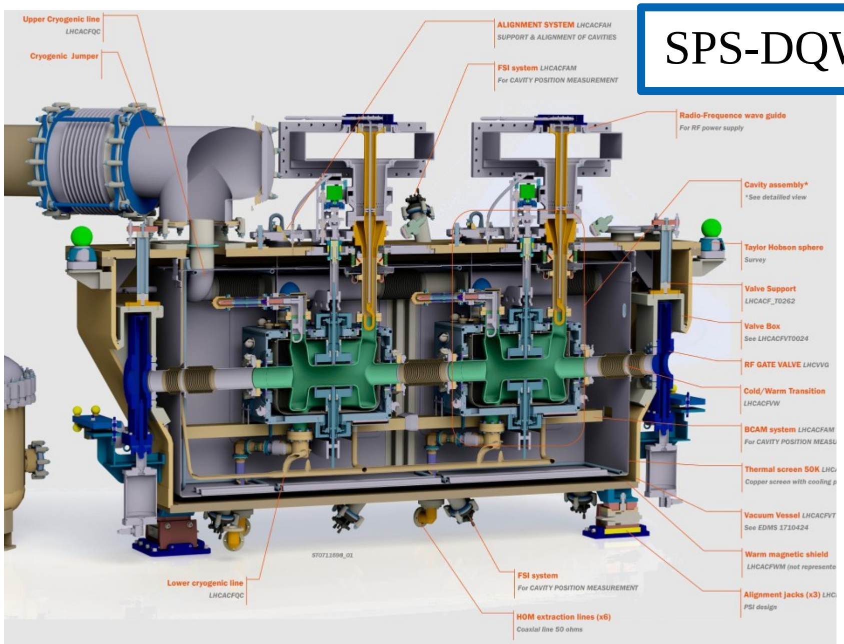

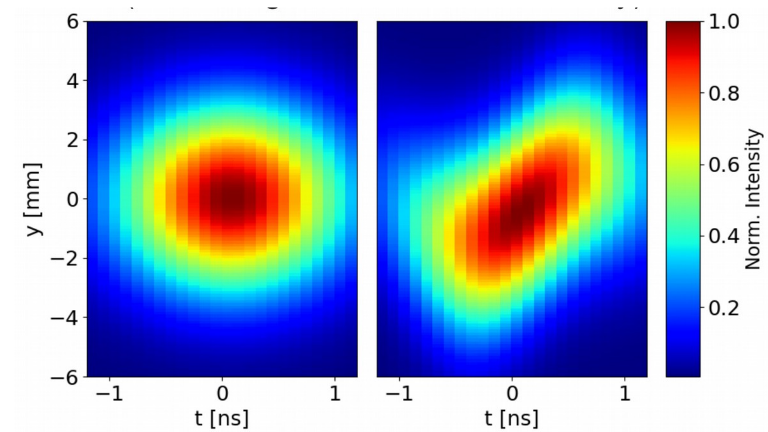

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L.R. Carver, et al., in Proc. IPAC'19

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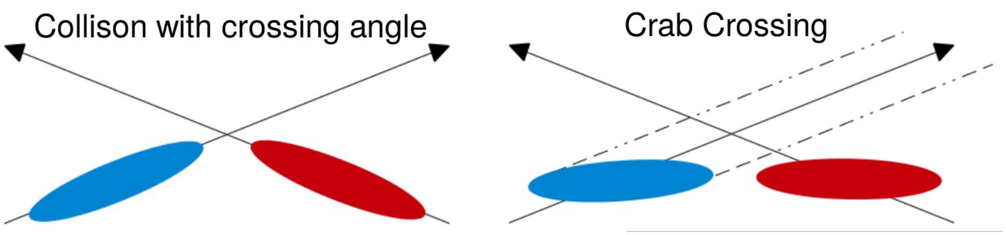

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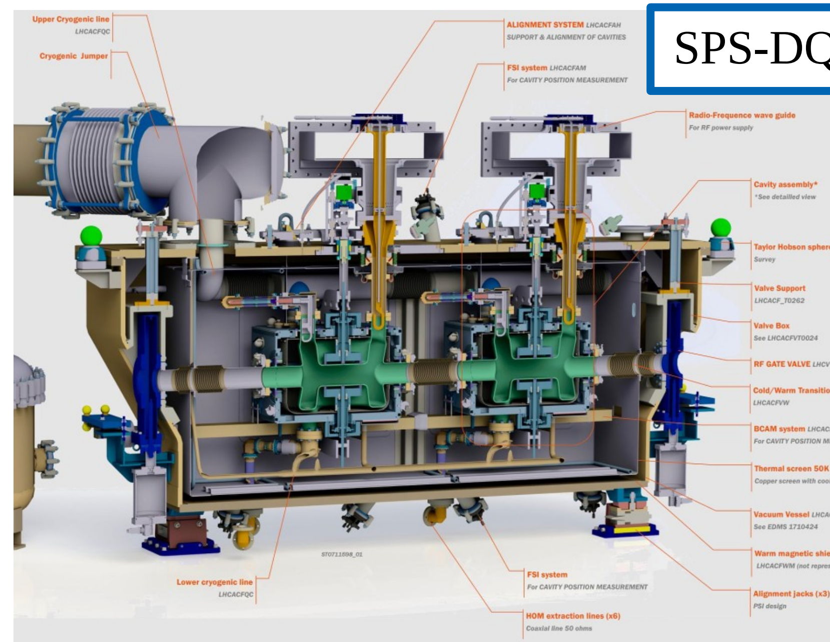

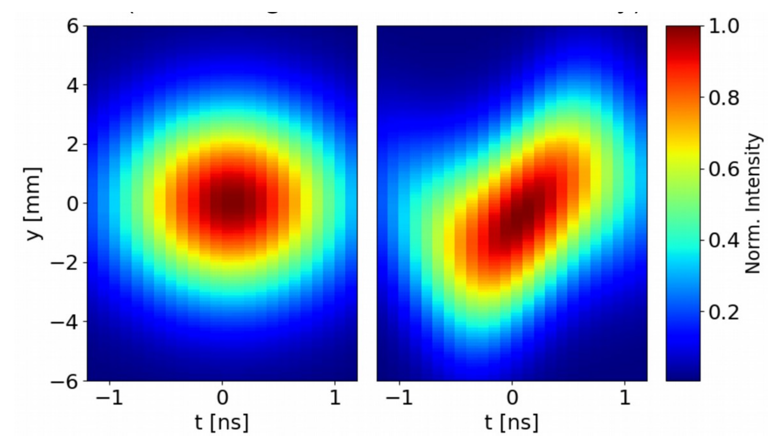

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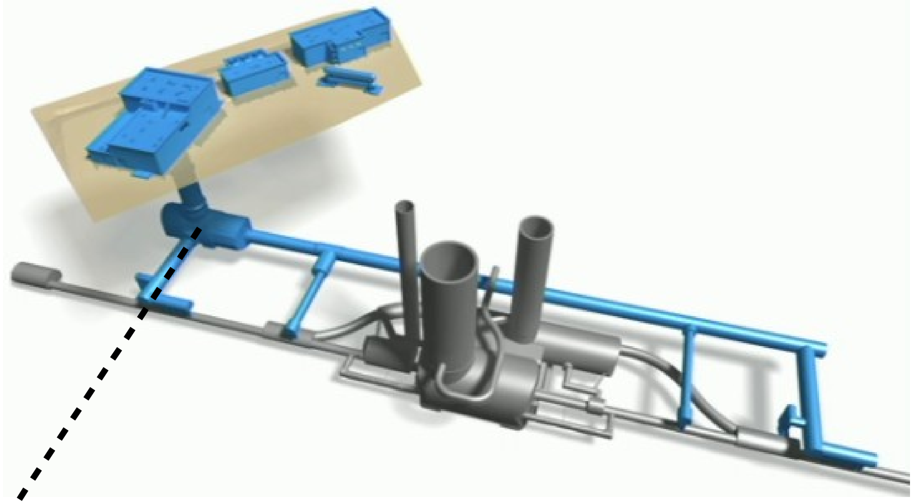
  - RF multipoles
  - HOM
- RFD test in prepartion

R. Calaga, et al., in Proc. IPAC'18
L.R. Carver, et al., in Proc. IPAC'19

Civil engineering work

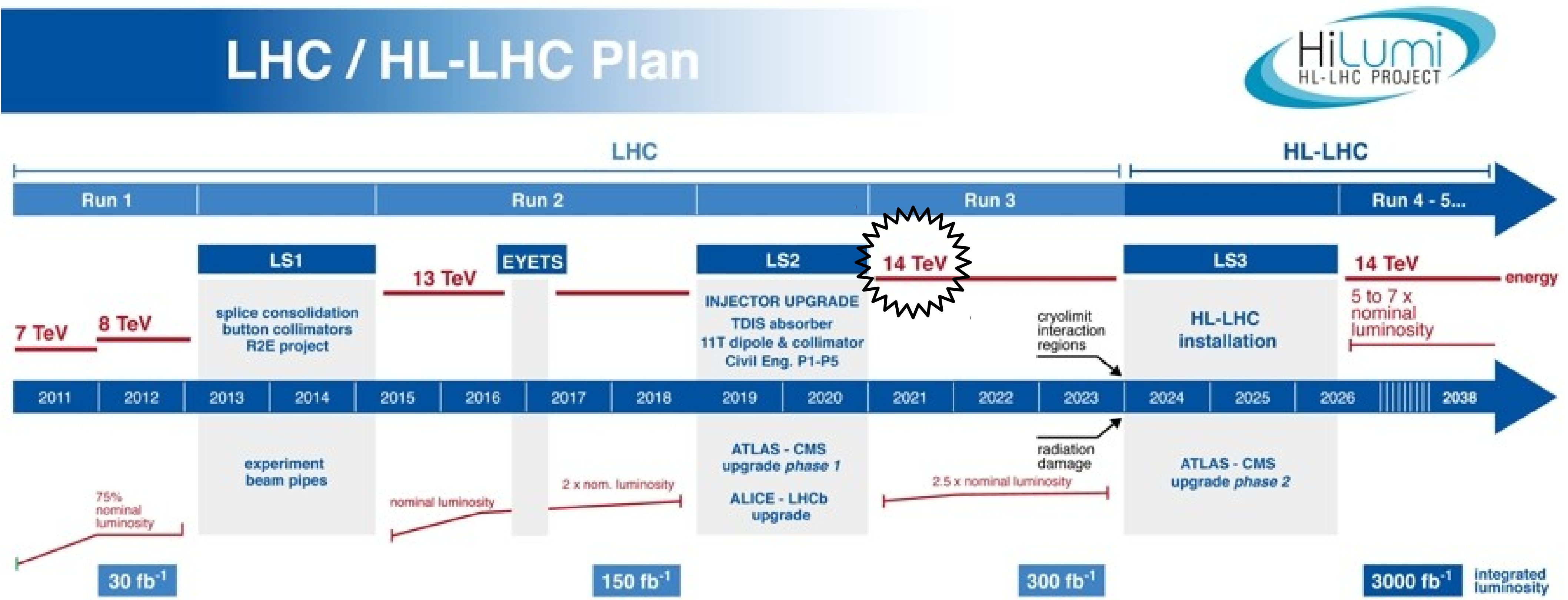
- New surface buildings and an additional radiation free service tunnel in IRs 1 and 5
  - Crab cavity RF power supplies
  - Power converters feeding IR magnets via superconducting links
  - Cryogenic plants dedicated to the experimental area



Run 3


LHC / HL-LHC Plan
HiLumi HL-LHC PROJECT
LHC
HL-LHC
Run 1
Run 2
Run 3
Run 4 - 5...
LS1
EYETS
LS2
LS3
7 TeV
8 TeV
13 TeV
14 TeV
14 TeV
energy
splice consolidation
button collimators
R2E project
INJECTOR UPGRADE
TDIS absorber
11T dipole & collimator
Civil Eng. P1-P5
cryolimit
interaction
regions
HL-LHC
installation
5 to 7 x
nominal
luminosity
2011
2012
2013
2014
2015
2016
2017
2018
2019
2020
2021
2022
2023
2024
2025
2026
2038
experiment
beam pipes
ATLAS - CMS
upgrade phase 1
ALICE - LHCb
upgrade
radiation
damage
ATLAS - CMS
upgrade phase 2
75%
nominal
luminosity
nominal luminosity
2 x nom. luminosity
2.5 x nominal luminosity
30 fb-1
150 fb-1
300 fb-1
3000 fb-1
integrated
luminosity

Run 3


LHC / HL-LHC Plan
HiLumi
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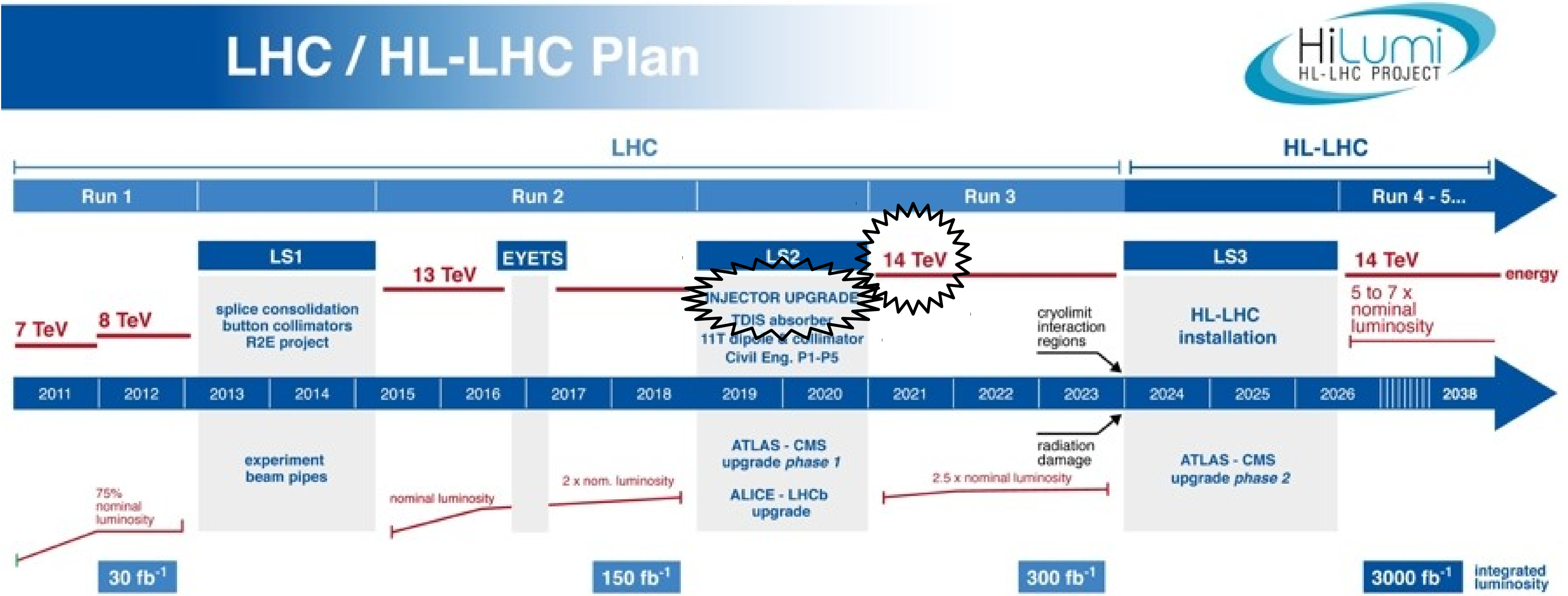
5 to 7 x
nominal
luminosity
2011
2012
2013
2014
2015
2016
2017
2018
2019
2020
2021
2022
2023
2024
2025
2026
2038
experiment
beam pipes
ATLAS - CMS
upgrade phase 1
ALICE - LHCb
upgrade
radiation
damage
ATLAS - CMS
upgrade phase 2
75%
nominal
luminosity
nominal luminosity
2 x nom. luminosity
2.5 x nominal luminosity
30 fb-1
150 fb-1
300 fb-1
3000 fb-1
integrated
luminosity

Run 3



LHC / HL-LHC Plan
HiLumi
HL-LHC PROJECT
LHC
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Run 4 - 5...
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7 TeV
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14 TeV
14 TeV
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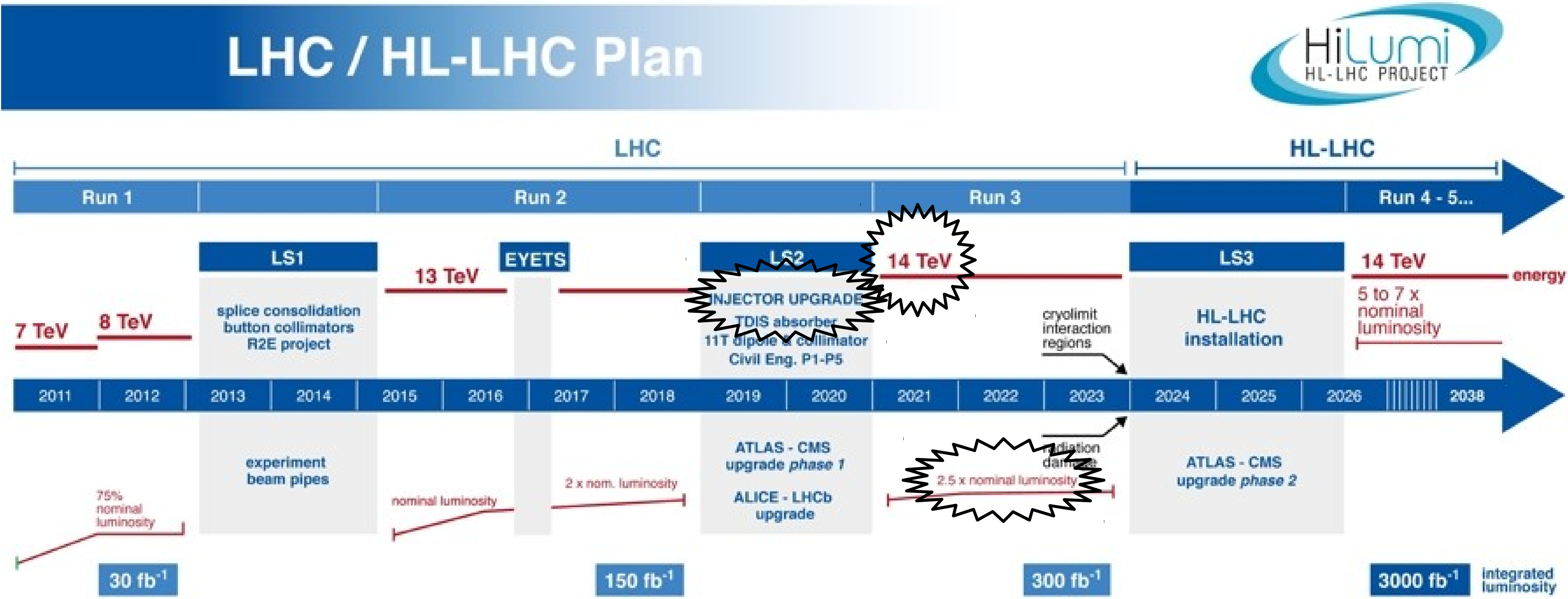
2011
2012
2013
2014
2015
2016
2017
2018
2019
2020
2021
2022
2023
2024
2025
2026
2038
experiment
beam pipes
ATLAS - CMS
upgrade phase 1
ALICE - LHCb
upgrade
radiation
damage
ATLAS - CMS
upgrade phase 2
75%
nominal
luminosity
nominal luminosity
2 x nom. luminosity
2.5 x nominal luminosity
30 fb-1
150 fb-1
300 fb-1
3000 fb-1
integrated
luminosity

Run 3


LHC / HL-LHC Plan
HiLumi
HL-LHC PROJECT
LHC
HL-LHC
Run 1
Run 2
Run 3
Run 4 - 5...
LS1
EYETS
LS2
LS3
7 TeV
8 TeV
13 TeV
14 TeV
14 TeV
energy
splice consolidation
button collimators
R2E project
INJECTOR UPGRADE
TDIS absorber
11T dipole & collimator
Civil Eng. P1-P5
cryolimit
interaction
regions
HL-LHC
installation
5 to 7 x
nominal
luminosity
2011
2012
2013
2014
2015
2016
2017
2018
2019
2020
2021
2022
2023
2024
2025
2026
2038
experiment
beam pipes
ATLAS - CMS
upgrade phase 1
ALICE - LHCb
upgrade
radiation
damage
ATLAS - CMS
upgrade phase 2
75%
nominal
luminosity
nominal luminosity
2 x nom. luminosity
2.5 x nominal luminosity
30 fb-1
150 fb-1
300 fb-1
3000 fb-1
integrated
luminosity

The High Luminosity LHC baseline

High integrated luminosity **300 → 3000 fb^{-1}**
with limited pile-up increase **27 → 130 events / bunch crossing**

High intensity **1.15 → 2.3 10^{11}p**

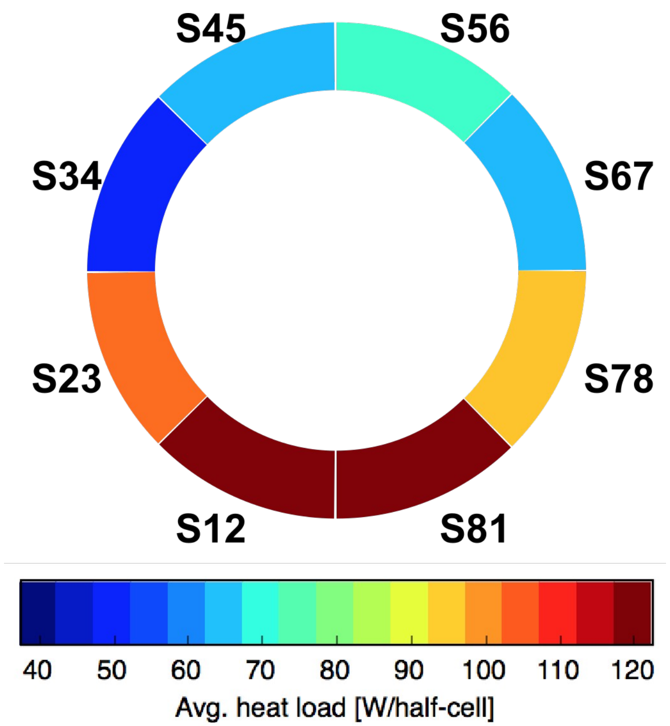
Low β* **55 → 15 cm**

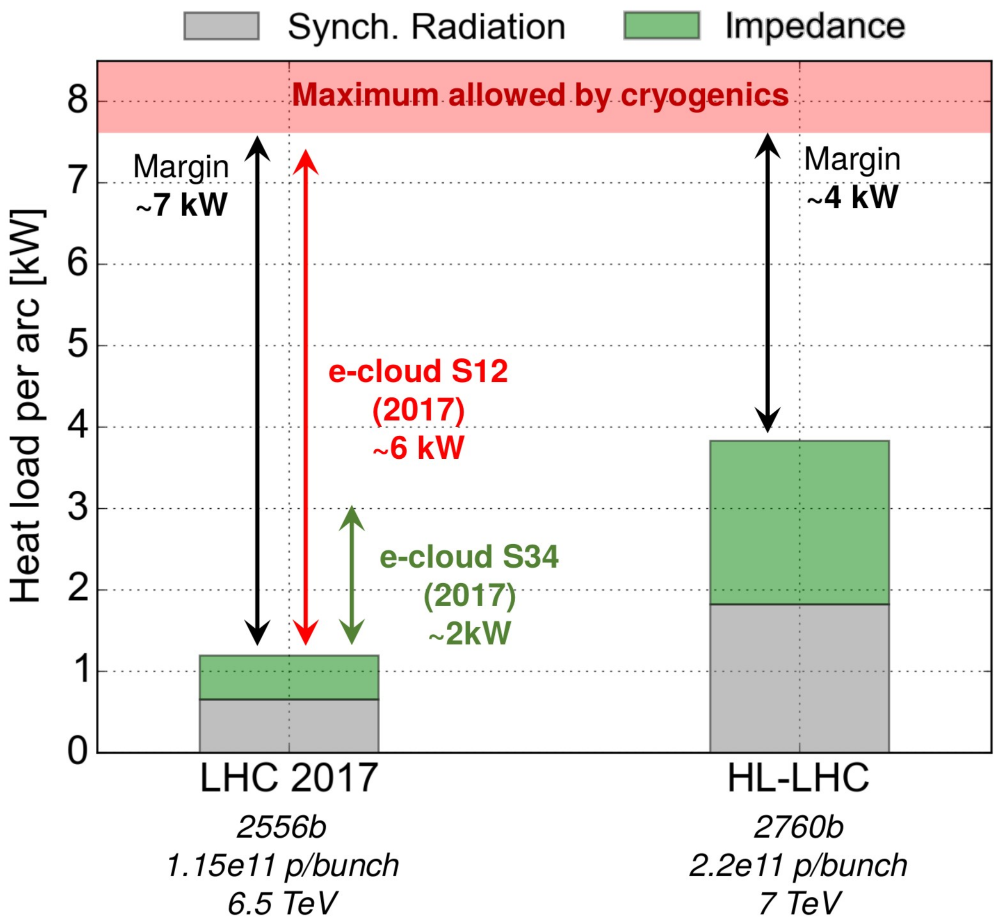
- Injector upgrade (LIU project)
- Collimation
  - Mitigation of losses in the DS (11T dipoles)
  - Low impedance collimators
  - Local cleaning in the IR
- RF full detuning scheme
- aC coating of the beam screen in the IR
- New cryogenic plants

- ATS optics
- Large aperture triplet
  - Beam aperture + shielding
- Crab cavities
  - Bunch shape monitors
- β* luminosity levelling

Available (at least partially) for run 3

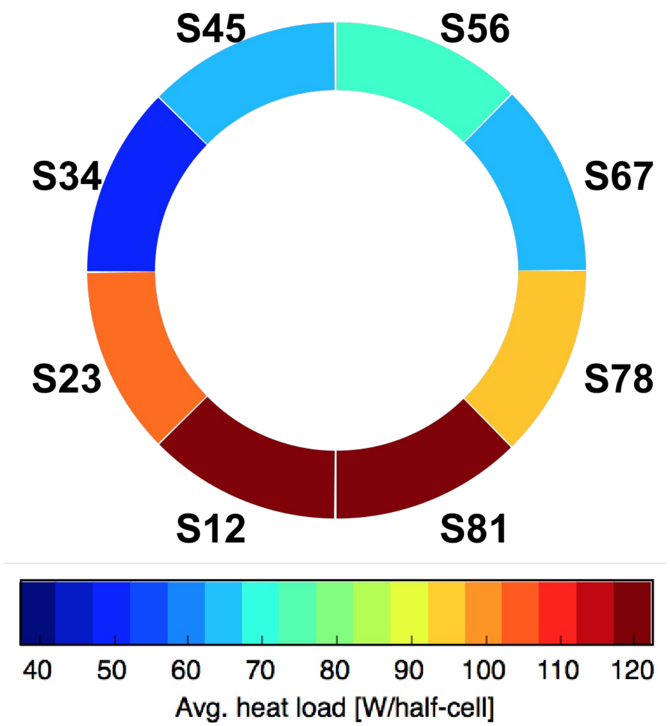
Heat load

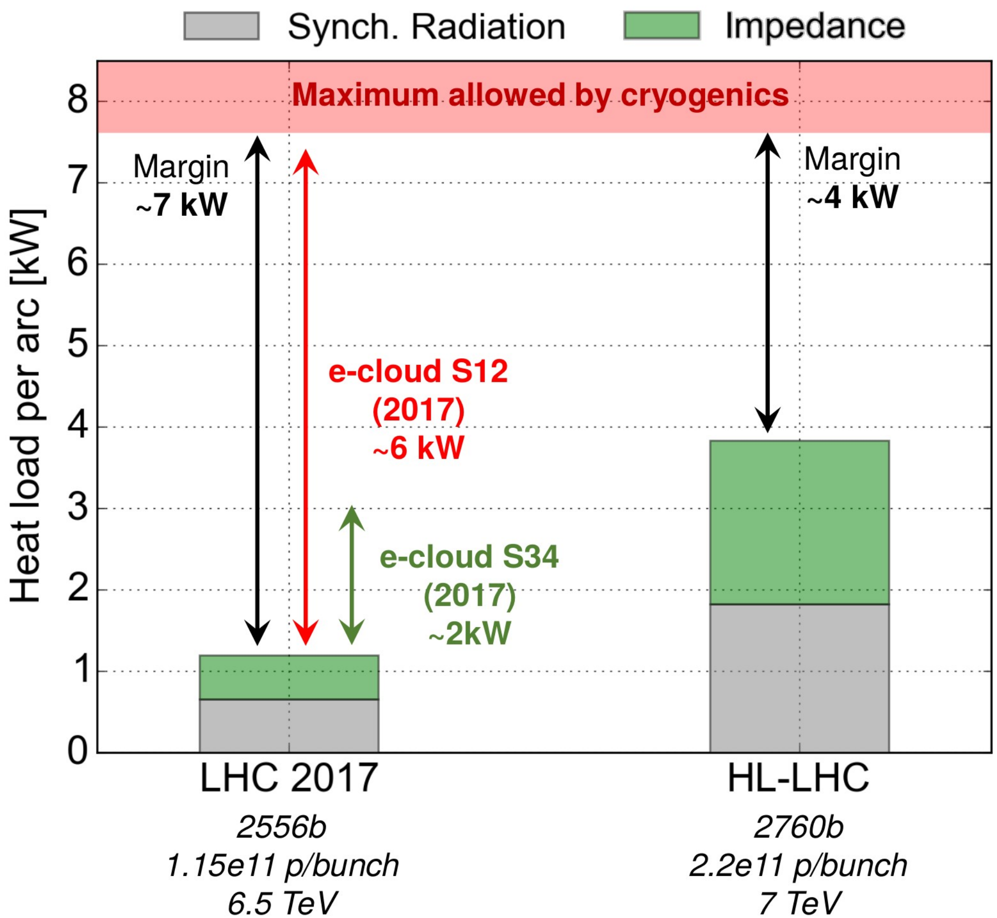




G. Iadarola, et al., Chamonix 2018

Heat load



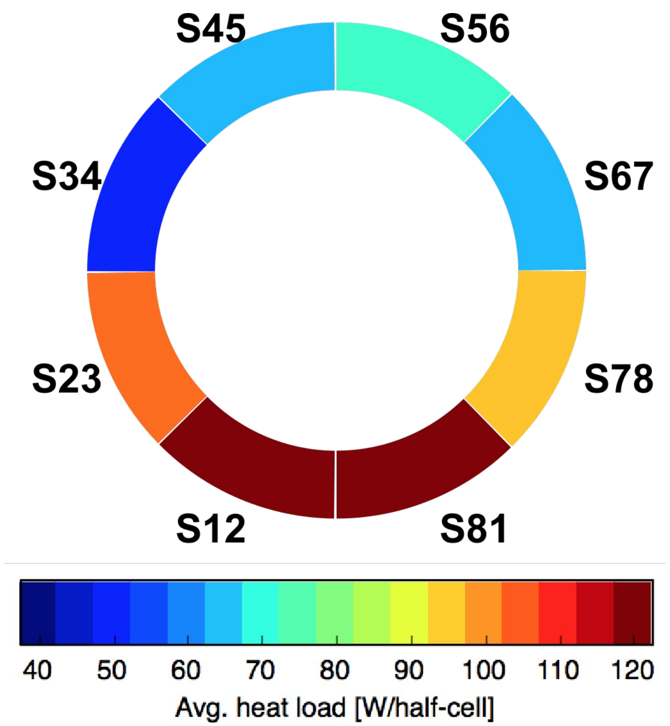


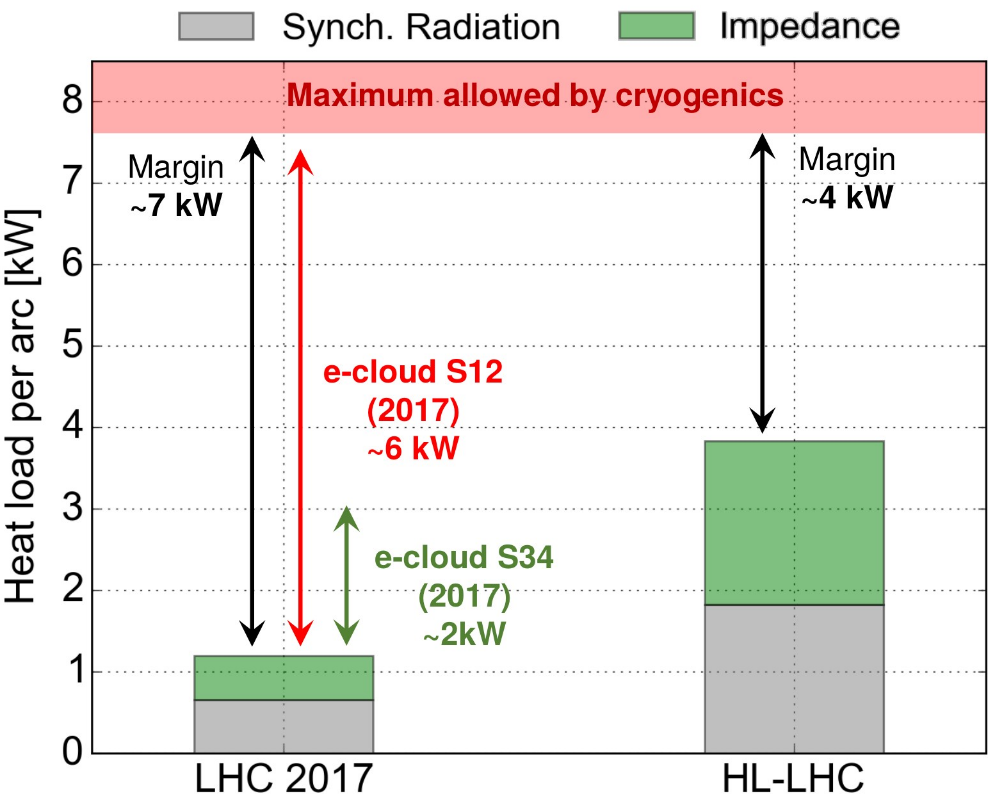
- In run 2, a high heat load was observed in all arcs due to **Electron cloud build up** featuring large differences between cells and sectors

    → The chemical properties of beam screen samples from good and bad cells are under study to understand the underlying cause

G. Iadarola, et al., Chamonix 2018

Heat load





2556b
1.15e11 p/bunch
6.5 TeV

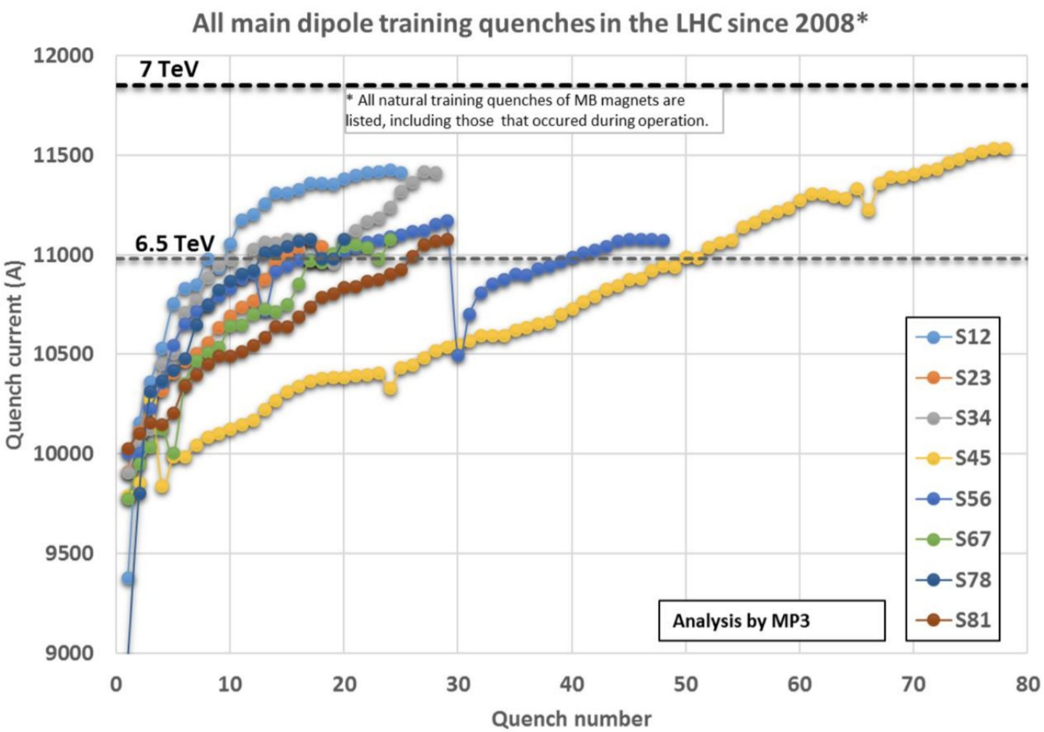
2760b
2.2e11 p/bunch
7 TeV

- In run 2, a high heat load was observed in all arcs due to **Electron cloud build up** featuring large differences between cells and sectors
  → The chemical properties of beam screen samples from good and bad cells are under study to understand the underlying cause
- Mitigation with amorphous carbon coatings are foreseen in the interaction region (triplet, matching quad.)
  - The heat load can be adjusted empirically using mixed filling scheme (25ns, 8b4e, 50ns) but with a cost in performance (pile up limit)

G. Iadarola, et al., Chamonix 2018

Top energy

- The training of the arc dipoles takes longer than expected



→ About 600 quenches are required to reach the design energy (> 3 months)

https://mp3.web.cern.ch/

Top energy

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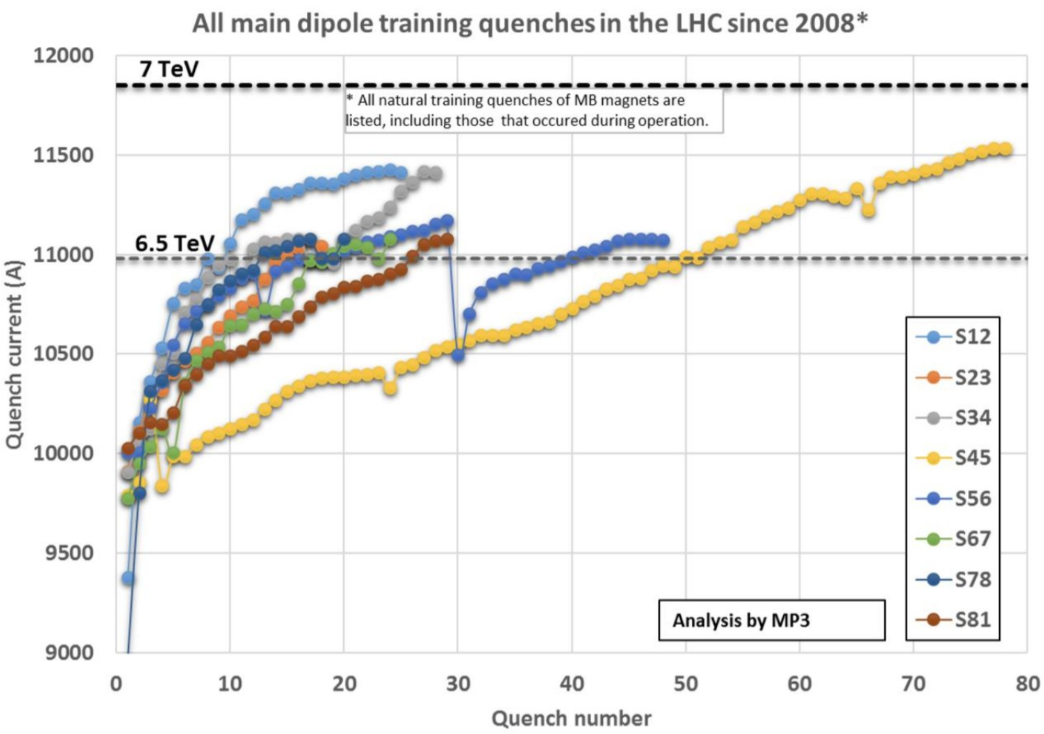


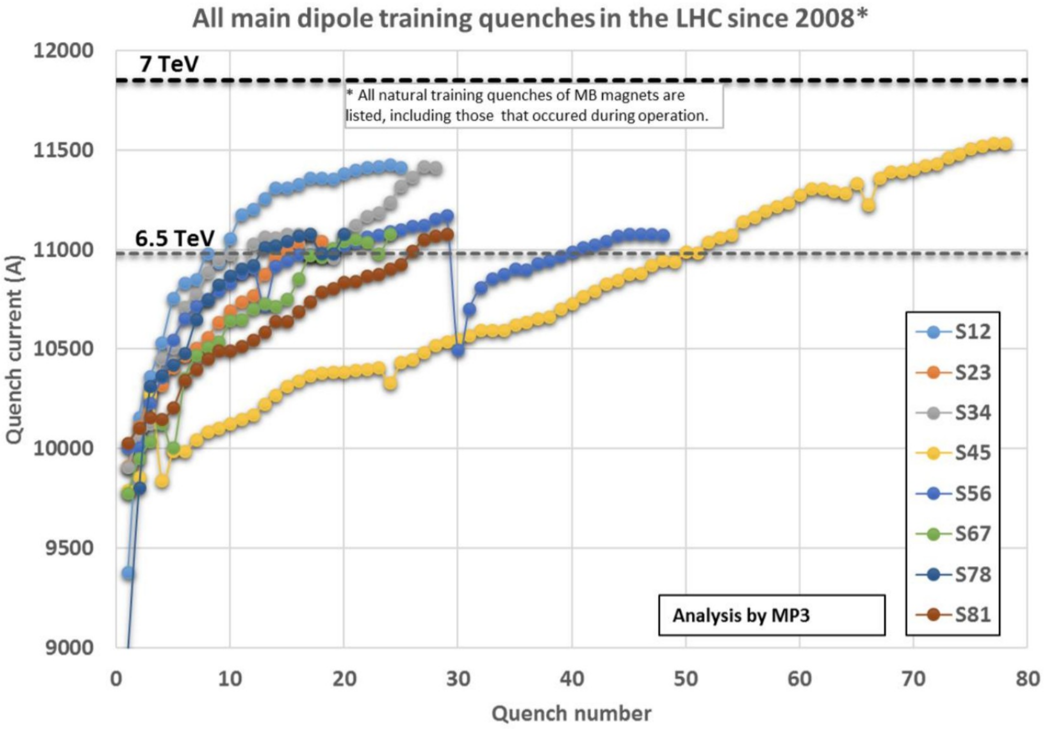
→ About 600 quenches are required to reach the design energy (> 3 months)

- The details of the training strategy for run 3 are under discussion

https://mp3.web.cern.ch/

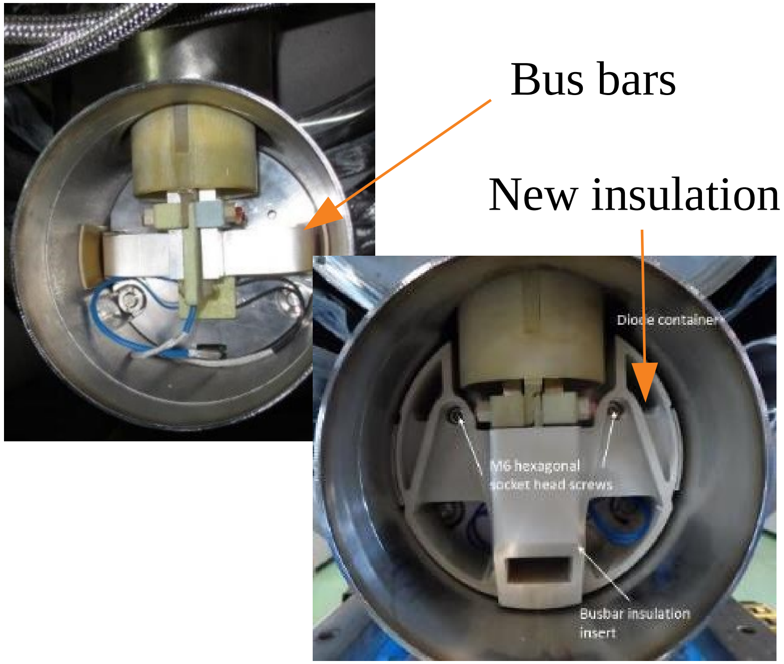
Top energy

- The training of the arc dipoles takes longer than expected



→ About 600 quenches are required to reach the design energy (> 3 months)

→ Consolidation ongoing to mitigate the risk associated with shorts in the cold diode



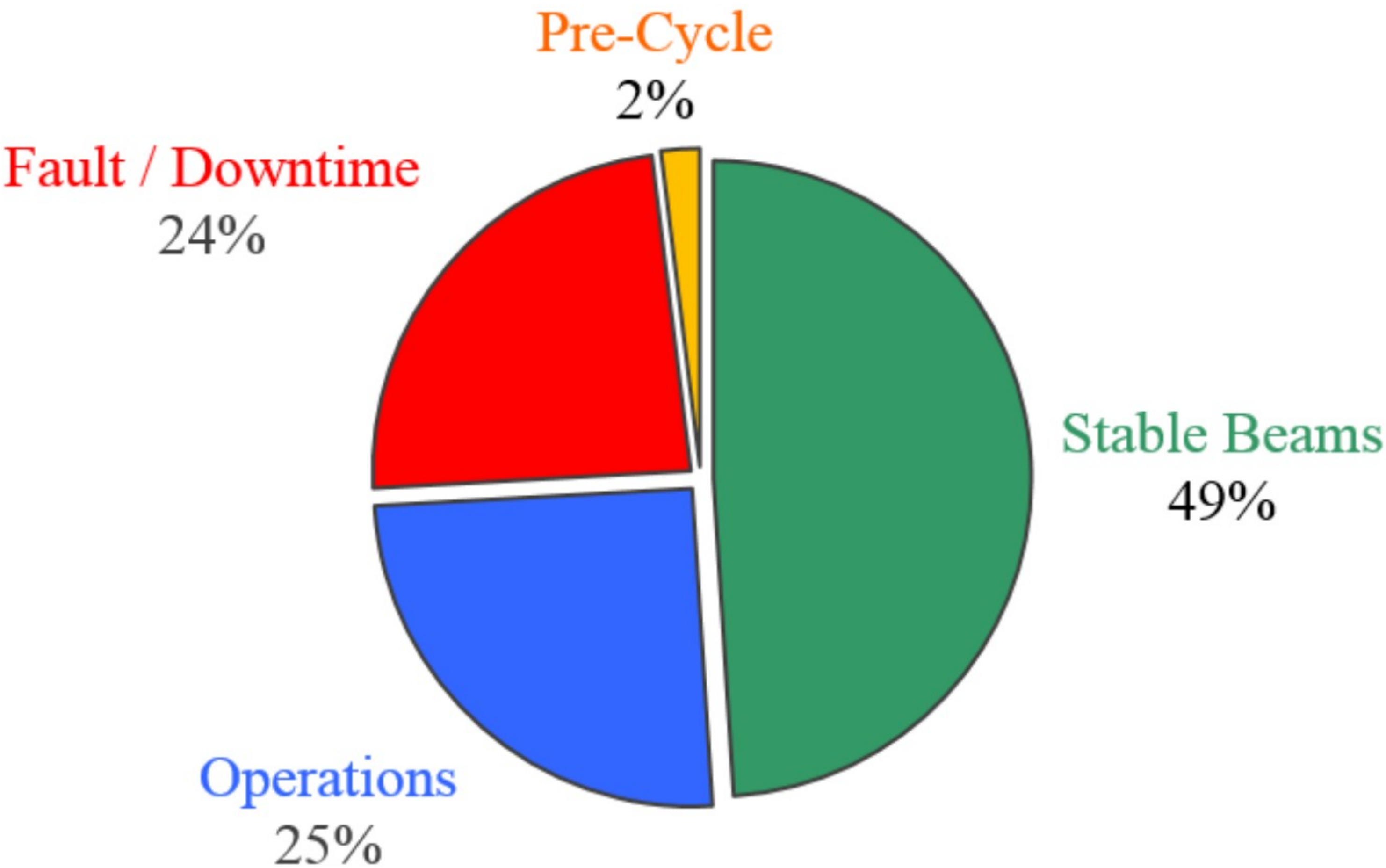
- The details of the training strategy for run 3 are under discussion

https://mp3.web.cern.ch/

Summary

- The LHC has outperformed its design luminosity by a factor 2 and is getting ready to reach the design energy of 7 TeV
- Operation with heavy ions exceeds expectation in terms of integrated luminosity and beam types (Pb-Pb, Pb-p, Xe-Xe, Pb^{81+})
- LHC's remarkable stability, reproducibility and reliability allowed for a high availability (49% data taking) in pushed configurations:
  - Optics design and correction with low β* **55 → 25 cm**
  - Tight collimation scheme **6 times better cleaning**
  - Yet maintaining **flexibility** (luminosity levelling, special runs, mitigation of non-conformities)
- The beam quality delivered by the injectors is a key to achieve the high performance **ε : 3.5 → 2.0 μm**
  - The installation of the LHC Injector upgrade is currently ongoing
- HL-LHC construction is ongoing
  - Magnets prototype testing are ongoing
  - Civil engineering work in points 1 and 5 are mostly on schedule
- Some HL-LHC technologies could be tested in run 2
  - DQW crab cavity prototype was tested with beam in the SPS
  - Beam based measurement of the new collimators' impedance
  - Beam-beam compensation using arc octupoles or wires

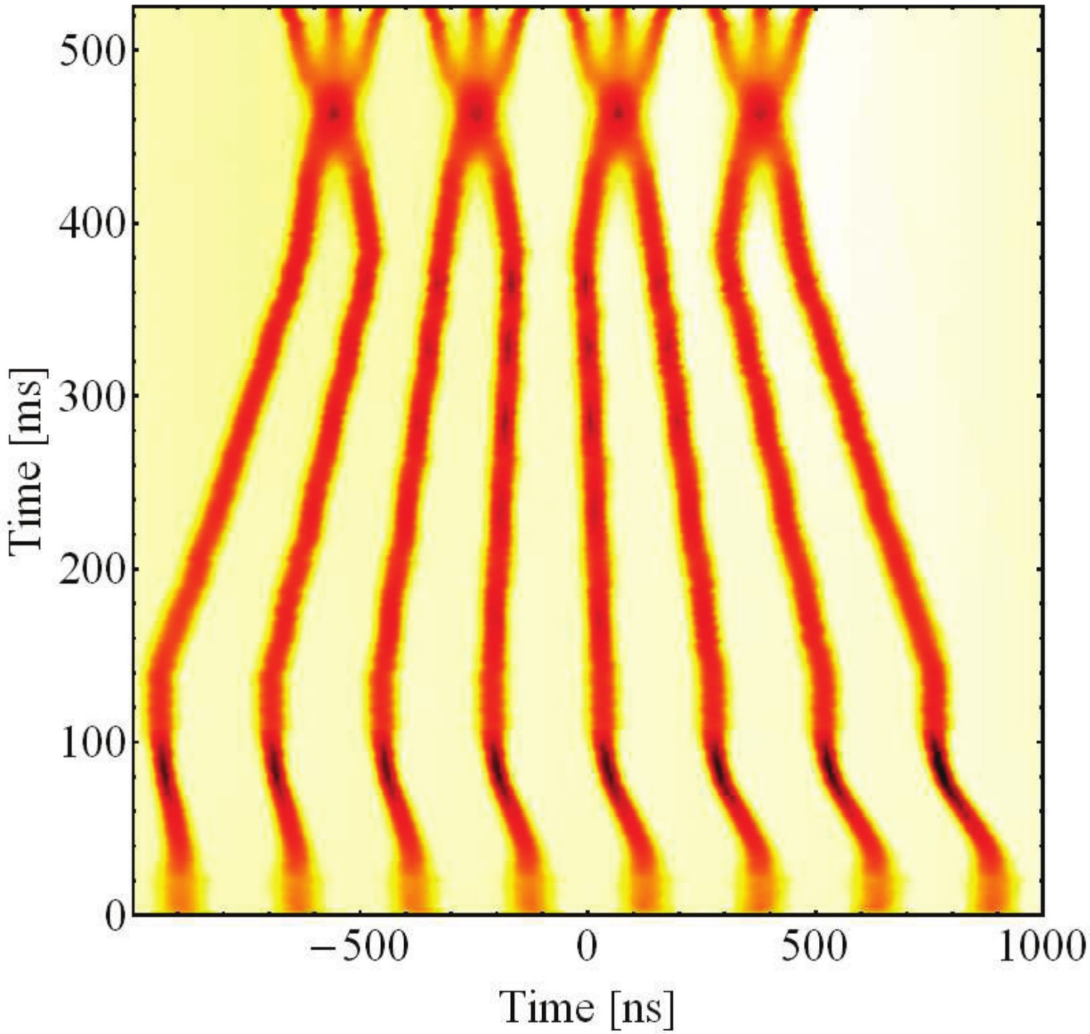
Availability (2018)



- High robustness of numerous individual components and operational procedures
- Fast reaction to issues (People and machine's flexibility, on-call specialists)

A. Apollonio, et al., in Proc. IPAC'17 and B. Todd, et al., in Proc. Evian 2019

Batch Splitting and Merging Scheme (BCMS)



Triple splitting h = 21

Bunch pair merging h = 7

Batch compression h = 14

Energy ramp from 1.4 to 2.5 GeV (longitudinal acceptance)

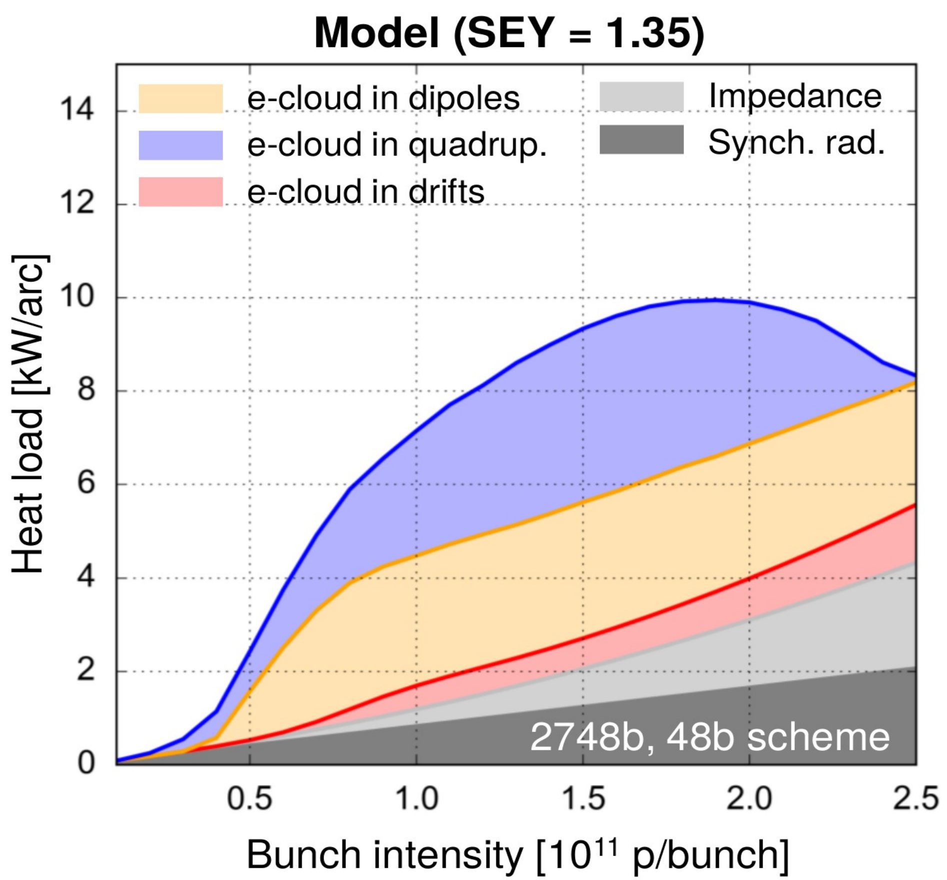
Two injections of 4 bunches from the PS booster on h = 9

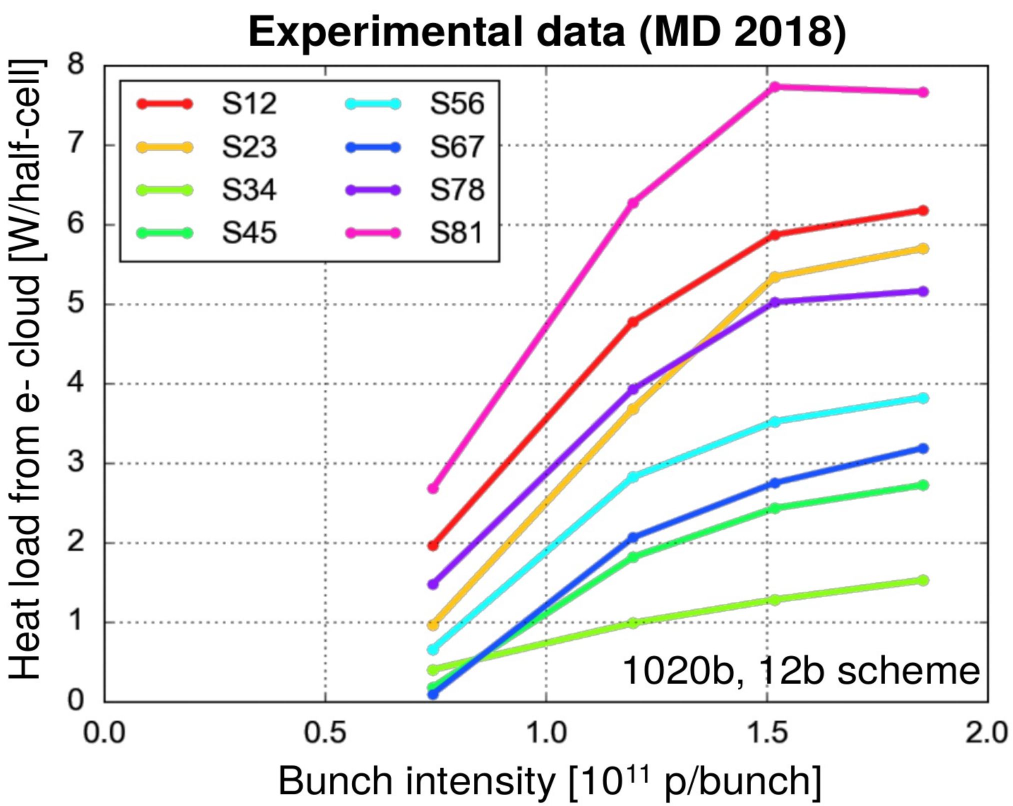
H. Damerau, et al., in Proc. IPAC’13

Dependence of e-cloud heat loads on bunch intensity

- **Higher bunch intensity** will become available from the injectors during Run 3
- With the available models, **simulations foresee a relatively mild increase** of the heat load from e-cloud above 1.2×10^{11} p/bunch
- Direct **experimental checks were not possible in Run 2** (RF limitations in the SPS)
- At end 2018, **trains of 12b with high bunch intensity** became available from the SPS

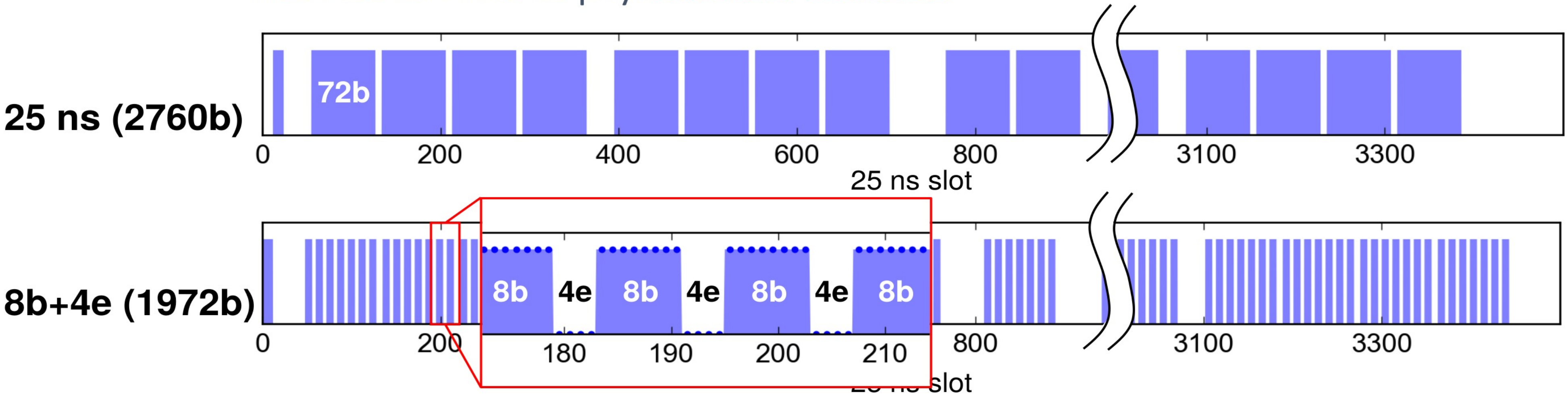
  → Tests done during LHC MD4 **confirmed ~flat dependence above 1.5×10^{11} p/b!**



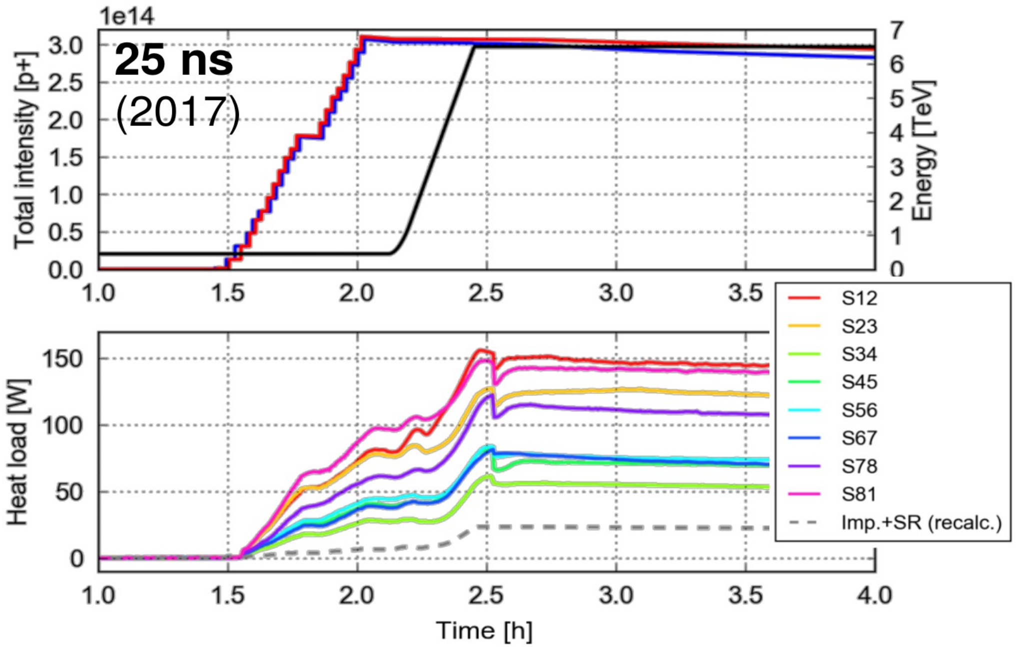


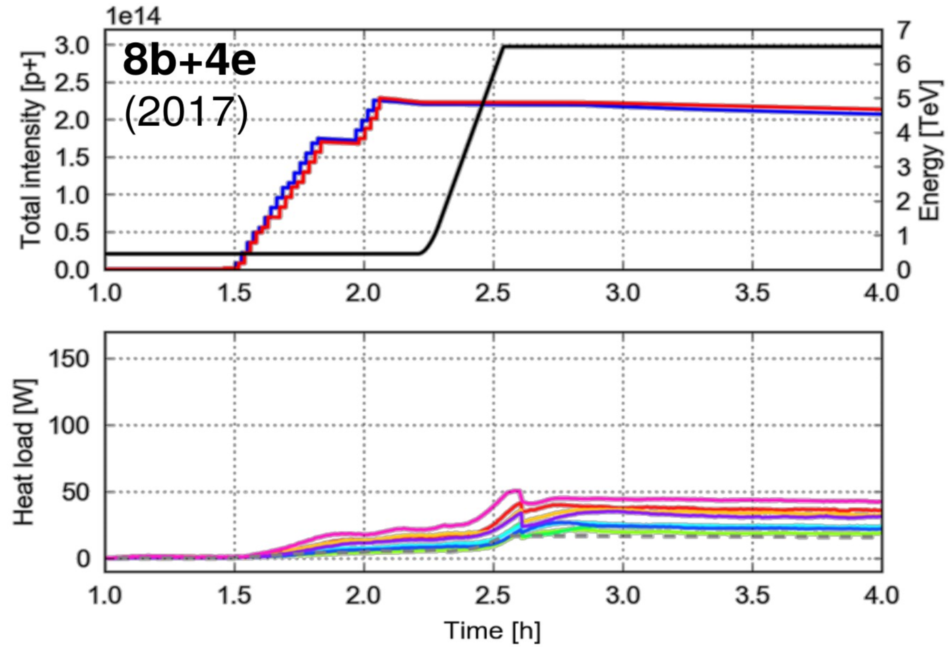
Arc heat loads - 8b+4e backup scheme

8b+4e: filling pattern conceived to **suppress the e-cloud** and hence reduce the heat loads. Price to pay: **30% less bunches**.



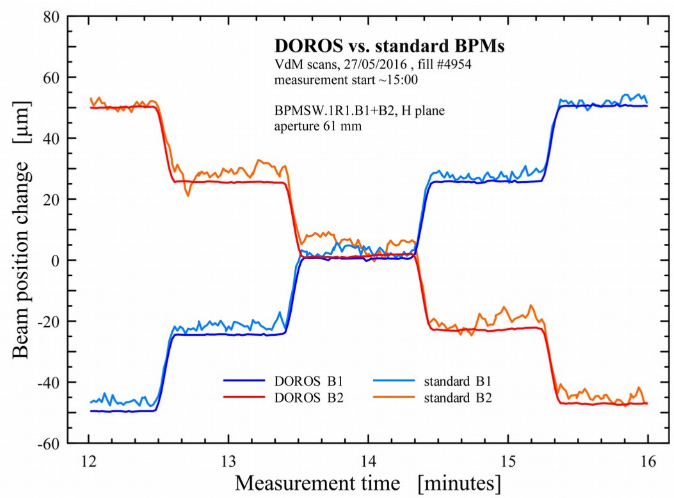
→ Successfully **tested** in the LHC in **2015** and **used in operation** in the last part of the **2017 Run** (to mitigate fast losses in 16L2)



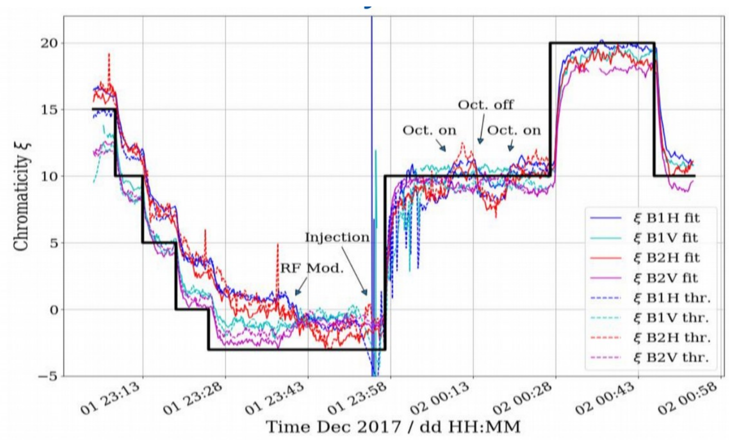


Operational and experimental diagnostics potpourri

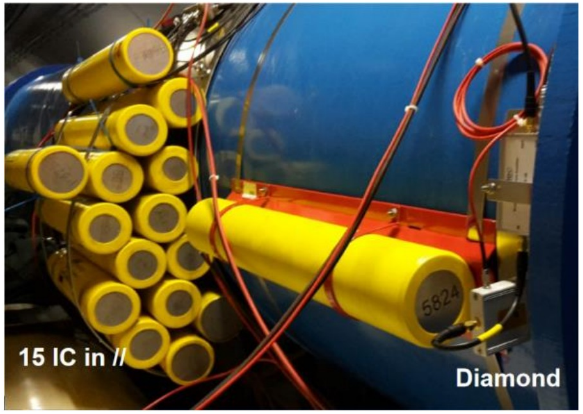
- High resolution position measurement using DOROS (M. Gasior, et al., Proc. IBIC'16)



- Non destructive chromaticity measurement using Shottky monitor (T. Tydecks, et al., Proc IPAC'18)

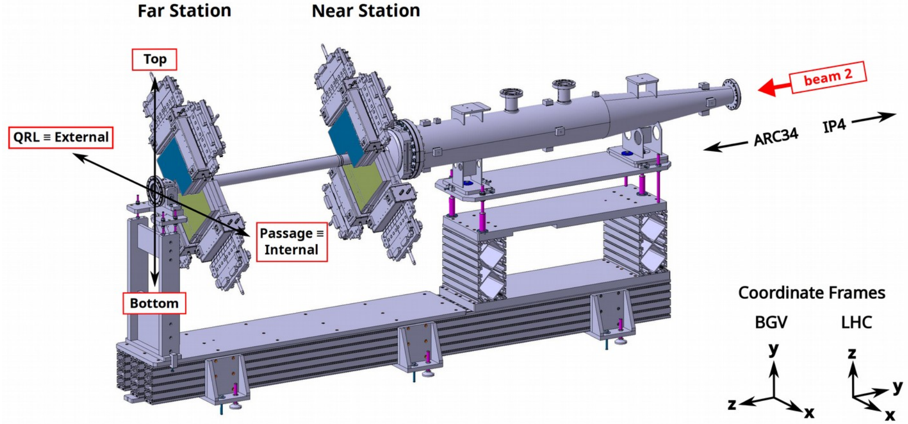


- Stacked ionization chambers for for high sensitivity beam loss measurement at 16L2 combined with single bunch loss measurement with a diamond based BLM (M. Krup, et al., Proc. Evian Beam Operation Worksop 2019)

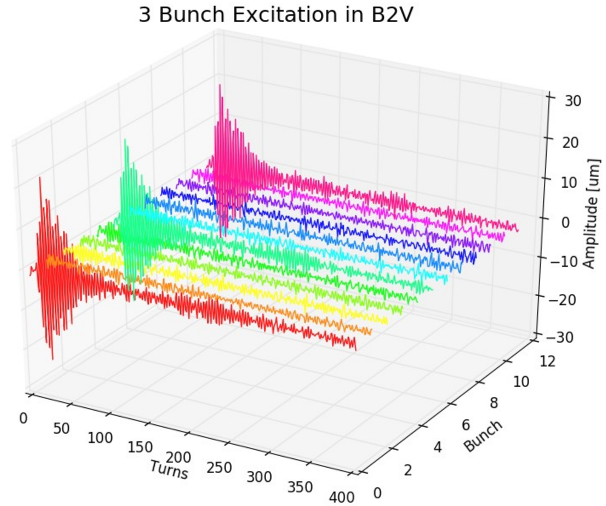


- Transverse halo monitoring using the synchrotron light monitor as a coronagraph (G. Trad, et al., Proc. IPAC'17)

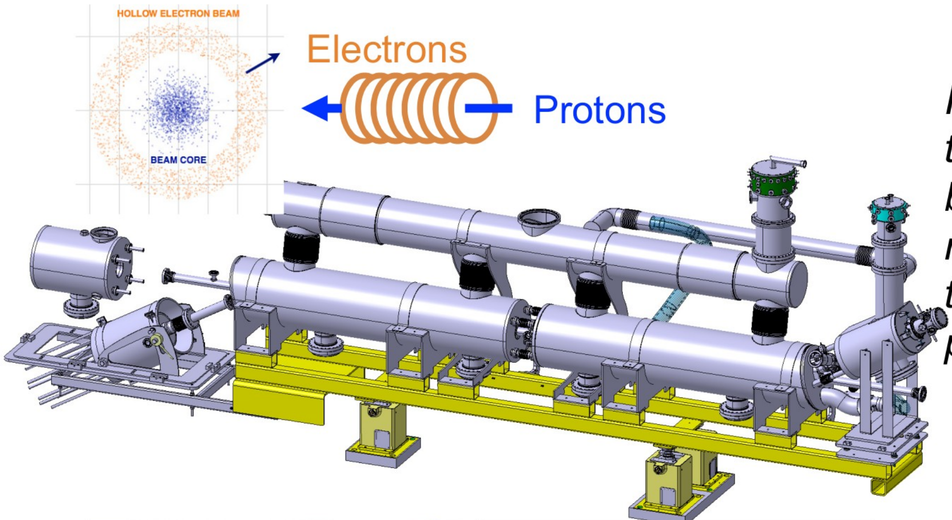
- Beam gas vertex monitor for non-invasive transverse beam size measurement (A. Alexopoulos et al. Phys. Rev. Accel. Beams 22, 042801 (2019)



- Single bunch kick and turn-by-turn position measurement for tune and coupling measurements as well as instability analysis (M. Soderen, et al., Proc. 2019 Evian Beam Operation Workshop)



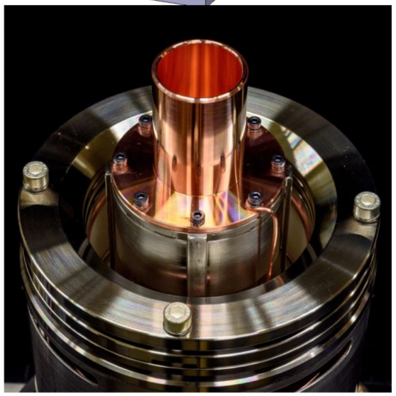
E-lens in HL-LHC for halo control - 30 MJ in the halo



It would allow controlling actively the halo, through a hollow electron beam (overlapped over three meters to the proton/ion beams) that selectively excites halo particles.

HL WP5: S. Redaelli, D. Perini, A. Rossi et al.

Cathode



Electron gun

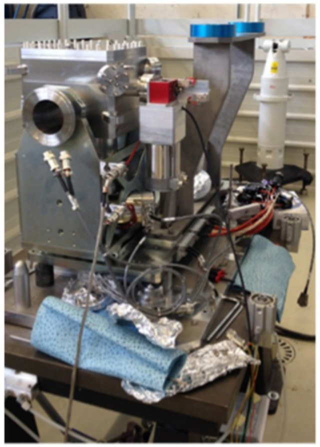
Design nearly complete.
Surpassed target e-beam current of 5A, now final cathode design (smaller) under test at FNAL.

Ready to built it, heading to integrated into the baseline.

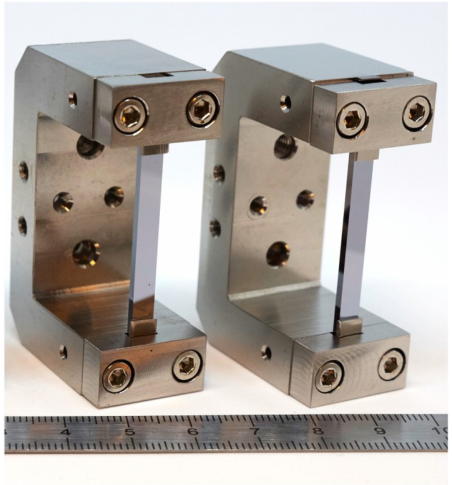
Test on crystal collimation (for baseline)

<u>Scope</u>: further improvement of ion cleaning after 2016 re-baselining.
***Studying if, for ions**, this can be an "adiabatic" upgrade of the IR7 system.*
*2017: **improved by up to x60 collimation cleaning** of Xe beams!*

Courtesy EN/SMM

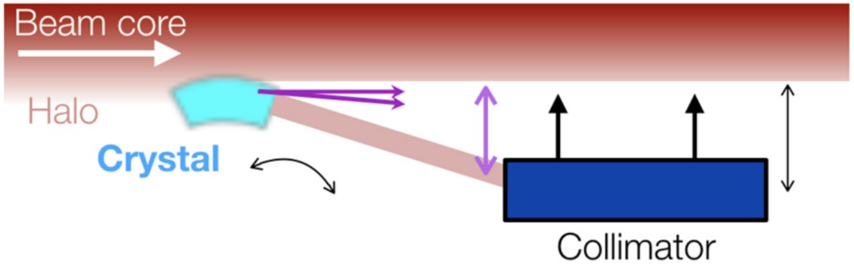


Courtesy UA9 collaboration/PNPI



4 mm = 50 μrad,
or 10 x 15m long LHC dipoles
or 300 T at 7 TeV

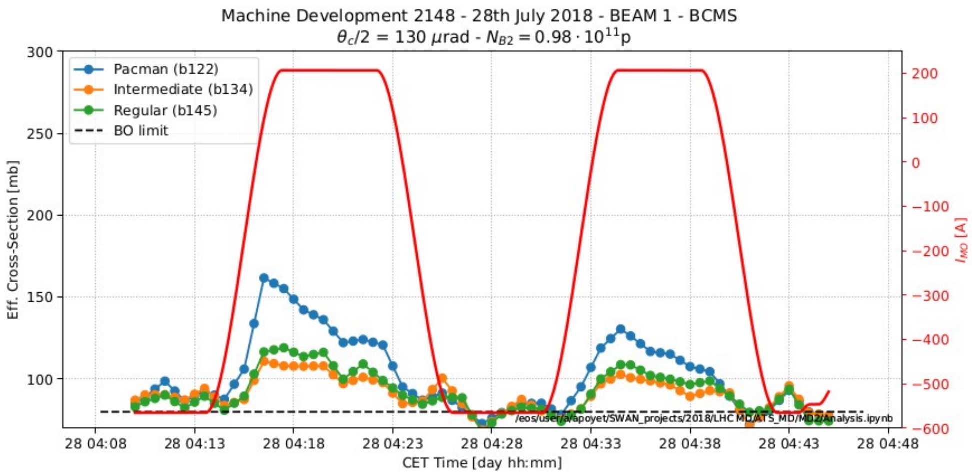
Two goniometers installed on B1 in LS2; two more on B2 in 2017, upgraded in 2018.
4 operational crystals for collimation.

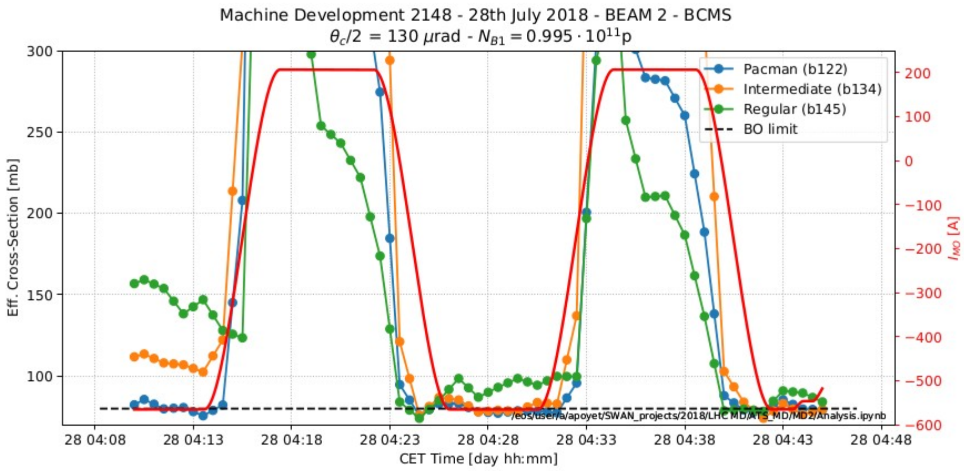


HL WP5: S. Redaelli, S. Gilardoni, M. Calviani al.

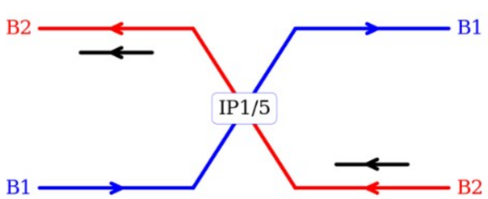
Global and local long-range beam-beam compensation

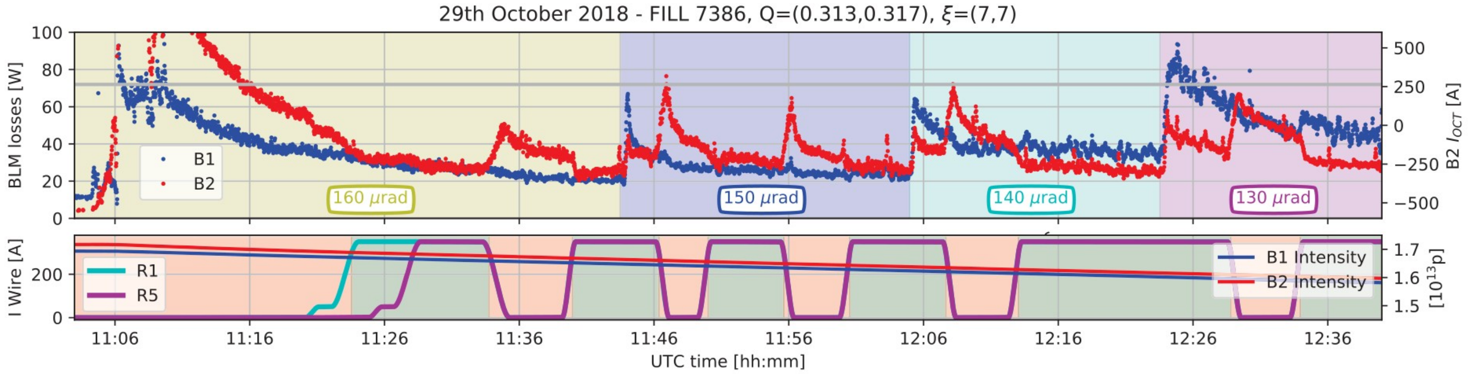
- A significant improvement of the beam lifetime was obtained when operating with a low crossing angle
  - Using the arc octupole with the ATS (S. Fartoukh, et al., CERN-ACC-2018-0032)





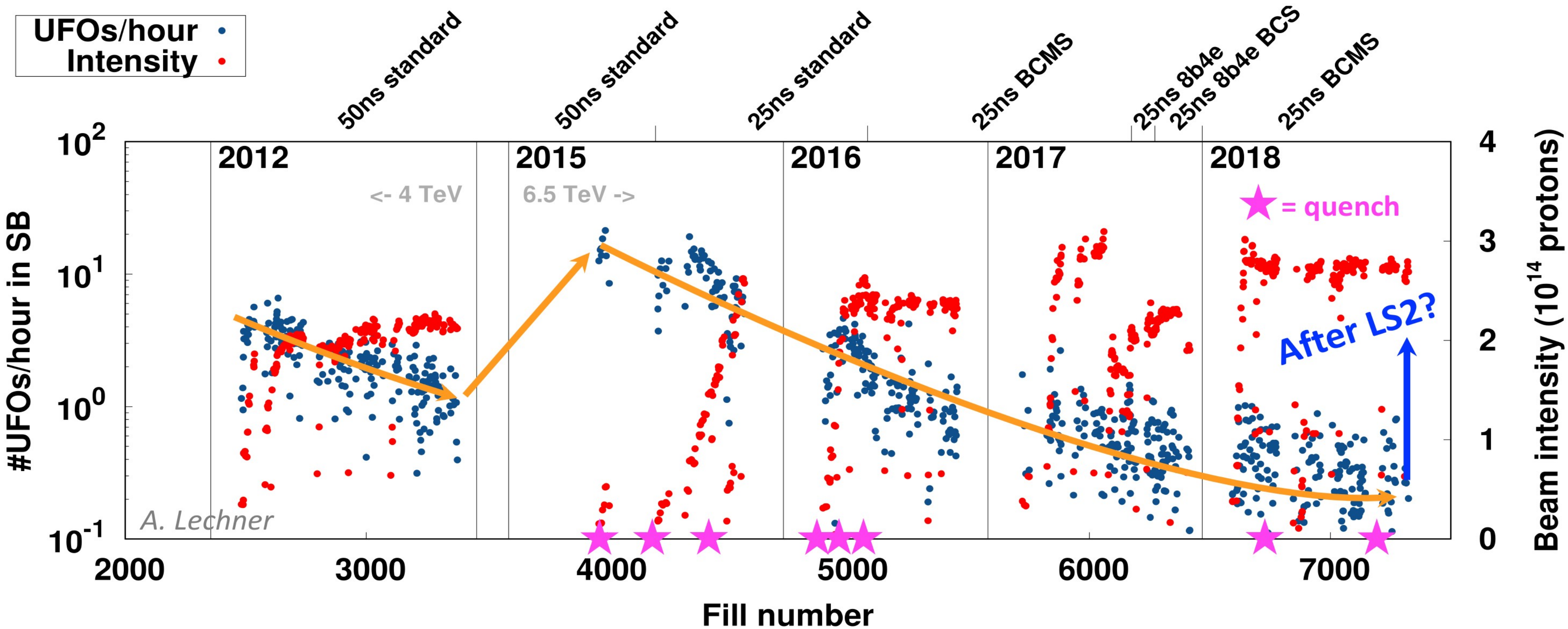
  - Using wires embedded in collimator jaws (G. Sterbini, et al., in Proc. IPAC'19)





UFO rate evolution

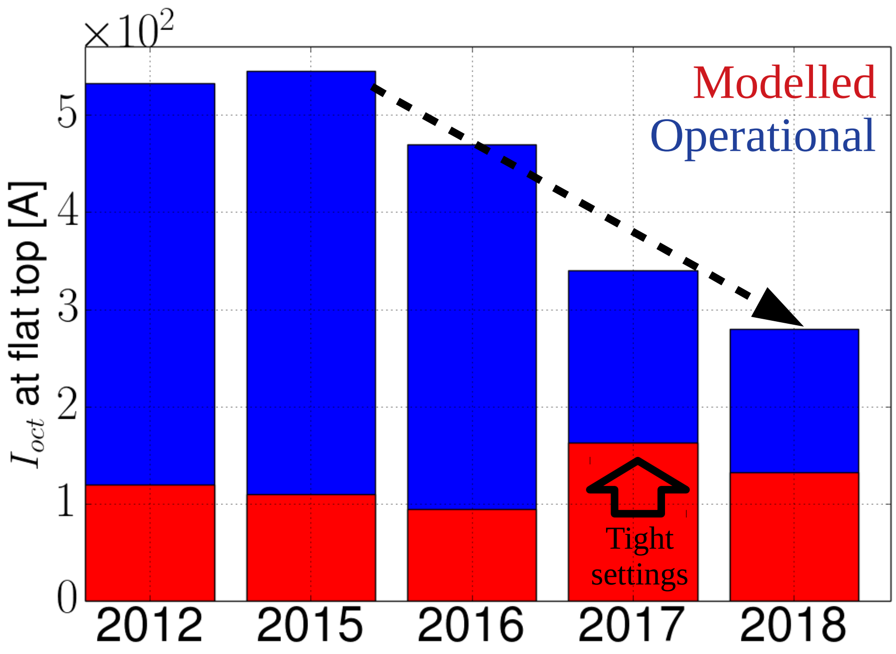
UFO rates in the arc (cells ≥12) in stable beams

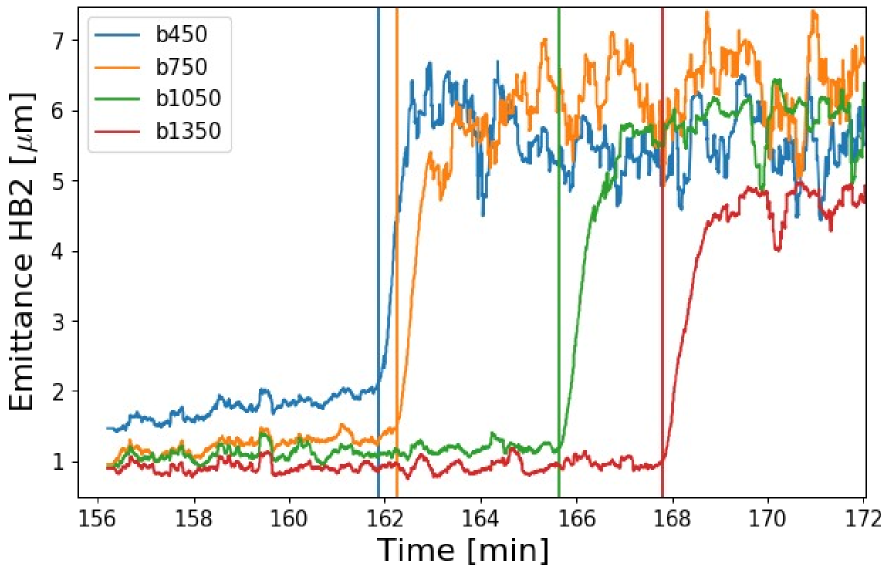


- Clear **de-conditioning** during **LS1** and **re-conditioning** during **Run 2**

  → **Similar UFO rate** as end of Run 1 achieved **after about 1.5-2 years**

Collective instabilities



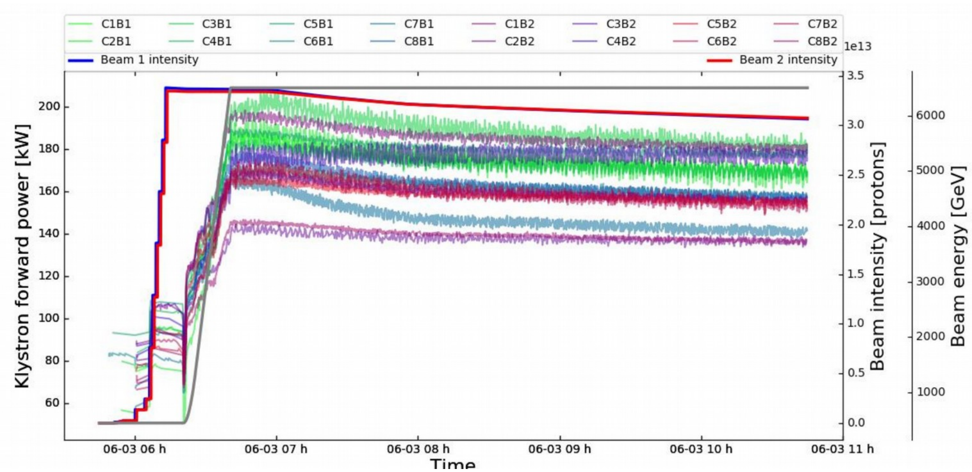


- The large additional octupole current required for Landau damping observed in run 1 (2012) could be lifted with better optics control (linear coupling, triplet non-linearities)
- The impact of noise on the beam stability seem to play a role in the long term stability of the beam

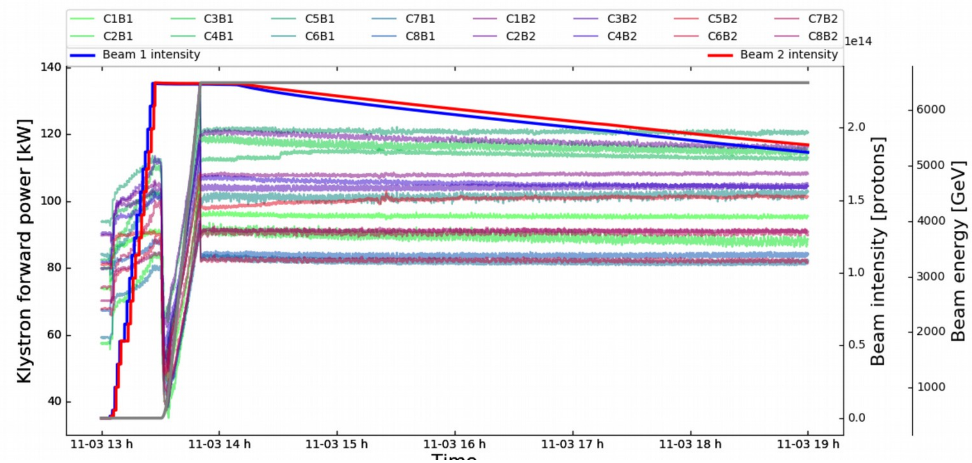
X. Buffat, et al., in Proc. Evian 2019

RF full detuning scheme

Half detuning scheme



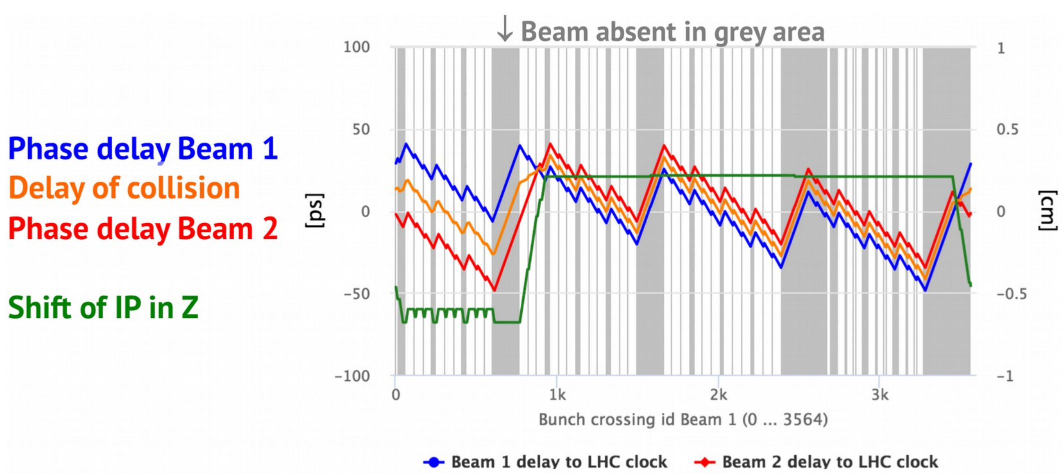
Full detuning scheme



- The RF detuning scheme allows for a reduction of the RF power required to compensate transient beam loading

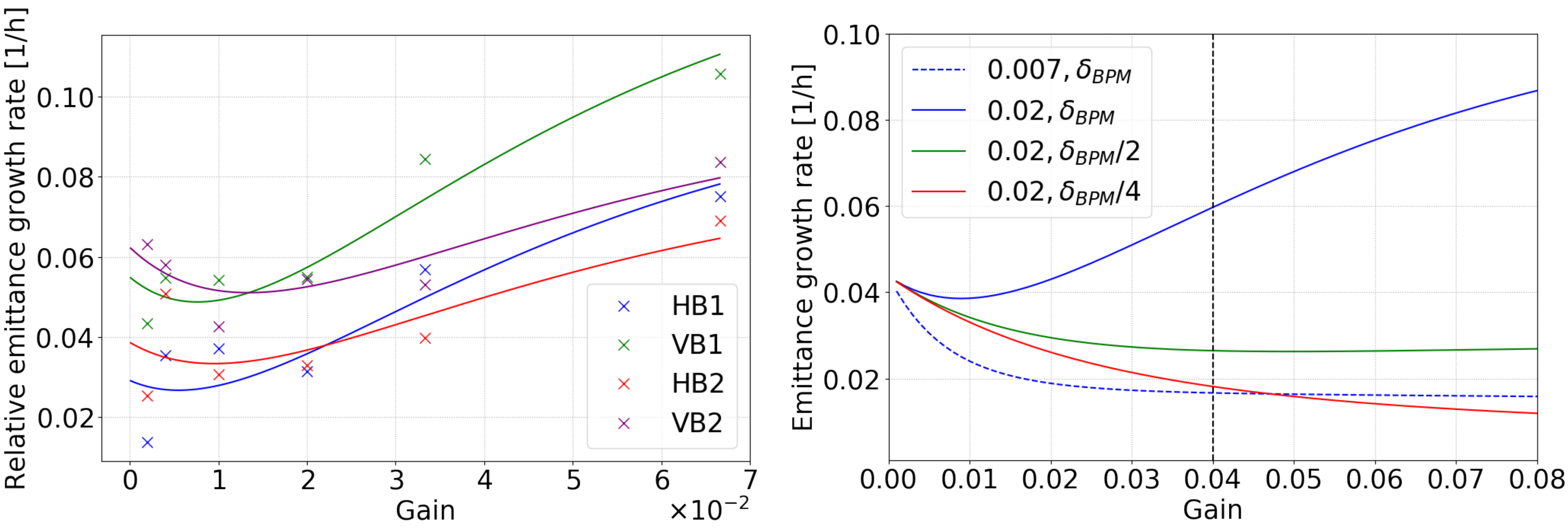
  → Required to cope with HL-LHC bunch intensity

  → The resulting bunch-by-bunch RF phase shift remains acceptable for the detectors



H. Timko, et al., Proc. Evian 2017

Noise studies



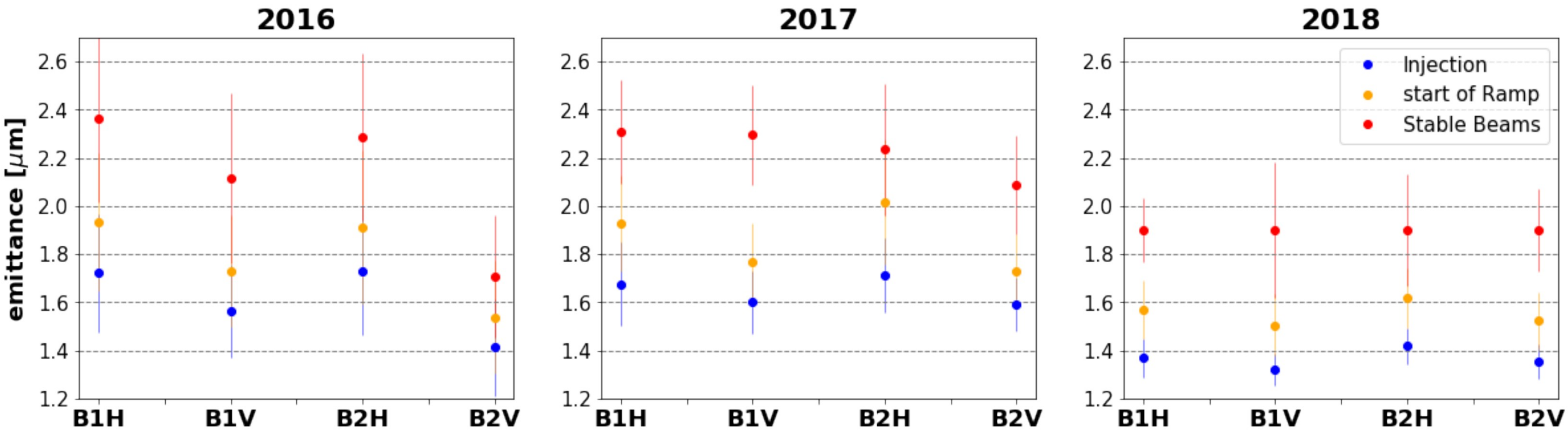
- Emittance growth rate in collision with single bunches featuring HL-LHC brightness showed that the feedback noise dominates becomes the dominant contribution

  → Currently the LHC feature a much lower beam-beam tune shift, such that the suppression of the emittance growth by the feedback remains dominant

  → An improvement of the feedback pickups noise floor is required for HL-LHC

X. Buffat, et al., CERN-ACC-NOTE-2018-0005

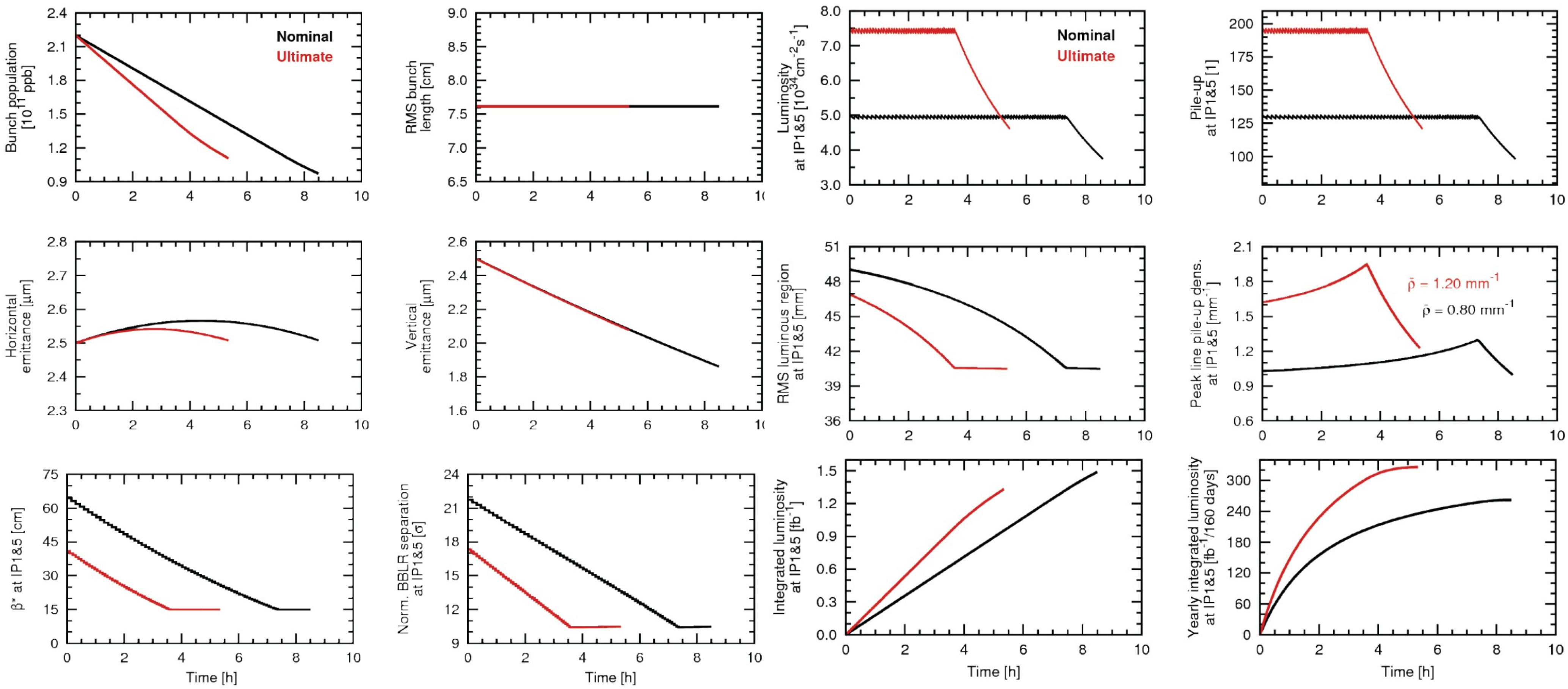
Emittance preservation



- The injected transverse emittances are not fully preserved through the cycle
  - IBS does not fully explain the emittance growth at flat bottom → **electron cloud**
  - A significant growth (~0.5μm) seem to occur during the ramp, its cause is currently unknown

S. Papadopoulou, et al. In Proc. Evian 2019

HL-LHC beam parameter evolution



E. Metral, et al., CERN-ACC-NOTE-2018-0002

LHC cycle example